

Neur 320 The Neuroscience of Human Nature

Numbered Transcript

NEUR 320 • Dr. Baland Jalal

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Lecture 1: The Constructed Self

At a glance — Jalal opens on human nature and the neuroscience of the self — how the brain constructs our sense of who we are.

- 1.1** In this course, I'll be talking about human nature and the neuroscience of human nature. Initially, we'll talk about the self, what the self is, how the self is constructed in the brain. Then we'll talk about the social self; we all have a social sense of who we are and how we deal with other people. We'll then move on to talk about emotions, basic instincts, attraction, how does attraction come about? Complex emotions, jealousy, love, imaginations, beliefs. Then we'll head towards things like consciousness. How do these tiny wisps of protoplasm inside the brain give rise to what we call consciousness, awareness, the sense of I, the sense of me, Baland? Or sense of Heavenly, or Brance, and so forth?
- 1.2** But really, it's interesting. You're dealing with this three-pound mass of jelly resting between your shoulders. From it, all your poetry, pity, infinity, Shakespeare, all of that arises from that. It's pretty unique when you think about it. It's just this lump of jelly-like substance that you can hold in the palm of your hands. And so how does that come about? What's going on inside your brain? And in this particular lecture, I'm going to focus on two aspects of the self.
- 1.3** First is unity. So unity of consciousness. We feel like a single agent in space and time. We don't feel fragmented. We feel anchored in this body. But I will also talk about how the brain in itself is divided. So we have two hemispheres, and these hemispheres are actually, they have their unique personalities. One of them is very unique in its approach to the world, and the other one is completely different. And how does that come about in the brain? What's going on? Why are these two hemispheres so different?
- 1.4** And mind you, the two hemispheres of the brain are linked by a little bridge called the corpus callosum. So these are bands of fibers connecting the two hemispheres, allowing them to communicate. If I was to cut the corpus callosum and divide your brain, you would literally have two personalities inside your brain. You'd be two people.
- 1.5** And you can communicate with each hemisphere. So that's possible. So if I was to watch a screen, so I have a screen in front of me, and I see a flashing, let's say, I ask my left hemisphere—let's say it's cut now, the corpus callosum is cut. So my two hemispheres cannot communicate. And I flash, let's say, I ask, do you like chocolate or vanilla? And ask the right hemisphere, it might say chocolate. Then you ask the left, it might say vanilla. So you have literally two people inside of you existing within you.
- 1.6** But it's interesting, the fact that the human brain, it's made up of all these parts. So you have the frontal lobes, the CEO of the brain, the prefrontal region. Then you have the

temporal lobes where emotion is processed. Right, Naomi? Emotions and things like recognizing faces and meaning and things like that comes about in the temporal lobes.

- 1.7 Then you have the parietal is right here, parietal lobes. In this part, you have things like creating a sense of a body representation in space. Where am I in space right now? I feel anchored in this body. If there's a stroke or damage to the parietal lobe, I feel like I'm floating out in space, for example. So that's one thing that can occur.
- 1.8 In fact, these right parietal regions, if I have a, let's say, oxygen deprivation in that part the region, that could account for things like near-death experiences, some people will say. Imagine you have a sense of a self, a body image, right? That neurologically sense of "you," the feeling that I occupy this space right here. Now, if there's deprivation of oxygen, or stroke, or trauma, you have a feeling like you're outside yourself. You're watching from outside.
- 1.9 And so some people argue, in fact, that near-death experiences are about you lying on the hospital bed and feeling like—there's some oxygen deprivation in that region of the brain—you're feeling like you're floating outside. You're seeing things from the outside. So that's actually an argument. It's very interesting. Though others have argued that REM-near-death experiences, and things like that, are more about the REM state and people that are highly inclined to have REM interruption and rapid dream-like tendencies of REM sleep. So that's another argument. But the right parietal is important for that part. Is that clear? Okay.
- 1.10 The final lobe is the visual cortex in the back of your brain, Tony. Right here. It has to do with vision, seeing. And there's about 30 areas in the brain involved for seeing alone. It's the most important sense for survival. So there's a huge area of the brain dedicated to vision alone.
- 1.11 So you have these four lobes. Now, if I was to cut any of these out of your brain, if I were to take away the frontal lobes, you would not survive. Temporal, you won't survive. Occipital, you won't survive. Any of these big parts of the brain. But interestingly, if you cut the corpus callosum, the middle of the brain, the center, and divide the brain, you will survive. You can do that. So that's very interesting. In fact, you have people surviving with one hemisphere alone, which shows you that the two hemispheres are essentially duplicates of each other. They are duplicates, copies of each other.
- 1.12 Initially, as kids, as children, the functions of the two hemispheres are very similar. They do basically—the left does what the right does and so forth. There's not much special—what we call laterality of function. They're not very specialized. But as we age, each hemisphere gets more specialized with time.
- 1.13 The language is in the left hemisphere. So there's an area in the temporal lobes that has to do with understanding of speech, conversation, and language. It's called the

Wernicke's area. Right in front of it, you have the Broca's area. Broca has to do with speech.

- 1.14 One of my friends, his wife had a stroke. Affected the Wernicke's. A very smart woman, very, very smart. Suddenly, she couldn't understand any language. You would talk, she couldn't understand any language. It was bizarre. It was like seeing a schizophrenic, really. Here you have this person, normally in other respects, smart, everything, but they just can't understand language at all. If the Broca's area is damaged, you can't produce speech. You understand everything, but you can't produce it. That's the problem there.
- 1.15 A lot of these initial experiments were conducted by Roger Sperry in the 1950s—so we'll just have a little bit of history here—who found that you can sever the corpus callosum and you would have patients survive. They could love, dance, play, play chess, have fun. But you cut the—
- 1.16 By the way, you might ask, why would you cut the corpus callosum to start with? What's the function of that? Why would you go ahead and split the brain in two? AJ? And the reason is that it was often used for things like epileptic seizures. So if you have seizures from one hemisphere spreading to the other, you would cut it in two in order to avoid the spread of seizures. That was the point of that. Does that make sense? So that was the reason for that.
- 1.17 Now if I, and interestingly here, when you then cut the corpus callosum, Larry, when you cut it, right? And you ask the chap, "Draw a square with your right-hand," they'll do it in a circle with the other hand at the same time. Can you do that? It's very hard, but they can do it. It just shows you that the whole brain gets split in that way. So they can literally do it at the same time.
- 1.18 And then you flash the word banana on this side. To be clear, the right visual field sends information to the left hemisphere and the left to the right hemisphere. So when I flash the word banana, it's going to go to the right hemisphere. If I flash the word apple, it's going to go to the left. So when they flash these two at the same time, literally, you'd have banana, right-hand would reach for the banana and the left for the apple. You would have that at the same time. You and I would be like, "What's going on? Let me just—" You know. They'd go at the same time, they would do it. That's very interesting.
- 1.19 Now, in one clever experiment, they showed the chap—this was a chap, he had his corpus callosum severed. They showed him his left hemisphere got a—his left visual field, they showed a snowy scene with a snow shovel and stuff. So that was the left visual field going to the right hemisphere. Is that clear? Snowy scene going to the right hemisphere. Then the left hemisphere, they show chickens. So they show chickens to the left hemisphere over here. Now they ask the chap, point to the object you saw. And then there's these images, and the left will then point to the left hand. So right hemisphere, shovel, and chicken, and so forth.

- 1.20** But then you ask the guy, you say, you ask him, "Why did you choose the snowy scene? What's going on with the snowy scene? Why did you choose that?" You ask him. Guess what happens? Remember, the snowy scene was shown to his right hemisphere, left visual field to his right hemisphere. The snowy scene is going to the right, but the left has no— it knows nothing about any snow shovels and snow, nothing. It only knows about chicken. Is that clear?
- 1.21** And then you ask the guy, you say, "Why did you choose the snowy scene?" Given that language is embedded in the left hemisphere, you know what it says? It says, "Oh, for the chickens, you need a snow—you need a shovel to clean the chickens." So the point is it's confabulating. It's making things up.
- 1.22** And that's a key part of this lecture. The left hemisphere is involved in storytelling, confabulating, building narratives. And sometimes when you have information that's unclear, that's ambiguous, it will literally make stories up. So because it knew the fact that, look, there's something about snow here, but it doesn't know what's going on, so it just confabulates. It makes a story around that.
- 1.23** Or, for example, the right hemisphere might get the message "run." The person starts running. So he starts running. And then you ask the left hemisphere. You say, "Why are you running, Joe?" Then Joe might say, "Oh, I'm trying to lose some weight, doctor. That's all. I'm trying to get fit." It has no information about running or snow shovels. The left hemisphere is engaged in confabulation, building stories, creating narratives, creating an internal sense of "me" that's consistent in space and time. So that's one of the left hemisphere's functions. So that's very important.
- 1.24** You can think of the left hemisphere as the place where language comes about. Detailed registration of events I talked about in my previous course. So detailed features of the world, language, even things like numbers, calculation, seems to be the left. Right hemisphere: big picture, overall sense of global impression of things. The right hemisphere is involved—a lot of emotionality in the right, so there's a lot of emotion processing.
- 1.25** And interestingly, while the left hemisphere is involved in, what you might say, building a stable sense of self, you know what the right is doing? It's always suspicious about the world. It's always questioning. Can this be? Is this right? Is this interpretation of the world correct? Or is something wrong here? So it's always questioning, and it's highly suspicious. So what I'm trying to say here is that I have a view of the world. I see myself, Baland, I am a human being. I'm a lecturer. I sometimes wear a suit. I have beliefs, right? I have certain ideas. I am in Scottsdale, Phoenix, and so forth. These are beliefs. These are ideas. Even more stable, in fact, things like I am a good, for example, I'm a good son, I'm a good friend.
- 1.26** So internal stable beliefs you want to have about yourself. You don't want these beliefs to be shaky. If your beliefs about the world were shaky all the time, that'd be a huge

problem. So you want these ideas, these concepts, the fact that you feel like you know you're a good mother, you're a good wife, you are a smart individual. You have all these ideas about yourself, and you have these—you don't question them. If you were to question who you are all the time, imagine what would happen to you.

- 1.27** In fact, these things come about in the left hemisphere. And if you were to question all those beliefs, what disorder does that look like when you think about it? When you're always questioning the self and your beliefs about who you are and the world? Which disorder do we know where you're always questioning who you are? What about depression? Think about it. In depression, the right hemisphere seems to be overactive, and so your beliefs about who you are as a person and the world is completely unstable. Am I being too complex here or is this clear? Is it good? Very good.
- 1.28** Left hemisphere in depression is underactive. In the right hemisphere is overactive. If I have damage to the left, right here, to the left hemisphere, specifically in the prefrontal lobes, right here, the right hemisphere, the right prefrontal lobes will become hyperactivated. This could sometimes lead to things like catastrophic reactions to small things. I might get depressed, sad for no reason. We see that in patients, actually. Patient has a stroke in that part of the brain, and you see them completely depressed, sad.
- 1.29** The stroke hits the right and the left prefrontal becomes hyperactive. And in this case, what you see is mania. The person thinks he can buy a Rolex or a Ferrari and he can't even pay rent, for example. Or he might laugh at a funeral, or he might not sleep and become excessively—I don't know what manic people do. He becomes manic in all its flavors.
- 1.30** And this is very interesting. Think about it. It's just a slight damage, and you see this personality change in this drastic manner. This shows you the division of labor between the two hemispheres, how the right has its job, the left has its own job, and so forth.
- 1.31** By the way, artistic expression, things like appreciating art, having a sense of artistic eye. When you think about it, when you look at a painting, it has certain dimensions. An artist will use certain colors. They know how to emphasize certain colors. They know to tone down a color here, not to overdo it with too many lines and stuff like that. That sense of artistic proportion seems to be in the right hemisphere. That makes sense, right? It processes big picture stuff. It processes global impression. So that seems to be in the right hemisphere. So it shows you the dimensions how the left and right hemispheres have different functions in that way.
- 1.32** Okay, for this one, we need to be—we might take some notes for this one, okay? I'm going to avoid to become too technical, but it's important to have this. In the brain, this is the cortex, it has six layers. So six layers deep. Higher sensory processing occurs in that region of the brain. Higher functions like beliefs, ideas, cognition, working memory.

- 1.33** There's something called working memory, which is the memory I use when I want to remember a six-digit number in my head. That's working memory. So it's like I'm thinking about it right now versus deeper long-term memory. The working memory, the one that requires a lot of focus and attention, that's in the frontal lobes.
- 1.34** Now, you have a strip on the frontal lobes right here called the postcentral gyrus. It's a strip of cortical tissue—cortex—cortical tissue. The postcentral gyrus is involved in processing my sensory information from my body going to my brain. So that's very important. Next to it, just behind it, is a structure called the SPL. SPL—superior parietal lobule—SPL. Big name, but what it's doing is very interesting. It's creating a sense of body image, a sense of self, but in the more abstract sense of who I am. I feel like I'm occupying this body right here. I feel like that. The first region is doing what? Actually, sensory information. The other region is doing what? More abstract sense of self. That's two structures in the brain.
- 1.35** Just next to that, you have the temporoparietal junction, TPJ. So postcentral gyrus, SPL, TPJ. I'm going very slow because I know I don't want to confuse you guys. TPJ. In the TPJ, that region, Heavenly, it's a key region for integrating sensory information from the world. So it takes things like vision, hearing, touch, and so forth, all the various senses, and integrates that into your personhood.
- 1.36** It is strategically located at the, it's literally called the temporoparietal "junction." I mean, it's a junction. It's a very strategically located region that really blends information about the world and uses that information, AJ, to create a sense of a self, to create you. And it chit-chats with the SPL, letting it know what the sensory world is telling you about things. And it's also chatting with the postcentral gyrus and the physical sensation map. Is that clear? So we're talking about three regions, one for integrating information, one for body image creation in the abstract sense, and one for the physical body sensory map. But there's more.
- 1.37** Next to that region is the insula, insular cortex. It literally has an entire map of all your organs And it's involved in processing bodily states, like when I'm sweating and when I'm physiologically aroused and all the sense of my heart beating and all that is being monitored. There's a map for that in the insular cortex, and it's conveying that to the SPL and the other centers, letting you know how you feel at the moment. Are you feeling agitated and vibrant or are you feeling sluggish? How are you feeling right now? So that's in the insula.
- 1.38** And then you have other structures in the prefrontal lobes that have to do with a more psychological sense of who you are. Like, "I know I am Larry. I know that I'm interested in knowledge about dreams, for example. I like REM sleep. I like to explore nature. I'm a good father. I'm a good uncle." You have certain concepts; that's in the prefrontal lobes. That's creating a sense of a personhood in the sense of a more psychological conceptual self.

- 1.39** Now, I have all these structures here, not to confuse you, but to let you know that they are all working conjointly to create a sense of a self in space and time, who you are. Psychological, I know I have a sense of who I am, my personality. Then that is conveyed to the abstract sense of a body image, and then my sensory map about the world, and then my physiological sensations, the insula and all that. So all of these structures are working conjointly to glue together the sense of a self.
- 1.40** There are more structures, but I just want to keep it simple for this lecture, okay? But that's how the sense of self is created here and now in time and space. Why you feel like you are, Naomi, sitting right there, and I feel like I'm Baland, then we're not blending. It's because of these structures creating an overarching sense of a self.
- 1.41** In certain patients, when you have damage to these regions, the sense of self gets muddled. It loses its sense of—you literally lose who you are. So if a stroke damages the right hemisphere, for example, there's damage to the right SPL, you might say, "Doctor, this arm doesn't belong to me, but it belongs to you, or my sister in Italy." Talked about that in my previous course, so you know about this, those who attended that.
- 1.42** But sometimes equally strangely, some of these patients who have damage to the right parietal region and the SPL will not only have paralysis of the arm, but they will say, "this arm is not paralyzed," so they will deny the paralysis. And I just want to be—I don't want to go too fast here. So when you have damage to the right hemisphere, sometimes you affect the motor regions of that part of the brain. So the stroke or the trauma can spread and it can kill a lot of cortical tissue, a lot of brain tissue. So if it affects, let's say, the motor regions, the arm is paralyzed.
- 1.43** But then if it also affects these body image structures, the SPL and so forth, you can get some of these strange ideas about the world, these delusions. You might say this arm is not paralyzed. You ask the patient, you say, "Joe, move your hand, Joe." They'll say, "I'm moving it, doctor. It's two inches from your nose." Okay? So they'll confabulate. Literally, we see this all the time.
- 1.44** Or they'll do like this. They'll take it and grab it like this and lift it. "Lift your left hand, Joe," you say, and then Joe will do like this. No, sorry. You say, "Raise your left hand," and they will do like this. "Are you raising it?" They will say, "Yes, I'm raising it." "Really? On your own? Can it move?" "Yes, it's moving right there." So they'll just do it like this and then move it with their left hand, completely in denial about the fact that their arm is paralyzed.
- 1.45** Some of these patients will engage in some of the most bizarre, delusional form of thinking. So all of us use what Freud called defense mechanisms, things like denial. So we'll engage in denial, of course. Once in a while, we might—something is blatantly true, but we deny it. So we have that. Or sometimes we might repress a certain memory. So it's something—we think we have a certain memory, but we're repressing

it. We don't allow it to come to the surface we pretty much repress a memory. That's what repression is, to make it very simple. Or we might confabulate, come up with strange explanations.

- 1.46** Confabulation, so that's when you make something up, like literally just make something up. You might rationalize. We all rationalize once in a while. A patient might, for example, do what when they rationalize these patients? They might say, you say, "Joe, lift your left hand." And they might say, "I'm tired of all these medical students, doctor, probing me all day. I don't want to do it anymore. I'm tired." Or one guy, he said, "I'm an Air Force pilot, doctor. I'm not accustomed to taking orders," he said. "I don't like to take orders." So they come up with these rationalizations.
- 1.47** But it's interesting how we all have these delusions, and they seem to be emerging from the left hemisphere. They tend to be a left hemisphere phenomenon. Meaning that this sense of self is constructed in the left hemisphere. And so to hold tight to that sense of a belief about who you are and avoid too many upheavals and too much shaking of the sense of self, it will come up with these defense mechanisms to protect it. So this is the idea, and we'll go more into that.
- 1.48** And so the way you can also, and this is also important to add here as a note, the left and the right hemisphere are always in a sense of a struggle. They're always competing. They're always wrestling for the share of attention. So while the left has these beliefs, the right wants to overthrow the beliefs, right? So that's one function between them, a difference. Right hemisphere, artistic, left, for example, math, and they compete. You get damage to the one hemisphere, the other gets more attention and can blossom more.
- 1.49** And this is, actually, your entire brain is made up of competing systems. So it's not only the left and right hemispheres. If you look at, for example, the emotional core of the brain, so you have the limbic centers behind your ears, emotional core. So if that region of the brain is overactive and you have a lot of fear, you have regions called the septum [septal area], also in the limbic structure that has to do more with pleasure and good emotions and positive emotions. That is constantly trying to inhibit the amygdala, preventing it from going overboard. Is that clear? So it's a system, it's an overall principle in the brain that systems are competing.
- 1.50** Now, I don't want to lose my thread here. I was talking about—Larry, remind me, what was I talking about before the two brains? Oh, anosognosia, right? These patients. These patients are very interesting. The fact that they will engage in obvious confabulations and denial and things like that. But it's a common thing we see in these neurological disorders where the left hemisphere is damaged. But you also sometimes see it if the the visual cortex is damaged. And so this makes it more complex.
- 1.51** So you have people with a stroke to the visual cortex or the visual part of the brain, and they literally cannot see. And then you do like this, "How many fingers is this? How

many fingers is this?" They'll say, "Four, doctor." And then you do like this. "What about this?" And then, "Two, doctor." And they say, "Are you seeing it? Are you sure, Joe? Are you seeing it?" "Yeah, I'm sure." I say, "Okay, Joe, can you walk over to the corner of the room to get yourself a glass of water?" And they will get up and then start walking and bump into furniture or something. And then you ask them, "Are you okay, Joe? Are you sure you can walk?" The chap is blind. He'll say, "Oh, have you changed the furniture of the room, doctor? You seem to have done some new design here."

- 1.52** So they'll say things like that to confabulate. Now, this is called Anton syndrome. It's another form of anosognosia. So it shows you how these types of delusions can vary greatly, but you have various forms of them. Do you know Oliver Sacks, by the way?
- 1.53** Oliver Sacks is one of the great storytellers of neurology basically, of all time. I mean, he's a great guy, and he's written tons of books about patients with neurological damage and things like that. There's a movie about his life where he is played by Robin Williams. And in this movie, Robin Williams is a very empathetic neurologist, and his patients have Parkinson's disease, so they can't really move. They have Parkinson's-like symptoms and Parkinson's disease and so forth. And they have all the symptoms of Parkinson's, masked expression, they can't move and all that.
- 1.54** And then he gives them L-DOPA. L-DOPA, that's the precursor to dopamine. People with Parkinson's lack dopamine in their brain. And so when he does that, they all become alive and suddenly start playing the piano and dancing and becoming active again and so on. It's quite a movie.
- 1.55** Anyway, this chap, this Oliver Sacks, he had prosopagnosia. And this is a condition where you can't recognize faces. So for example, he would walk about in New York and he would— once he walked into this man and he was like, "I'm sorry, I want to go." And then he realized, "My God, it's a mirror of myself." It's a mirror and it's himself that's right there.
- 1.56** He was a great guy, Oliver Sacks, and I was fortunate to know him, not to brag or anything, okay? It has a context. I did know Oliver Sacks. He came to the laboratory where I was working with my mentor, Ramachandran, and he would often come. It was quite cute because even though he was this global superstar in neurology, he was very shy. So he only let five or six of us come to his seminar. So only this short little room, and he would sit down and he would talk about the brain. Such a character, Oliver. But he would have some of these conditions and—
- 1.57** He talks in his book, he has one book called *The Man Who Confused his Wife for a Hat*. This is the title of his book. And he talks about a patient. He describes a patient that he saw that was lying in his bed, sleeping. And as he was sleeping, the patient kept throwing his own leg outside the bed. So he kept throwing it, and kept throwing it, and kept throwing it, Heavenly, kept throwing it.

- 1.58** And then, the nurses got all worried. What's going on? And Oliver rushed to the room to see him and said, "What are you doing?" He said, "There's this nasty thing in my room, this thing. I don't know what it is, but it's stuck. It's glued to me. I want to get rid of it. It's hairy, doctor. It's nasty. I wanted to be removed." And Oliver goes on and proceeds to say, "Well, it's actually your leg," and tried to calm him down, but he just wouldn't go for it. And eventually, he calmed him down.
- 1.59** So he had an example of these conditions, like somatoparaphrenia, it's called, one of these conditions. So it shows you how these delusions can vary. Actually, I pointed to the right leg. Often, you have it in the left side of the body. Because of the right hemisphere and its involvement in creating a sense of a body image, it's often on the left side, you see these delusions. You rarely see anything on the right. It's always, "My left arm doesn't belong to me. It belongs to Naomi." "My left leg is not paralyzed," for example, and so on.
- 1.60** How do you test these patients? How do you experiment on them? Trying to understand what's going on in the brain and how it all works. One way is to do the following. So you can ask them, first of all, "Move your left arm," right? And they'll say, "Oh, I can—doctor, I can see it moving," and all that. That's the first thing. You just want to do a basic examination first.
- 1.61** But then you say, for example, you might say, "Joe, can you move that table over there?" He'll say, "Yes, I can move it." "Can you lift it?" "I can lift it." "Can you lift it with your right arm?" "I can lift it with my right arm." "How much can you lift it with your right arm?" They might say, "About an inch." "What about your left arm?" Mind you, it's paralyzed. They'll say, "I can lift it about an inch and a half, doctor." So with the right arm, that's not paralyzed, they say, "I can lift it an inch high." The left arm, they say, "Inch and a half." What's going on? Larry, it is as if a part of them knows that it's paralyzed, that they're trying to hide it, and so they compensate for that, engaging in a reaction formation.
- 1.62** You know what reaction formation is? It's a Freudian defense mechanism where, let's say somebody is really homosexual, for example, and then he acts all macho and trying to be tough and all that. So that's called reaction formation. Or somebody who is very bad at soccer or something, and then he acts like he's—that's a very bad example. I don't know. Reaction formation, you assert the opposite. So the homosexual example is perfect. So people have this tendency, they act rough, and they do the opposite of what is actually the case. Now, this is interesting. This is interesting.
- 1.63** And then you can test them even further than that. You can say, "Okay, what if you bring in a cocktail tray, like a cocktail tray like this, and you bring it to him?" You and I, if I was to bring a cocktail tray and I had, let's say, glasses filled with water and bring it to you, what would you do? [student: "Drink it."] Yes, but if I gave it to you, you'd grab both sides, right? You'd grab both sides, right? [student: "Yes."]

- 1.64 Okay, what if I tie your left hand behind your back? What would you do? You know what you might do? You might go for the center of gravity and go like this, right? These patients, when you bring in the tray with water and cocktail glasses, you bring it to them. They always, always go to the side of it, and they go for the side of it as if their other hand is working. And then it falls down and the glasses topple and everything falls to the ground. And then they'll just sit there and go, "I had some water on here. That's okay. Nothing—" and sit there all happy and dandy as if nothing has happened. It's interesting.
- 1.65 Or you can do another task. You ask them to tie shoelaces. You can tie shoelaces. Or threading a light bulb into its socket. If I gave you \$10, I say, "Okay, you're going to do this within a time frame, let's say, two minutes, or let's even say less, let's say 20 seconds." You and I would probably go for the tying the shoelaces. It's pretty fast, right? Or actually, what would we do? You go for the shoelace. It's faster than threading the light bulb. Yes, we would go for the shoelaces. Now, Naomi, if your one hand is tied behind your back, what would you do? [student: "The light bulb."] The light bulb, right? Because you don't have two hands.
- 1.66 These patients, for some reason, always choose the shoelaces, always. So they will sit there and just sit with one hand and just over and over and try to tie the shoes. And it's not getting tied, but they sit there completely unfrustrated, completely happy, and they're just sitting there and trying to tie the shoelaces. Then one hour after you ask them, "Did you tie the shoelaces?" They will go, "Yes, I tied them. They're perfectly fine." Completely oblivious, completely neglecting the fact that they have this condition.
- 1.67 There's an interesting trick you can do to help with this condition. This was done by Italians in the 1990s. They thought, let's sprinkle some water into the left hemisphere, the left ear. By attacking and putting water inside here using a syringe, you activate the vestibular system. Now, this vestibular system is involved in a sense of balance. So a sense of balance and creating a sense of balance. That's also a part of creating a body image, where you are in space and your body image.
- 1.68 Now, if I put water inside your ear, a syringe, it activates the right hemisphere temporarily. So you have the patient sprinkle water into his left ear. Wait 20 minutes. You say, "Is your right arm paralyzed?" They'll say, "No." "Try to move it." They'll do like this. And you ask, "What about the left?" "Oh, it's paralyzed, doctor." So now they admit to it. And then you say, "What do you mean it's paralyzed?" "Well, it's been paralyzed the whole time I've been here. The entire three weeks at the hospital, my left arm has been paralyzed."
- 1.69 Completely rewriting the script now. And then you asked them, you said, "Didn't you say that your arm was paralyzed, that it wasn't paralyzed, that you could move it?" And they said, "No, I never said that." So they are denying their own denial now. Completely rewriting everything. And again, simply, you have temporarily revived the right

hemisphere where the stroke had made the damage, and so they engage in it. They become normalized temporarily.

- 1.70** At this point, the doctor might go home, and the doctor, you wait 24 hours, come back to the patient and you say, "Yesterday I was here. I saw you." "Yes." "And you told me that your arm wasn't paralyzed. Is that right? That you said—was it paralyzed or not paralyzed?" You're trying to question him, right? And he says, "Well, my arm—" so you ask him, "Move your right hand," and they will move it. "Left hand, move it. Try to move it." At this point, they'll go back to the original delusion, okay? And say that, "My arm is moving fine, doctor. It's fine." Then you ask them about yesterday, and they say, "No, I never said that. I never said that it was paralyzed yesterday." All these delusions in these patients are wild.
- 1.71** The fact that you can temporarily revive the right hemisphere shows you that it's really this imbalance between the right and the left hemisphere that's creating these delusions. The reason I'm saying this is there's still debate among scientists, neurologists who might say, "Well, look, the fact that these patients are saying that is really because they have neglect."
- 1.72** Neglect is another neurological syndrome where you have damage to the right hemisphere, and you ignore everything on the left. You simply ignore it. You pay no attention to it. So they wake up in the morning, look in the mirror, and they shave one side of their head on their face like this. Or if you're a lady, you put on makeup just on one side. Put it on. And then you dress, you completely neglect the left side. So you might put the sleeves, you might put the sleeve here, but the other sleeve is out here. Striking.
- 1.73** And then they sit down for breakfast and they only eat from half of the plate. Draw a painting. Draw half of—"Joe, draw a flower," and they draw half a flower. Completely. You ask them, "Is this flower complete?" "Yes, doctor, it's complete. There's no issues with it. It's beautiful, doctor." You sit down with them and you ask them about their lives. You play chess with them. Ask them about their affairs in life and career and everything. They're smart, have no issues, other than ignoring half of the world.
- 1.74** So for a long time, they thought, well, these patients who say their arm is not paralyzed by engaging in these denials are, in fact, people with neglect. They're just neglecting their left side of their bodies. But it turns out that this cannot be the case. It seems like this splashing of water into the left ear and reviving the right hemisphere shows clearly that that's not the case, that we have the right hemisphere and the imbalances between the two hemispheres.
- 1.75** Also, you might ask, with neglect patients, the way you can actually help them compensate for their lack of—ignoring their left side of the world is by simply moving their head. So if I have neglect, I simply look to the side, I start incorporating things from that part of the world. That's actually a trick. There was one lady, she had neglect

and she had this chair, this chair where you can roll around. She'd sit with that, and then every time she thought she might have forgotten something on the left, she would just roll like this, and then she would start incorporating that into the world. But it's clear that neglect does not account for it. It's these Freudian defense mechanisms that seem to be embedded in the right hemisphere.

- 1.76** My colleagues did an experiment. They said, "What if we make the paralysis more acceptable to the patient? Can we use a trick to make the paralysis just more acceptable to you?" What if I bring in, say, a syringe filled with water and tell you, "Larry, this is an anesthetic." Okay, so let's try this experiment. Larry puts out his right arm, put it into his right arm. Right? Right arm. And then you say, "Larry," you imagine now you have the syndrome, "Move your right arm." You're moving it. No problem. "Isn't that an anesthetic?" I say, "Yes." "I don't know. It's not working. Something is wrong." You might do that.
- 1.77** When you inject it into the left arm, and you do like this, you inject it into the left arm, you ask them, you say, "Joe, move your left arm," and they will say, "Doctor, doctor, I can't now. It's paralyzed." So all of a sudden, they will completely neglect three weeks of saying that it could move. And now, because you've made the paralysis acceptable to them, they will literally engage in, they will become normalized temporarily.
- 1.78** And then you ask them, "Well, didn't you just move your right arm? Didn't I just put it into your right arm? You said it wasn't working." Then they might say, "Oh, mind over matter, doctor." Okay? Or something like that. They might come up with some excuse for the fact that it's working on the left but not the right-hand. So it clearly shows you that it has to do with our belief systems and our ideas about the world and how that comes about in the left hemisphere and the right hemisphere.
- 1.79** And so really, by studying these patients, we are getting a glimpse into human nature. It's almost like it's like looking at human nature through a magnifying lens. Because we all have them; we all have these delusions, but to a lesser extent, because we don't have trauma to those regions of the brain.
- 1.80** Now, let me try to explain how the left and the right hemisphere work and how this competition works, and these beliefs come about. We said that the left hemisphere creates these persistent beliefs about who you are as a person and your beliefs about the world. The right is the devil's advocate, always questioning, always suspicious, always trying to shift the paradigm and make you change your view of yourself and the world. It's critical, and you have to have this balance.
- 1.81** Now, let's say I, Baland, let's say I have a— I'm getting married, let's say. I'm getting married, and I'm all ready for the marriage. All the guests are there, all dressed up for it and looking forward to it. And my fiancée is in the other part of the wedding thing, and she's ready to—the whole thing is about to get going. But then suddenly this guy comes

in and tells me, "This girl that you're about to marry has cheated on you." Okay? What do you do?

- 1.82** Your left hemisphere might say, it might engage in some of these defense mechanisms because it wants to cling to the status quo in the here and now. So they want to go, "Oh, that's not true. That can't be the case." Okay? So you engage in some of these defenses. You brush it under the carpet. You engage in denial. You engage maybe in denial. Could be denial here. Or you rationalize and say, "No, this can't be the case. My girlfriend probably just did something else, and it was misunderstood."
- 1.83** But then a second guy comes in and says, "No, your girl did really cheat on you." And a third guy, and a fourth, and a fifth. And the fifth one even brings a videotape showing whatever, right? Clear-cut evidence. What happens now? You can no longer engage in Freudian defenses because the evidence is overwhelming. And so at this point, your left hemisphere is overwhelmed by the right hemisphere's evidence, and the whole script is now flipped, and you literally have to revise the script of who your girlfriend is and your potential wife is.
- 1.84** And this is how the left and the right hemisphere works in all of us. Left clings to the beliefs about the world. And whenever we have some discrepancies, small hiccups in the system, we go, "Oh, I rationalize it. This cannot be my— my boss is probably a good person. He doesn't mean that. He had a bad day today." I engage in denial at various points to cling on to my beliefs about myself and the world. But if my boss was to scream at me five days in a row or a month later, or even, I don't know, then it might be a different scenario. Ultimately, the right hemisphere gets the upper hand and you flip the script. So this is exactly how we believe the right and the left work together in this way.
- 1.85** We all have these defenses and these left-right imbalances and the left being the preserver of the self and the right questioning. But when you then have the damage in the right hemisphere, you can no longer hold—the beliefs become—the suspicion, so to speak, is not there. There's no devil's advocate. You don't question anymore. And that's the whole point of this whole lecture, that in these patients, and you and I, we have these defense mechanisms. We ways of making consciousness run more smoothly.
- 1.86** I mean, think about it. If I didn't have these defense mechanisms, my behavior would be pretty unstable. Consciousness, the sense of I, the me through space and time needs to have a coherence and a flow to it and not being perpetually indecisive. So that's why the brain needs this clear direction. Damage hits, and this balance is completely shifted.
- 1.87** You might also ask, why do we have denial in the first place? What's the— beyond these defense mechanisms, what's the purpose of denial? And one theory that I just want to add here, in closing, is that it's been argued the reason why we have denial is, let's say there's this monkey, and he has access to the bananas. He knows where the

bananas are. Then the alpha monkey comes, and he wants to know where the bananas are. Now, if I was to lie and say, "I don't know what the bananas are," I might give that away because—

- 1.88** When you smile, for example, a real smile, you use certain eye muscles when you lie. For example, a real smile might have slightly lifted smile to the left because you use more the right hemisphere for emotionality. So it has slightly lifted your face to the left. You can clearly sense when somebody is lying, different muscles in the face.
- 1.89** So denial might serve the purpose of being able to fool the alpha monkey. You engage in the denial, you don't know where—you tell yourself or your brain completely represses the information and you say—the monkey comes in and monkeys don't speak, but you would say, "I don't know where the bananas are." And in that way, you wouldn't give any telltale signs. So that's one idea.
- 1.90** But I think an objection is, what is an objection to this theory, Larry? If I, the monkey, want to preserve my bananas, and the way to preserve them would be to engage in repression and denial, what's the problem with that? Would I lose access to that information and then I wouldn't be able to find my bananas myself, perhaps? That's one flaw of that theory that in order for you to preserve resources, you have to, forget it yourself, actually repress that information or engage in denial. Well, that would mean you'd also lose access to it. So that's one objection.
- 1.91** But I think there's some truth to that argument anyway. I think the fact that we engage in denial also because of the fact that it's helpful for basically lying. It's a form of camouflage, survival. I think that could be the case still. All right? Let's go over to questions now.
- 1.92** You said that in people that have a better ability to appreciate art, and stuff like that, that their right hemisphere is more active and bigger, question mark?
- 1.93** Potentially, yeah. It could have become hypertrophied.
- 1.94** In cases of the great artist like Van Gogh, he lost his mind toward the end. Do you think that his ability to appreciate art made him spiral? Or do you think he was just already going to go insane?
- 1.95** What I think is that in the lecture on dreams, in the course on dreams, we talked about how in a sense, the way we see the world, we see the world through a very hierarchical lens. We have the prefrontal lobes, the higher order brakes on the brain, quieting other regions all the time. So it quiets even the art regions, keeping it from flaring up too much.
- 1.96** When we are dreaming, for example, these regions that have to do with art, emotionality, instincts, get the upper hand and become hyperactivated. And so that's why your dreams look like a big art museum, right? Now, you can, and there's a lecture devoted to this, but if you use certain stimulation methods and quieting down the

prefrontal lobes, you can actually hyperactivate some of these sensory lower areas, and you become hyper artistic.

- 1.97** In people like Van Gogh and other people that lose their minds, and even certain creative people in the population, what I think is going on is that these regions that have to do with art, the art departments of the brain, there's a region called the inferior parietal lobule. Literally, it's like a little region right there, it has to do with conjuring up images. So we believe that it's involved in art. In fact, I mentioned previously, if you have a stroke to that part of region, you can't even dream. So art, dreaming is this overarching thread here of that.
- 1.98** But point is, I think people that lose their minds, their prefrontal cortex is simply not working, and they don't have a structured CEO to keep things in check. And so that's what I'm—so they have a form of—they're dreaming when they're supposed to be awake. Form of psychosis, almost. What did he die of? Suicide, madness. It's just kind of madness. Yeah. Psychosis, probably. A lot of these artists do. There's a lot of artists that, even without psychedelics, will jump from buildings and think they can—it's crazy. But you have psychosis in these populations. Great question.
- 1.99** I thought it was fascinating, you were talking about left brain, right brain. It brought me back to some things you mentioned, the previous lecture about genius. It is, could I—my hypothesis with my limited knowledge is that creativity is related to the right brain. Is genius intention on the left side? And do superior people in our culture— well, is it left brain-right brain, a tension between creativity and genius in the way that you understand it?
- 1.100** Yes, great question, Larry. I think that there are different types of genius, and we'll have a full lecture for this. It seems like the artistic geniuses have an overactive right hemisphere. These are the artistic, poetic geniuses. The left geniuses would be the calculating Rain Man-type. You've seen Rain Man? Calculating geniuses, left hemisphere. The left is good with numbers, calculation. The right is good with math and artistic expression. It seems to be that division of labor of the right and left.
- 1.101** But I think for true genius, though, and we'll talk about that, you need both. These savants, they are very good at a certain thing, but only one thing. And ideally, you want to be good at multiple things. And I think somebody like Einstein, for example, he did calculation, but he also did spatial calculation. So almost as if his two hemispheres both were hyper-developed. And in fact, they looked at his angular gyrus, the place where things like the images you create, images and math and things like that in the brain, it has several substructures. That part of the brain was hypertrophied in him. He had huge angular gyri. But yes, it goes into that.
- 1.102** I know someone personally, that went through some emotional traumatic things as a child, but there wasn't physical damage to their brain. Symptoms of that was epilepsy. They're now in their 30s or 40s, and they also have I would say, a development in their

maturity is also a condition. I'm curious to maybe have some understanding of what type of damage maybe occurs from those traumatic experiences and maybe what that causes to limit the development of someone's personhood intellect, things like that?

- 1.103** Yeah, it's a great question. It has to do with the the lower structures of the brain, the lower sensory region. These are more autistic, savant syndrome, and you can have that. So early epilepsy, even. So these savant syndrome—savants are people that have certain specialized skills. They're very good at a certain thing, but bad at other things.
- 1.104** And so what you often see is damage to the higher cortical regions, but the sensory regions, lower level, can flourish then. They are not hyperactivated. And so I think what you're seeing then is that, basically, the shift towards almost like they're dreaming when they're awake in the sensory realm. They're very sensory strong. They have certain skills, but they can't really do higher functions. I think that's in a nutshell, what's going on.
- 1.105** Why in childhood, for example? I'll talk about that, too. But I think the brain in childhood is extremely malleable. It's very plastic. It can change its shape quite a bit. That's really what I think is going on. If you have the damage early on, it's much easier for the brain to rewire itself. Although you can see cases of this also when you have damage in adult populations, for example. Great question, Sam, though. I'll go more into depth later.
- 1.106** When you described information competing left, rational— I don't want to say rational brain versus right, creative brain—does that also imply that it also functions competitively with information going, say, up and down the brain?
- 1.107** Absolutely. I mentioned this in the lecture, too, is that it's definitely these competing subsystems and systems are all over the brain. So the prefrontal lowers other parts of the brain. The septum [septal area]—the positive emotion part of the brain—the negative emotions. Oversimplifying. They process both, but you have that. And so left and right. It's something all over the—it's an important question, actually. Thank you.

Lecture 2: The Social Self

At a glance — The social self — identity in social context.

- 2.1 It's about the social self, the sense of who we are in a social context, in a bigger social. So it's going from first lecture, personhood, unity, consciousness in terms of the self, and then now talking about how the self is related to other people and how consciousness is built through interpersonal relationships.
- 2.2 I'll have to borrow a few conditions out there to help us illustrate how the social self comes about. I'll use a condition called autism. You probably all know it. It's a rising condition. Now, it's, I think, one in 100 people, and it's higher among kids, like 1:4, so much more. You have boys having it more. We'll talk a little bit about psychopathy. People are psychopaths. A very good example of how the social self can erode. We'll talk about excess empathy, depression, and I think that's about it for this lecture.
- 2.3 Why is autism important? Well, interestingly, it was actually discovered by two scientists working independently in the 1940s. They both called it autism, which means "self." So both working independently, both called it autism, which means self from Greek. Autism is interesting. It has various symptoms. It has what we might call sensory cognitive symptoms and motor symptoms. So these two categories of symptoms.
- 2.4 An autistic child might be in front of you. He's often withdrawn, tends to be very withdrawn, keeps to himself, preoccupied by certain things. So he might look at the dictionary and just read words over and over, just keep doing that. So hyper-focused on certain objects or playing with certain things or reading about certain animals. They have this hyper-focus. So this is one key symptom of the disorder.
- 2.5 Often, their speech is impaired, so they can't engage in speech like you and I. And if you talk to them, they might be like—they might not look you in the eyes, they look away, look down. They seem a bit flat in their expression as well. They have certain repetitive gestures that they make, like flapping, so they might do like this over and over, they keep doing that, so they have these symptoms as well.
- 2.6 In the social realm, in terms of social interaction and engaging in what you might call empathy or cognitive empathy, they are very poor at that. By cognitive empathy—we can divide empathy in two. There's cognitive empathy, which is the empathy when I peek into the brain of Larry and I'm saying, "Larry might be thinking this right now. This is Larry's agenda right now. He has all these—" I create a full theory of mind around you. That's cognitive empathy.
- 2.7 But you also have something called affective empathy, which is a more emotional type of empathy: feeling what you are feeling, putting myself in your shoes. In the previous course, we talked about mirror neurons. These are neurons that activate whenever I merely feel somebody else's sensations and emotions. So if I, Baland, was to inflict pain

on my hand, pain neurons in my brain fire away. If Sam over there inflicts pain to your hand—can you do it? We need to have an illustration—mirror neurons in my brain, neurons in my pain pathways, will activate by merely watching you inflict pain to your hand, a subset of my pain neurons. These are called mirror neurons.

- 2.8** We have mirror neurons in regions such as the ACC, and these are regions that are—think about the emotional core here. You have the emotional limbic region of the brain, very important here, emotional limbic. Then you have the frontal region. Between them, there's a bridge that helps us link emotion to cognitive thought. This is the center, the medial part of the prefrontal lobes. The lateral, the sides of the prefrontal lobes, are involved in cold cognition, calculated utilitarian type of cognition, while the medial is involved in more emotional decision making.
- 2.9** Structures like the ACC, anterior cingulate cortex, and a nearby structure called the orbitofrontal cortex, these two structures are bridges between emotion and higher-level thought and emotional decision making. So you have that bridge. The ACC is the emotion part of the—it's the empathy department of the brain in some respects. If you poke your hand with a needle, if I poke my own hand with a needle, the ACC lights up. If you poke your hand and I watch you, the ACC lights up. My ACC lights up by watching you poke. Make a flick. Can you do it? My ACC lights up; it becomes hyperactivated now. So there's an empathy type of projection here going on when I feel your emotions. That's emotional affective empathy.
- 2.10** Autism—there's this idea that autistic children, autistic individuals, have issues with cognitive empathy. They can't really see the world from your perspective, AJ. They can't really say, "What is AJ thinking right now? What is AJ up to right now?" That ability is not there. The fact that we have this ability in the first place is pretty profound. I mean, one of the problems with not having cognitive empathy is that autistic people can't lie.
- 2.11** Think about it. What is a lie? If I have to lie, I have to put myself in Amanda's shoes and think, "Would she know that I'm lying if I say—" Lying is a cognitively exhausting skill. Think about this for a minute. The fact you have to put yourself in somebody else's shoes and say, "What are they thinking? How can I construct this lie in a manner that is effective?" Autistic people can't lie. They can't, because they don't have that ability to see the world from your eyes.
- 2.12** By the way, autistic people—I learned this a few years ago. I was in Cambridge. I was meeting with a guy called Simon Baron-Cohen. Have you heard of him? He's a leading autism expert. You know him? You know Sacha Baron Cohen from The Dictator? His cousin's Simon Baron-Cohen. It's very funny. Here you have Sacha—he's a funny comedian—and then his cousin is a world-leading expert on autism. It's funny. The one is dressed in a very formal suit; the other is very eccentric, to say the least, in his movies and all that.

- 2.13** Anyway, I was heading out to meet him, and we were talking about developing apps for autism. How do you develop—in my work on OCD, and we'll talk about that a little bit as well, I've developed some applications to treat OCD. The way we want to treat OCD is by having—because we know that we have this mirror neuron system, Sam, what we can do is, if you have OCD, for example, and you have strong inclinations to want to go wash your hands all the time and you feel the world is dirty and all that, one way you could treat the condition is that if, let's say, I have OCD and I want to go wash my hands like 100 times a day, I merely watch an app of myself doing it instead of actually going to do the handwashing. If I do that several times a day, instead of going and washing my hands, I might learn that not washing hands is not deadly and terrifying. I unlearned the fear that I have to go wash my hands all the time. Is this clear, this concept?
- 2.14** Basically, I'm tapping into the mirror neuron system. Me and my colleague, Ramachandran, developed these apps, and others at Cambridge as well. I want to acknowledge them in case they watch. But these apps tap into the mirror neuron system by allowing you to really watch a video of yourself washing hands and you feel like, "I'm actually doing the behavior by simply watching."
- 2.15** And we thought, with Simon Baron-Cohen, we thought, "Maybe we can do an app where somebody with autism watches videos of themselves, or it could be even other people, displaying certain facial expressions," because autistic people have difficulty with understanding social expressions. If they look at your face, Sam, they can't really decode, "Are you happy right now? Are you sad? Is he smiling?" They can't really decode all that. They don't have the machinery in the brain to do that. There's a region of the brain called the fusiform face area— saying it very slowly, FFA. That region of the brain is involved in recognizing facial expressions. Now, it has other functions, too. It's involved in direct classifications of objects—it knows this is Jane and Mariah, this is a cat and a dog. So direct classification, it has that job, too. But one of its key jobs is facial recognition.
- 2.16** Now, you can imagine if you have autism, might that part of the brain be big and strong and active when they look at faces, or might it be quite low in activity? Exactly, right? Atrophy, perhaps low activity. That's what you see. Autistic individuals, their face recognition area is very low in activity. It's very low. On the other hand, there's a connection between the face area of the brain and the emotional core behind the ears. It's a region called the amygdala. The amygdala, Brance, is the emotional core of the brain, right? There are connections there from the fusiform face area to the emotional core, the amygdala.
- 2.17** Now, people with autism have low activity in the face area, the fusiform face area, but high activation in the amygdala. High activation. So whenever they see a face, not only can't they decode it, but their emotions get all flared up and they get all fiery and terrified and scared. Is that clear? So there is that. They are socially puzzled. That's what we call it. Richard Davidson calls it socially puzzled. We all have this—many of us,

like me, perhaps, I'm a bit socially puzzled at times. There's actually a spectrum. So even though autism is out here, it's the outmost socially puzzled, it varies in the population.

- 2.18** You seem to be socially intuitive, for example. I've spoken to you many times, Naomi. You have very good conversational skills and all that. But Larry over here is very—no, I'm just kidding. This is a joke. Forgive me, okay? You are very socially adept. You have social intuition that's high, characterized by what? High activity in the fusiform face area and low activation in the amygdala. See that pattern? It's the opposite of what? Autism. It's the opposite of autism. Autism is what? Low activity in the fusiform face area. I can't decode faces. High activation in the fear center of the brain—I have fear. Socially intuitive person like Larry and Naomi, they have what? High in the face area, and they have low in the amygdala. So there is that pattern.
- 2.19** I'm going to come back to Sacha and Simon and all those guys in a minute, but I want to add here that for a long time, Richard Davidson, for example, thought, "Could it really be that people with autism have low activity in the fusiform face area because they simply have an underdeveloped fusiform face area? Or could it be that they're merely looking away all the time because of the high emotionality in the amygdala, the almond-like structure behind your ear?" Is it that the fear centers are so active and fiery all the time that I simply, whenever I see a face, I look away? That could, of course, lead to atrophy over time, but it's not like there's something inherently wrong with the face area. That's an argument.
- 2.20** He did some brain scans, and he actually saw that people with autism, whenever they saw faces in the scanner—you would see that pattern of the fusiform area being low in activation and the amygdala being high—but they would always look away. There were kids in the scanner, and they would lie there and look away each time there was a face in the scanner. So now it's up to debate: Is it really the fusiform face area that's low in these people, or is it actually the amygdala that's causing the whole thing, the whole trouble? We don't know for sure yet, but I think it's an interesting thing.
- 2.21** What's even more interesting, I think, for you and I, especially for me, is that you can actually train your fusiform face area and your amygdala to be active in a pattern that makes you more socially intuitive. So by going out and interacting with people and training your fusiform face area to have more calmness and feeling more relaxed around people and feeling more socially blended with people, that is something that can make your fusiform become more active over time and make your amygdala go down. So it can be trained, what I'm saying. That's the point.
- 2.22** Anyway, back to my story. So I was driving, I was on the train. We use the train a lot in Cambridge. So I was going on the train to meet Simon Baron-Cohen, Trinity College, Cambridge, go through the grass, walk on the grass, go into his office, sit there, talk about the app. "Could we develop an app?" I was telling him. And he was telling me, "Could we develop an app where people with autism watch facial expressions, and then

they go to a café. Then when they are in the café and they feel puzzled, what do they do? "I feel so puzzled right now. What is Lizzie thinking? What is she thinking right now? Let me take out an app. Is she smiling? Is the lecture interesting? Maybe she thinks the lecture is okay. Maybe it's fine. Yeah. Okay, I can understand that." And then they feel better and feel more relaxed. So that could be.

- 2.23** This is an app we thought about building. This is still something we haven't pursued, but it is an interesting idea that we could potentially— somebody might build that, or we might do it. We don't know. We need more funding for science; we don't have a lot of funding in science.
- 2.24** Now, the engagement—my point was also, on my way to this meeting with Simon Baron-Cohen, I had to do a lot of my work on understanding autism. I had to have some knowledge. You see, I don't want to come across as ignorant when I go to see an expert. So Baland had to brush up on the knowledge; I had to really do my homework. When I learned that people with autism have something called systemizing—it's quite embarrassing. When I went to the meeting, I kept saying systematizing, even though it was—I was trying to sound like I know a lot about autism, but he was okay, he was forgiving.
- 2.25** But systemizing—they have this trait, which is they systemize things. They organize the world in categories all the time. They're like machines, almost, in the sense where they have that. And that explains also why they like dictionaries, they like programs, they like computers and computer games, and things like that because of the whole systems. They like systems approach to things. That's a huge part of it. A more socially intuitive brain is less like that. They're more in tune with people and social interactions and, "What is my friend thinking now?" and "What is my mother thinking of me doing this?" and praise from my mother and all that social stuff. They're very poor at the social things. They are more into systems and programs and that.
- 2.26** I asked as we walked past—this was actually before we went into the office—I asked him, I said, "Simon, can you tell me about your theory about autism? What's going on?" He said, "Yes, we think that in the fetus—" This is called the extreme male brain theory. Tony, within the fetus, a child, any child, if there's overexposure to things like testosterone, too much of that, it could lead to an extreme male brain. So the way that they see autism is almost like you have an extreme male brain and an extreme female brain. These are just categories for saying that females tend to be better at social intuition and social interpersonal while males tend to be more interested in objects and systems and programs. This is in the crude categories. Autistic children tend to have an extreme male brain where the testosterone has made their brain extreme towards the male part of things and just completely neglected the other regions. So that's one idea. So he explained this to me.
- 2.27** You also have another interesting— within the autistic brain, you have a lot of—so, on the neurons, you have something called, on the dendrites, you have these small

protrusions that look like mushrooms, called dendritic spines. And dendritic spines—these mushroom-like bottoms—on the neurons, on the dendrite—the dendrite is what receives connections from the axon on the different neuron, on the separate neuron. They have tons of these dendritic spines, as if their brain is hyperconnected in many ways. So they have a hyperconnected brain. Does that make sense?

- 2.28** That explains also why they have a lot of other symptoms. So not only the flapping and those motor symptoms and the social symptoms and all the cognitive ones, but they also have hypersensitivity to light. They can't tolerate light very well. See, I'm familiar with that. I know how that feels. That's not fun, being overwhelmed by light. So they have that. They can't tolerate light. Sounds—they might have a sound sensitivity as well. They have a sensitivity to touch. If I touch your leg, Larry, they don't like touch or maybe certain material on the skin, like a shirt or something like that. If it's silk or if it's too rough or something, they don't like that. So they have these problems with texture and things on their skin. So you have all these symptoms that are part of this disorder. The disorder doesn't only—we think perhaps this overconnected brain could explain it.
- 2.29** Imagine having autism. Imagine not being able to understand what somebody else is thinking, feeling alone, just wanting to look at phonebooks all day, feeling lost, maybe, and then having sounds and light all overwhelming you. Would you want to stay home? When I had that light sensitivity, trust me, I wanted to stay home all the time. I just wanted to stay inside because when you go outside and there's light, it's overwhelming. Imagine having this. You know what they do? They often mask it. They mask and hide their symptoms, causing things like depression, OCD, high suicide rates, a lot of suicide in that population. It's quite sad that you see that in that group of people, all these symptoms. It's associated with a lot of symptoms like that.
- 2.30** Now, one explanation for especially their social symptoms is that they have a mirror neuron dysfunction in their brain, that their theory of mind modules are not working optimally. So this is one theory. And certain research has been done. You put gel on electrodes on the brain called an EEG; you measure the activity of neurons. It appears that from some research that people with autism have problems with mirror neurons. So mirror neuron regions of the brain seem to be dysfunctional in people with autism.
- 2.31** The mirror neuron system—by the way, I should add this, the mirror neurons were discovered in the '90s. And initially, there was a lot of excitement about these mirror neurons. It was like, "Oh, wow, how great. What are these mirror neurons all about?" I remember everywhere, you look online and, "Oh, people fall in love—mirror neurons." "Love from a mother to a child—this is mirror neurons." "You learn how to learn language—this is mirror neurons." You watch somebody else speak, and then you look at him and you learn. You don't even need hundreds of years of practice. In the adaptive process, you don't need several generations. You can just watch somebody and instantly learn from them. Is that clear?

- 2.32 So mirror neurons explained everything. But then over time, it became clear that this was probably not the case. Maybe mirror neurons don't explain everything, but they seem to be very important. They seem to be important, and they seem to do something profound. But we have to make this little caveat here about how important and how widespread their function is.
- 2.33 By the way, you have mirror neurons or theory of mind in certain birds. Did you know that? There are certain birds, when they fly—the corvid family—when they fly out, they won't hide their food if they know another bird is looking. Is that interesting? It seems they have mirror neuron systems or theory of mind, at least. It's quite fascinating. But of course, they would be underdeveloped in birds and stuff like that, but there's something there.
- 2.34 Mirror neurons—frontal region seems to be important for mirror neurons. The frontal region and then a region called the temporoparietal junction. We learned about that before. Remember, with the whole sensory blending, to create a sense of a self? These two regions tend to be involved in cognitive empathy. Whereas emotional empathy tends to involve more of deeper structures in the brain within the limbic emotional core for the emotional affective empathy. That's the more emotional centers of the brain that are involved in that. The ACC would probably be more in the emotional department of empathy, although some studies have found that it's the bridge between emotion and cognition, so it's in between, and there's several studies on that. It's not clear. So there is this overlap, there is this distinction, and in some cases, overlap.
- 2.35 By the way, the theory of mind is interesting. The theory of mind, in the cognitive sense, is in the temporoparietal junction here, on the frontal lobe, we said. If you have damage to the temporoparietal junction, guess what can happen? We mentioned you can float outside yourself and all that in previous talks. But even something called the self-other distinction, knowing that I'm me and you are you. It's called the self-other distinction. This is something that's crucial, knowing that I'm me and you are you. People with schizophrenia, for example, have problems with that. A simple damage to that region of the brain, and you can no longer know that you are you. So this is a part of the theory of mind circuit.
- 2.36 Now, when I was growing up, there was a lot of psychopaths in my area, in my neighborhood. Sociopaths, probably even better description. So a psychopath is somebody that is cold, callous, unemotional. They are very good at cognitive empathy, knowing what you're thinking and what you're up to and what your intentions are. Very good at that. But affective, emotionally, completely flat. Nothing. Here we have a condition that serves as a mirror to autism. It serves as a mirror because here you have somebody who is a complete opposite in a sense.
- 2.37 When I grew up, I met a lot of people like this. I met a lot of people that were very callous, very cold, criminal. Psychopathy affects about 1% of the population; about 25%

of people in jails, for example, might have psychopathy. Psychopaths lack loyalty, for example, manipulate a lot. You can say the cold, completely calculated psychopath is—have you seen "The Godfather"? Have you seen Michael Corleone in the final scene? This is the scene where he's sitting there and then they're about to baptize—I think it's his sister's daughter or something, I don't know—but his family member is being baptized, and he's supposed to be the godfather. And at the same time, he's hiring—he's having all these murders arranged at various places. And he's sitting there in the church completely silent, completely cold, and just nothing. That's an example of psychopathy.

- 2.38** The autonomic system of somebody with psychopathy—you have the sympathetic and the parasympathetic branch of the nervous system. The sympathetic is the stress system. When you feel aroused, you feel agitated, you feel emotional, you're sweating when something terrifying potentially is about to happen. Then you have the relaxed parasympathetic nervous system when I'm sitting at home and I'm meditating and all that. That's the parasympathetic nervous system. Their nervous system is wired completely differently than you and I. They're completely cold, flat, and they don't have these emotional upheavals and reactions. They don't have that, the psychopath.
- 2.39** Another example. I don't want too many gangster movies, but have you seen "Goodfellas" when they're having spaghetti? This is a cool scene. They're having spaghetti, these three gangsters, in one of the gangster's, in Joe Pesci's mom's house. They're sitting there—it's a funny scene—they're sitting there, and the mother is talking to them about. They're pleasing the mother, "Mother this and that," and they're being very sweet to the mother. And then, in the meanwhile, in the trunk, there's a dead body. He's half dead. He's been stabbed 20 times, and he's trying to get out of the trunk while they're sitting, slowly having their spaghetti and just having fun with the mother. It's a cool scene in the sense that it really illustrates what psychopathy is all about.
- 2.40** The sociopaths are the folks who— their autonomic arousal and their stress system can actually become agitated. They get agitated at times. It's more of an acquired form of psychopathy. It's acquired, it's learned. It's often if you have—there's genetics, so we have genes and epigenetics and things that can change who you are at the genetic level. If you had a lot of stress and traumas in your childhood, you grew up in an environment where there was a lot of crime and violence, you learned that violence is good. Being aggressive and physically violent is a good thing. That can lead to the acquired psychopathy called sociopaths. That's how you differentiate. They are much more impulsive. They're more aggressive in the impulsive sense. The psychopath is different. He plans it out. He knows how to plan his murder, very cold and callously. "I'm going to plan this and that."
- 2.41** A psychopath is the person who, let's say, a girl breaks up with me. I get offended. A psychopath doesn't feel hurt. He gets offended because of his ego. He plans how to get

back to her. Obviously, he was dumped because he was a bad person. Then he plans, "How do I get back to her? I get back with her if I'm very nice, if I'm very romantic and how I was in the beginning." He knows how to do it. Then he gets back to her and he's with her for a year. And then on the anniversary, he says, "I'm going to dump you now because you dumped me and all of this was just a plan to get back at you." This is the psychopath. Everything is planned and calculated. No emotions, huge ego—narcissistic features. There's no guilt. There's no shame. All of these things that make us human are not in them. Think about this for a minute. It's quite profound. They don't have that. Sociopaths have that. They have a little bit of that because they haven't transformed completely.

- 2.42** There seems to be a genetic component to psychopathy. Specifically, genes involved in serotonin and interpersonal connections, so they have lower serotonin in the system. And then oxytocin, which is a neuropeptide involved in—we call it the love hormone, so involved in connection, pair bonding, and things like that. And so you might have diminished receptors in a region called the ACC. Remember the ACC? The empathy part of the brain? They might have diminished receptors for empathy in that region.
- 2.43** And you know what the annoying thing about this condition is? It's a personality disorder. You can't treat it. It's stuck for life. If you have psychopathy, you're pretty much stuck with it for life. Something you can't really treat. You can try to manage it, but you can't even manage it, probably. How do you manage a psychopath? He just outsmarts the therapist. He knows how to think and how to act with the therapist. What do you do with a psychopath? It's a terrible condition, and yet you have it. It's quite horrible in that way.
- 2.44** Okay, this is an important part of this. So psychopathy—it's a bad thing, right? It's pretty horrible, isn't it? We don't like it. Is it always bad? Are there instances where being a psychopath is a good thing?
- 2.45** If you're a spy.
- 2.46** Could be. But what if you are an athlete at the highest level and you have to train each day and you have to be hyper-focused, no emotion ever? You want to be the best. You have a big ego, grandiose. You want to be the best at your game. It's literally like, "I'm going to be cold. People don't matter to me. I'm going to be the best of the best." It might be important too but is it good to be a psychopath? What about if you're CEO of a company? No empathy. Just fire these people. Fire these guys. Being a psychopath is like—think about it, there's no emotional—you approach the world completely emotionless, while all the functions of the outer part of the brain up here, the logical, all that part, the DLPFC region, are hyperactive.
- 2.47** You know what everybody else is thinking, but you have no emotions. The amygdala part of the brain, the limbic structure, is completely flat. No emotion, no activity of the emotion there. Yet, this is active, this part of the brain. This part of the brain is also

thinner. All the emotional parts of the brain are thinned out, literally shrinking. But it's highly adaptive. The slogan, "Just go do it"—Nike, and all that—no commercials, but literally, "Just do it" is their—they never hesitate. They never procrastinate. They just do things; they get things done.

- 2.48** They fire the people they need to fire. They've found that actually a lot of people in business, CEOs, business type, high-performing athletes, have psychopathy. This is well known because imagine if you come from a high socioeconomic background and you have these genes for psychopathy, you're not going to go be a criminal. It doesn't pay off. You're going to be high earning CEO. You're going to be a businessman or something. Why would you go do crime? It's only the people in the poor neighborhoods that have aggression, where aggression pays off. Basically, intelligence is the key variable. So if your intelligence is high and you have psychopathy, you definitely go for other things than crime. But if you have low intelligence, crime is the way to go.
- 2.49** A number of experiments have been done to look at psychopathy and how all of us process emotions. And so there's this trolley dilemma—it's called the trolley dilemma: a trolley is moving, and you are on a bridge, and you see the trolley is about to hit five people and kill them. But then there's another track where there's one person on the track. Now, you have a switch you can flip. By flipping the switch, the track will change, and instead of killing these five people, it's going to kill that one person. The question is: what would you do? Would you flip the switch and have this innocent person on the other track die, or let destiny unfold and let the five people die? What would you do? Anybody? Would you flip the switch? It turns out most people would actually flip the switch. They'll say, "Okay, I'm going to flip the switch and five people won't die." Fine. It makes sense, right? It makes total sense in terms of numbers, regardless of how you look at it.
- 2.50** But what if you are standing on the bridge, and then before the train even comes close to hitting the five people, there's a fat guy standing right here. You can push him over, he falls down, and he dies, and the train hits that guy, and he stops the train from killing the five people. Would you do it in this case? Would you kill that poor chap, or no? It's the same as before. What's the difference? Here you're flipping a switch. Here you are pushing a guy? What would you do? You see the dilemma, right? You see the whole—what would you do? Would you push the guy? You would push the guy, wouldn't you? No? It's interesting. Most people wouldn't push the guy. Why is that the case? It's the same dilemma as before. It's not different. It's one life versus five lives. Why would you change now and not want to flip? Why not save the five people?
- 2.51** You know what the difference is? The first example is what's called cold cognition. You're simply using the outer part of the brain, the DLPFC logic. There's no emotional contact directly with somebody, so it's much easier to just flip the switch. When you're dealing with, you have to actually physically push somebody over, you're dealing with

what? Hot cognition. This involves medial centers of the brain. Remember we said in the brain, you have the front medial regions involved in decision making, emotional decision making that involves emotions? That's what's going on here. You're using that part of the brain. When you're using that part of the brain, you are less likely to push him over. It becomes an emotional thing.

- 2.52** That's why modern warfare today is killing so many people; we have become desensitized, insensitive to the killing because it's all through drones and planes and tanks and whatnot. Whereas before, when somebody was being killed in a direct physical combat, we felt it much more because we engaged the more medial structures of the prefrontal cortex. That's important to know this distinction.
- 2.53** There are other examples of this, other scenarios you can use that really shows the difference between these structures and the activation of these structures. If I was to give AJ, if I was to say, "AJ, I'm going to give you \$1,000"—that's too much—\$100 right now versus \$110 a week from now, what would you take?
- 2.54** Am I guaranteed the \$110?
- 2.55** I can't say; I won't say anything. This is it. These are the parameters.
- 2.56** I would take the \$100.
- 2.57** Brance? [inaudible] I feel you, brother. No, I feel you. Right. Okay. Most people would take the \$100, right? There's something special about the here and now. If you have immediate gratification, why delay it for 10 bucks? I'll just go for the \$100 now, right?
- 2.58** But AJ asked the same question: "What if I'm going to give you \$100 53 weeks from now or \$110 53 weeks from now? Or 52 weeks from now, I'm going to give you \$100, and 53, I'm going to give you \$110. Which one would you—what would you do? Why? You're contradicting yourself, brother. Or are you? Are you really contradicting yourself? It's still one week, right? But because it's far in the future, you don't have the same immediate gratification needs.
- 2.59** The first scenario involved the impulsive reward part of the brain. This is called the nucleus accumbens. It's involved in things like globus pallidus and similar names, meaning that in those parts of the brain, you have things like impulsive gratification, pretty much. When you're an alcoholic and you get alcohol, those parts of the brain become hyperactive. These are the impulsive reward pathways of the brain. When you are engaging in the first scenario, AJ wanted his 100 bucks, that's because he's using those parts of the brain. That's been shown in brain scans. Whereas when you are doing the 53 weeks versus 52, you're using what? Utilitarian, [lateral], DLPFC, prefrontal sides, no longer the inner parts, no longer the reward parts. It's all calculation. So you're more like the psychopath.
- 2.60** By the way, the scenario before with the psychopath— you ask him the same question about pushing the poor chap over. Guess what he'll do? "It doesn't matter. I'm going to

push him over." They always push the guy over. So you and I, and most people, will never push the guy over when we have to touch them directly. But the psychopath has no issue. They'll say, "Well, it's one versus five lives; it's pure math. I'll push him over." That's what they do because those centers of the brain, the amygdala, can become big, but they can also shrink. The cells there can become very small, and that's the psychopathy scenario.

- 2.61** I want to talk a little bit about one part of the brain that I love. This part of the brain—I mean, when you study the brain, you know what happens over time? It's almost like these structures—when I first studied biology, I heard all these Latin names. I thought, "What the heck is this? Superior this and inferior that." It didn't glue in my head. But then, over time, I learned that these parts of the brain have specific functions, and they almost became like little friends. Really, because you know, wow—
- 2.62** There's actually a part of the brain—it's called the orbitofrontal cortex—it actually does, when I pay attention to the world and I'm detecting threats in the world, it actually becomes activated. If I have OCD and I see threats everywhere, it becomes hyperactivated. It's called the orbitofrontal cortex. Orbitofrontal. It's this little thing here. If I zap it, if it's hyperactive in OCD, I zap it to inhibit it using transcranial magnetic stimulation, it becomes smaller. Imagine this. If I take an SSRI to dampen the activity of it, it becomes smaller.
- 2.63** There was a guy who had OCD. He had severe OCD, and it's hyperactive in OCD because you imagine the whole world is dangerous all the time. Everything is terrifying. Everything is contaminated. He took a gun and shot himself through the head. He wanted to kill himself—living with OCD is very hard, very, very hard. So he wanted to kill himself, execute himself. So shot himself. The bullet went through his brain. It took a part of the orbitofrontal cortex, and by accident, his OCD was cured. Imagine. Can you believe this? He became an accidental brain surgeon. This shows you how these brain structures—what they're doing.
- 2.64** Now, it turns out, OCD also has excessive empathy at times. Because if things are threatening and the orbitofrontal cortex is involved in threat and threat detection, but it's also part of the social network part of the brain because it's in between emotion and prefrontal cognition, so it's in there in the middle— psychopaths have very small orbitofrontal cortex; it's almost flat. No activity there. Is that interesting when you think about it? It's quite interesting. In fact, if you have damage there, say you don't have OCD and you have a stroke to that part of the brain, you might acquire psychopathy. You become a sociopath, just by a structure. Think about this.
- 2.65** Think about if Naomi—you're a nice girl, you're here, you're studious. Imagine overnight if your daughter completely changed just because of a stroke. Think about this. A damage to that region of the brain and complete personality change. This is powerful how the brain works and how certain parts can have significant, profound

ways of changing our personality. That's what I meant about brain structure being important for things.

- 2.66** What I find interesting is that sometimes being hyper-stressed or being sleep-deprived can reduce the activity between the prefrontal and the emotional core, the medial prefrontal and the emotional parts of the brain. So we know that. Literally one night of sleep deprivation can extremely make the activity very, very low. And that mirror is actually the psychopathy brain. When you are hyper stressed—have you noticed when you are very stressed and you have a thousand things on the agenda, you become less empathetic. Have you noticed that? You have less empathy, you're a little bit colder, you're a little bit more to myself and more selfish. Literally, when you are stressed and hyper stressed, your prefrontal becomes less active, the front part of the brain. You have less activity in the prefrontal. And over time, it will literally shrink and the connections with the emotional core—you become more impulsive.
- 2.67** The point of all of this is: even though you have the brain in psychopathy that's like that, wired like that from birth, you can, in some cases, temporarily have a somewhat similar brain, or over time, if your brain shrinks in the prefrontal regions, even start to mimic some of these conditions. So we have to be careful how we sleep and how we affect ourselves in terms of stress and so forth.
- 2.68** So no empathy is a bad thing. Having cognitive empathy is a good thing. Having emotional affective empathy is also a very good thing. Is there something we could call hyper-empathy? Could you be too empathetic? Is that possible? Is that a good thing or a bad thing to be too empathetic? Overly empathetic, having excess empathy? People who are depressed, their ACC, anterior cingulate cortex, show hyperactivity. We talked about in a previous course that even though the ACC is involved in empathy and empathetic reaction, it's often a more negative tone to the ACC and the activity of the ACC.
- 2.69** If you stimulate the ACC, you might even feel like a person is floating outside that's evil or a vague sense of a person. When you stimulate the ACC, it's often a vague person. It's not a person. Then you have to go back to this region, in the body image region. But it's often evil. So there's social cognition in the ACC, but it's often a negative tone in the ACC.
- 2.70** In depression, you find that the ACC is hyperactive, as if a part of the depression profile is having excess concern for other people. Not merely about boundaries and boundary setting, but it's really about having excess empathy. "I feel the pain of the world too much. I feel your pain a little bit too much. I feel what others' pain is like, and because of that reason, I feel it in my own skin, and it hurts me." That's part of the depression profile, to have excess empathy.
- 2.71** You have a lot of oxytocin receptors, this pair-bonding hormone in the ACC. A lot of receptors. Receptors, as I mentioned— I have mentioned, but not in this course,

perhaps. You have receptors. These are like little parking spots on the surface of cells, allowing other neurons to connect with that particular neuron and cells within that overall structure. You have a lot of doorways, a lot of parking spots for oxytocin in the ACC. So they can flood this region of the brain.

2.72 There's a creature. It's quite cute, actually. It's— what do you call it? It's like a hamster-like creature called a prairie vole. These animals will pair bond for life. They have this pair bonding ability. They will find a partner and stay with them for life. They tend to have a lot of oxytocin receptors in the ACC. And so when you block these receptors, they won't nourish their partners. Before, they're very nourishing because they have a lot of oxytocin. But because you remove that, you block the receptors for oxytocin, you find that they don't want to become nourishing and nurture and take care of their stressed-out partners. It's showing you directly how it affects us in that way and it affects both animals, but also humans, potentially.

2.73 Let me mention a case. I think let's end on a case. There's this guy, Charles Whitman. As detectives, as the true detectives we are, we want to try to figure out what's going on in his brain, what's going wrong. One day at the Texas University, Austin, he went to the Tower. He brought with him ammunition—a lot of ammunition there, guns. Nearing the Tower, the receptionist, he kills her with his rifle. He goes to the Tower and starts shooting at people. The first person he kills, actually, is a pregnant lady. Can you believe it? A pregnant lady, and then bystanders who come to help her, he kills them too, killing 13 people overall. Eventually, the police gets to him. They are able to get him at the Tower, and he's finished off.

2.74 And at this point, they're like, "What happened? Who is this guy? Why would you do something like this?" They found out that the night before, he actually killed his mother and his wife. But digging further, they see this guy was completely normal. Nothing unusual about him. He wasn't violent. His IQ was 138 or something. He was a very smart, caring person, but yet he committed this crime on this day. They looked at his record, they looked at his track record and his medical history, and it turned out he had gone to the doctor and said, "Doctor, I have these violent urges. I want to commit crime. I feel violence inside of me, and I have no clue why. In fact, when I die, you have to look inside my brain to see what's going on."

2.75 So eventually, they looked inside of his brain, because now they had his brain, he was dead. And he had a tumor growing on a region of the brain called the hypothalamus—if you have seen my previous courses, you know the hypothalamus is involved in things like aggression, violence, also things like sleep and other autonomic functions, but definitely aggression. That tumor was pressing on his amygdala as well. And so that explained his rampage that day. Again, showing us how neuroscience—you have small lesions, you have tumors, and you have complete personality changes. Sometimes you become a monster and kill your whole family, family members and people in society. So I wanted to end on that. Thank you.

- 2.76** You think there is so many more people born now with autism than in 1960?
- 2.77** A great question. I think potentially you can better detect it now. I think that's a part of it. So people probably had it before, but they didn't really go to the doctor and they didn't diagnose it. So underdiagnosis, I think, is probably a part of it. But I think there's more. Things like toxins during pregnancy, environmental toxins—yeah, perhaps, vaccines you're saying? Yeah, it could be. Who knows? I think there's a lot of these factors that play in. I think environmental factors from toxins, from environment, chemicals, maybe even vaccines. Who knows?
- 2.78** Within the last 10 years, we've seen cases in the US spike in regards to violent crimes as well as autism, depression, are big epidemics, at least of what I hear in Western civilizations. Is there research or different types of treatments being done in general health care that is shedding light on discovering possible symptoms or trauma or conditions of people's brain activity that could possibly prevent cases like what you ended with? That's my question.
- 2.79** Prevent depression and all these disorders, you're saying?
- 2.80** Or maybe shed light or bring awareness of it from an observation of the brain perspective prior to things escalating.
- 2.81** Okay, so preventive measures you can take before autism develops or before depression or any these is that what you're saying?
- 2.82** Correct.
- 2.83** A great question. Yeah, I think preventive medicine is important. I think it's a key part of the future. Now, autism is a different category because it's so developmental in the brain and the fetus and things. But I think mental disorders, certainly, there's a huge epigenetic component—epigenetics means that you have genetics and then you have epigenetics, meaning the environment can influence which genes get turned on and off. If I have genes for depression, I have genes for another anxiety disorder, let's say, if I make sure I have a good environment, I can buffer and make sure I don't get the actual depression or the mental disorder. Those are the things you can do.
- 2.84** Now, in terms of more neurological disorders like Parkinson's, dementia, and all that, gene testing could be important, particularly, because if you know early on that you're predisposed to potentially having dementia, Alzheimer's, then you can do lifestyle changes early. So genetic testing could be something one could do. But for a psychopathology, for depression, I wouldn't do gene testing for that. Looking at your family might be a good idea, so seeing how many family members you have with these conditions could tell you, "Okay, I have to live a life that's not too over the edge and make sure that I do therapy if I need so before I tip over the cliff."

Lecture 3: Emotions in Action

At a glance — Emotions in action and the brain systems behind them.

- 3.1 Yesterday we talked about a few things, and once I got back to the hotel, I had some extra notes. I'll put it here. Extra thoughts.
- 3.2 In the final lecture, we talked about this man who went up to the tower, Austin, Texas Tower, and he killed people. So he went up there for Cameron, who wasn't there. He went up to the top of the Austin Tower. He had ammunition with him, and he started killing people, starting with a pregnant woman and killing 13, wounding 33 people. They looked inside his brain. They found that he had a tumor. The hypothalamus was pressing on his amygdala. The amygdala is what? Emotional core. So he had that. I think this condition reminds us that, first of all, brain structures affect behavior and interpersonal behavior in profound ways. So that was one key point.
- 3.3 In this lecture, we'll talk more about emotion and how emotion comes about in the brain. Basic things like attraction—how is attraction? How does that emerge? What is things like laughter? Why do we laugh as human beings? And I think what we also learned from that case of that guy was the emotional core of the brain, that the way the emotional core of the brain interacts with various systems affects our social interaction in profound ways. It could be aggression—you become impulsive, you start killing people.
- 3.4 Yesterday, we talked about the fusiform face area. Remember that, AJ? What is the fusiform face area doing? Recognizing faces, right? But more than that, it also is involved in what? Direct classification of objects as well. It has that function. Now, interestingly, if there's a decoupling so that the emotional core of the brain, the amygdala, and the wires going to the fusiform face area, if the wire is cut, guess what happens? You have a very interesting condition called the Capgras delusion.
- 3.5 In Capgras delusion—let's say this chap, his name is Joe. He's been in a car accident or a motorcycle accident, even more severe. So there's a huge blow to his head. He comes out of the hospital. He talks to people. He seems fine. He plays chess, watches YouTube videos, is social with people. He's completely fine in all respects, but he has one quirk: each time he sees his mother, he says, "This is not my mother. This is an imposter."
- 3.6 The mother is trying to soothe him. She goes, "Oh, Joe, it's me. How are you doing?" And he's like, "It can't be you; it's not you. You're an imposter." He keeps telling the doctor, he might say, "Dr. Jalal, my mom is an imposter. How do I tell her? I know it's not her. She's a fake mom. It's a duplicate mom. It's not my actual mom." But everything else is fine.
- 3.7 Same with the dad. So the dad is a fake dad. It's not the real dad. This is interesting. In this case, clearly, there's a decoupling from the face recognition area of the brain and

the emotional core. So each time he sees the mother, there's no emotional warmth, there's no glow, there's no sense of, "I feel this is my mom. I'm supposed to feel good inside. It's my mom, it's my dad." So for that reason, for that reason alone, she must be an imposter. And again, it shows how emotion is a part of our whole landscape of creating a sense of a self.

- 3.8** When we go about in our daily lives, we use our emotions. The insula part of the brain and the emotional core, the amygdala—specifically, the insula, in this case, is the place that picks up bodily emotions and feelings from the body, things like sweating and things like that. It picks it up, sends a signal—the insula, it sends signals to the emotional core, the amygdala, which conveys that information to the rest of the brain. But if that link is cut, then you end up with a situation like this where your mom is an imposter.
- 3.9** Guess what, Cameron? The mom will call Joe, she'll go: [phone dialing sound], calling Joe. Joe is picking up, he goes, "Hi, mom, it's you. How are you doing? Oh, I missed you. How are you? I've been with my fake mom all day. Now, the real mom is back." What do you think is going on here? Why does the chap, each time he sees his mother, go, "This is a fake duplicate mother," but then when he speaks to the mom on the phone, she's the real mom. What is that about? Any guesses?
- 3.10** Right, exactly. This is exactly what it is. The emotional core, the amygdala, has a separate circuit going to the auditory regions. The voice does trigger and evoke all the warmth. It's just not with the facial recognition. So it shows us how it all can be coupled like this. And so you have a dissociation, a clear example of, in one case, she's not the real mom, but in another case, she is the real mom.
- 3.11** There's an even stranger condition. In this condition, you go around, let's say, you're in a mall, and everybody you see looks like your Uncle Joe. So this is Joe. "Oh, no, this is my Uncle Joe." There's a cashier, the registrar, "Oh, this is my Uncle Joe." And the parking guy, the parking officer, "This is my Uncle Joe." So what might be going on in the brain here? What do you think?
- 3.12** Well, if the fusiform face area, the face recognition area, is hyper-connected with the emotional core, you go about the visual landscape and everybody looks like all the faces evoke emotions. Your brain is trying to make sense of that. So it goes, "That face, it feels familiar. I can't account for all the emotions I'm feeling. Therefore, it must be a familiar face. In fact, it looks like my Uncle Joe now." Is that clear? Because of the emotional reaction that each face is triggering, for that reason, the face must be—the brain pretty much makes it look like your Uncle Joe in this case.
- 3.13** Occasionally, what you will find as well is if the signal is going from the emotional core, the amygdala, to the other nearby regions, face area and other areas as well, you might have a patient who looks in the mirror and says, "I'm dead." They see themselves as dead. You interview them, you say, "What do you mean you're dead?" They'll say, "I'm

dead." "Really? Okay. So what if I poke you with a needle?" Their hand starts bleeding and you say, "Do dead men bleed?" They say, "Yes, dead men do bleed after all." So the delusion will persist, even though it's clearly a delusion. They see themselves as a dead person and everything else is fine. Think about this. Everything else is okay. All their activities in their life, their school activities, their job—it's just this one quirk. Again, it shows us how the emotional core, how it's hooked up to various centers; it shows you really how that affects our interpersonal relationships.

- 3.14** We saw that very clearly as well with the autism example. Remember, we talked about being socially intuitive and socially puzzled, and if the activity of the face area is high and the amygdala is low, that's a good thing. And I think one practical lesson we can learn from that is that we can all train that, and that's very important, meaning it can flip. So we can become more like the autistic individuals by not engaging with people. Nowadays, people are very—including myself, we tend to be very much on the screen all day and watch videos and not really engage with people.
- 3.15** I think what can really happen, given the plasticity of the brain, the brain is hyperplastic, changing all the time, especially when we pay attention to things. I think a key part of this course is to go, "Oh, I want to apply some of this." For me personally, I need to train those interpersonal skills sometimes. I can be really awkward at times, socially. Larry, you're very good socially. But I can be awkward sometimes. When you read too much and you have too much work on computers and you watch screens, you become a little bit clumsy socially.
- 3.16** I'm very awkward. When I go out, I can definitely feel like— after COVID, I was literally a robot. Have you seen "The Big Bang Theory?" What is his name? Sheldon? I was Sheldon. I was literally having a conversation, and then, in the middle of a conversation, I was going, "Oh, I'm thinking about a scientific theory." I was completely Sheldon. But I've come out of Sheldon now, out of the shell, and become a little bit more social just by interacting and titillating those emotional core structures a bit. And you can really do that.
- 3.17** The amygdala activity can certainly go much down when you learn that faces are not really terrifying and people are nice and warm and just humans like you. I would encourage everyone to really use those circuits. Having heart and emotionality is a key to success in life. Trust me, no matter where you are, even if you are hyper-skilled and you have all this knowledge and you are hyper-smart, unlike me, but if you have a lot of good things, say some talents, it won't get you very far without heart, without being able to talk to people and being able to communicate. So that's one key. I really want to emphasize that. Go out and have deep conversations with people and learn those social skills if you've unlearned them. Not AJ; he has social skills.
- 3.18** I think this also makes me think about the whole thing— I talked about psychopaths yesterday and how psychopaths are very cool. Sometimes it could be a good thing to be a psychopath. You can be very precise, you can focus on your work in a very

professional manner, you fire people that are in the way, and you are hyper-focused and "Just Do It"—a Nike slogan type of mentality. And that's good for professional athletes to get you far. But I think if you're not a psychopath, which I hope none of you are, then having a heart is a big thing.

- 3.19** The greatest athletes out there—look at Messi, how much heart he has. Did you see him when he lost the Copa América to, I think, Chile or something, in one of the finals? The man was crying. You saw the heart. He's not like a robot who's just training. He has all this heart. And then when he won the World Cup against France, he was crying with happiness. And I think that's a part of it. The true greats, like Muhammad Ali, have heart. That's more than just focused training. I think being a psychopath can get you so far in life, but the real top of any discipline comes when you have a heart as well, and you can train that. If you become rigid and you become narrow, you can become much more human. I think that's where the real skill is, and it can be trained.
- 3.20** I think the way to train is important, to add a caveat here. It's through brain plasticity. Knowing the basics of brain plasticity is important because it's not only important for the direct application of this course, but for anything you do in life. How do you plastically rewire your brain to alter and shape human nature, make human nature more—you have a human nature that's innate, but you want to shape certain aspects of it so you're not completely tied down by your biology. I think knowing plasticity is a key thing.
- 3.21** These are steps I apply in my life all the time. One key step is certain chemicals. There are the chemicals of neuroplasticity. One is dopamine, the other is acetylcholine. It's not very complicated, actually; it's very simple: dopamine and acetylcholine. Dopamine, Sam, is involved in what? Reward. An anticipation of pleasure. "I'm anticipating a great meal. I'm anticipating a great party, a great soccer match." That's dopamine. The other neurochemical is acetylcholine. It's involved in focused attention. These are the two key ingredients. If you want to change your brain, you change it when you are hyper-focused on something because it's so immersive that you can't—you have to be focused on it. At the same time, it's pleasurable. There's something there, some anticipation of joy, of pleasure. There's this reward there for your brain.
- 3.22** Let's think of examples of that. If I'm playing a very immersive video game, very immersive, grabs my attention, and it's pleasurable. Would that be sufficient for neuroplasticity? Yes, it would. In life, whenever we do something, whether it's engaging in flow activity, things that immerse us, that is the key to neuroplasticity. When you go about in your daily lives and are doing certain things, random things, those won't necessarily change your brain, but when you actually engage in things that are pleasurable and have the attention component.
- 3.23** Beyond that, you need things like neurofertilizers, things that will make your brain more plastic naturally in a more biological protein sense so the brain is more like a kid's brain. That fertilizer we call BDNF and GDNF. You get that by exercising, for example. So

exercising is a great thing to do. Fast walking—when I mean exercising, specifically cardiovascular exercise has seemed to be the key. And things like fast walking can do it as well. The reason is when we walk fast, the brain goes like, "Why is this chap—why is Baland walking this fast? Why is he walking? We're supposed to be in a cave worrying about lions and tigers, not walking around."

- 3.24** I mean, think about food scarcity, Larry. There was a lot of food scarcity, so you don't want to use up those calories. You really want to save them, right? You want to save the calories. You want to save all the energy. Energy is a saving mode. By walking, the brain has a concept called anticipatory proliferation. Specifically, it's expecting that the person is wandering to a new place, it's going to a new location. This new location—that's the only reason we used to travel, walk for a long time for, say, half an hour, an hour, when we were going to a separate place to live or going to hunt.
- 3.25** For both cases, you needed neuroplasticity to know the landscape, to know the spatial layout of the land, to know where the lions and the mates and everything are in the landscape. You needed plasticity for that. So when we mimic that and walk about in our daily lives, walk around Scottsdale, fast walking—ideally, it has to be a bit fast—that can lay the grounds for the fertilizer, for the BDNF, brain-derived neurotrophic factors. So these are the keys for neuroplasticity.
- 3.26** All right, so let's look at the limbic emotional core of the brain. There's this guy, let's call him Joe. Suddenly, out of the blue, he developed pedophilic tendencies. His entire life, normal. Normal tendencies, nothing strange, nothing unusual. But suddenly, he was watching all this child pornography online. He decided to visit a brothel for the first time in his life—and this is a completely normal career guy. Nothing is unusual about him. Then his young stepdaughter, he starts hitting on her. Insane. What's going on?
- 3.27** The mother, the wife to Joe, alerts the authorities. He's brought in to the police station, arrested, and he's put in jail. There, he starts hitting on people. Can you believe this? On the prison guards, he starts hitting on people. Completely disinhibited, completely—he started hitting on people. And then, this is a few months in, he has to go for sentencing and to meet the judge and all that. But the day before, he complains of a severe headache. His head is hurting. "I have so much pain in my head." So they scan his brain.
- 3.28** As they scan his brain, they see that in the orbitofrontal cortex, there is a tumor there, and it's pressing on his hypothalamus. So it's affecting the orbitofrontal cortex and the hypothalamus. Orbitofrontal cortex involved in social decision making, impulsivity, it's part of the medial part of the frontal lobes. And the hypothalamus is involved in things like aggression, sexuality, sleep, and regulation of heart rate and things like that. It's a very key structure. So he had this tumor. He was brought into surgery. They removed the tumor, and the chap's sexual tendencies went back to normal. Can you believe this?

- 3.29** So he was let out. He went out. For six months, he was living a normal life. Nothing unusual about him. But then, about six months later, the tendencies emerged again, and he became the sexual monster, the deviant he was before. Once more, they brought him to the doctor, did a scan. The tumor had regrown. They removed it, and he became normal again. Can you believe this?
- 3.30** This is a key part of this course. The brain is very modular in its structures. It has all these structures. You have a structure for language, you have a structure for sexuality, for sleep, you have a structure for decision making, you have a structure for art, you have a structure for vision, you have a structure for hearing, seeing faces, and when these are damaged, profound alterations occur to your personality. I think that is what this course teaches us, that the brain works in this way.
- 3.31** Now, this makes you think. This makes you think, Cameron. This guy, for example, was he truly a pedophile inside and the prefrontal cortex was just inhibiting his sexuality? Was that really the case? How can you actually be a pedophile just because a part of the brain is knocked out? How is that even possible, Sam? Tell me. How could that be? That is a question for you and I. I don't have an answer.
- 3.32** Was he truly a pedophile? We don't know. We have no clue. We just know that we can't really say that he was a pedophile. What is a pedophile? Pedophilic tendencies, we find that in a certain amount of the populations. There are actually things in the brain that are different— pedophilic brains are different; they have certain structures that are different. Can you believe this? I believe the IQ is lower in pedophiles. I believe that, on average, their height is lower, so their body image schema might be different, wired differently, and we'll get into how body schema in the brain, when it's wired differently, can affect our sexuality.
- 3.33** But it really raises this question: is the brain inhibiting a real monster? Or how do you define what lust is and what sexuality is in the first place? What do you think, Cameron?
- 3.34** How do you define sexuality?
- 3.35** Yeah, the point is: was he truly a pedophile inside, and then the prefrontal cortex was just inhibiting that activity, and then perhaps he is guilty? He could be a pedophile any day when his prefrontal— what I'm trying to get at is: how do we understand this in a broader philosophical perspective, what sexuality is and what it means to all of us and how these brain structures affect it and so forth? We can chat about it later, but I'm saying this as seeds for thinking.
- 3.36** In 1935, James Papez, a neuroanatomist, identified a structure that was a ring-like structure like this, inside the brain, that was involved in emotion. One way that they came to know about this structure was through the rabies virus. The rabies is a virus— how do you pronounce it? Rabies. Sorry, my English is not good—so you have the virus: it enters when you get a dog bite. It goes to the peripheral nerve all the way up to the

brain. In the brain, it targets the limbic structure, this limbic ring, and the person becomes aggressive from the rabies. He becomes aggressive; he wants to bite people.

- 3.37** It's interesting because the virus makes you aggressive. It enters your saliva, specifically so you can obey its command and bite others and spread the virus in that way. And then the next person will have the virus like a bullet; it will go to his brain, to his limbic structure, affect it, and then it keeps spreading, like zombies, almost. Think about it. Here you have a creature, you have something that's a million times larger than you and you're controlling it. That's a very smart move. A very smart virus. From knowing about this virus, they could understand that, really, it's in the limbic structure that things like aggression and violence and even impulsive tendencies can emerge.
- 3.38** There are key links between the amygdala, the core fear center of the brain, and a region called the hypothalamus, which we just mentioned, the center for fear, aggression, sleep, and all that. There are a lot of wires going there. So whenever my amygdala is aroused and I see something in the world, I can send command to the hypothalamus. The hypothalamus is almost this structure that—it has so many command centers.
- 3.39** First of all, it can signal to a nearby structure called the pituitary, which helps us release hormones. It sends signals to the glands, and we release hormones in the body. That makes sense, because whenever you have a potential danger out there in the world, something dangerous, you perceive it, first of all. Interestingly, there's actually connections from the amygdala. So if the amygdala is here and the recognition area, the fusiform, is here, you have centers—whenever you see something in the world, going from the visual part of the world, you see it, it goes to the amygdala, and then it bypasses—if it's threatening and there's a lot of activation, it bypasses the visual recognition area.
- 3.40** Normally when I see things, it goes something like this: I see it, it's sent to a visual recognition area—I know it's a table, it's a chair. Then I send signals to the amygdala, which says, "Oh, this is terrifying. Be scared of this." And then you send signals to things like the hypothalamus, and you say, "Oh, release hormones now"—cortisol, adrenaline, all these actions that we know. So hypothalamus is also involved in fleeing, fighting, feeding, and sexual behavior and all of this. So it has all these actions that it can make you do, plus hormones. So the key point is: occasionally, when the thing you're seeing is terrifying, it will bypass the recognition area and go straight to the amygdala.
- 3.41** Let me illustrate this. If there's a very charming guy or girl, you might notice, even before you've actually recognized them, your heart is beating a little bit and you're sweating. Or if there's a predator, even before you know what it is, you have started to sweat and become physiologically aroused. We know this from the laboratory: when you flash images of emotionally evocative objects, you literally will have arousal before the person even knows what it is. But I guess my key point here is that hypothalamus is involved in these various things. It's like when you have the emotional core activate

emotionality, it goes to the hypothalamus and says, "What do I do with this? What do I do with these emotions?" "Release hormones. Run, fight, engage in various behaviors."

- 3.42** And I think a key part of this is to know that we differentiate between instincts in the brain. Is fear an instinct, for example? What about anxiety? Is that an instinct? No, right? Okay. Absolutely. So fear is an instinct. An instinct is innate. You're born with it. It's not learned. There's parental instinct. There's an instinct to be social. If I lock somebody up in—what do you call this place where we keep them for 24 hours with no people?—solitary. What happens to him? He goes crazy. So humans have social instinct, parental instinct, instinct to reproduce and so forth. These are instincts. These aren't learned. Attraction is actually instinctual. There's a huge instinctual component to attraction.
- 3.43** But anxiety is learned. Did you know babies, small babies—you can watch this on YouTube—infants, three months, you give them a big snake like this, they'll play with it. They just play with it. The big snake is going around. Can you believe it? It's around the baby and they're playing with it like it is a very cute little animal. No anxiety, nothing. Anxiety is largely learned. Anxiety is a learned mechanism, often conditioning.
- 3.44** I learn, for example, that certain animals—I go out and play as a kid, I get stung by a certain animal, let's say a bee, like an insect, and I get fear of bees. Each time I see a bee, I have anxiety. Now, in some cases, it might generalize to whenever I see something that looks like a bee, I get anxiety, too. That's when we get into the pathological realm. But that's what anxiety is about.
- 3.45** But of course, anxiety can also be very adaptive. Anxiety has an adaptive component, too. Or am I wrong? Is it always bad to have anxiety? What do you think? Am I right? Is it good? Sometimes. Pathological anxiety, when I'm terrified of everything and I walk about in life and my heart is racing when I'm just driving down the freeway, I feel like everything is terrifying—that's not good. But if you have some anxiety, let's say before an exam, before an important interview, that can make you more—first of all, moderate levels of stress makes your cognition more sharp, you prepare more and so forth. So it has an adaptive component to it.
- 3.46** Unlike basic sensations and instincts, emotions orchestrate a more sophisticated and a more organized behavior—a set of behaviors and processes in the body. Anxiety will often initiate fight-and-flight reactions. Hormones are released. Psychological parts of the brain are activated as well. These are all coping mechanisms.
- 3.47** But if my hand is burned, I immediately scream and remove my hand. That's pretty much it. It is a sensation. A sensation is much more simple, for example. That could be pain we experience and so forth. Disgust is another sensation. But emotions are more—they have a more orchestrated response and they last for longer. So when we have anxiety, it stays in the system for longer.

- 3.48** Often we find that the prefrontal, the rational part of the brain, and the emotional core speak different languages, these two centers. For this reason, they often reach different conclusions. I know I shouldn't be scared of this thing, this interview, that I have, yet I cannot help it. I have anxiety. Why is that the case? This is because—really, they came about for different purposes in nature. Anxiety came about for immediate survival. It's emotionally wired while the CEO of the brain is rational and calculated. So we have these different aspects and functions, really.
- 3.49** Let's talk about attraction for a little bit. How does attraction come about in the brain? What is attraction about? First of all, in attraction—the regions of the brain involved in attraction are regions like, first of all, the prefrontal cortex. There's a rational decision making. It's a rational, psychological thing. I've maybe learned that certain features are attractive. So even though you have innate structures that are important in the attraction response—and we'll see that, the psychological component is huge.
- 3.50** Let me give you an example. Have you seen "Coming to America" movie? You've seen some of it? These guys, these two princes, are coming to America from Africa. They're actually princes. And then they meet these girls. They're out there to get married in America. So they meet these two girls. And one of the girls is not attracted to them because they present themselves as servants, as poor people. Then when the girl learns that the guy is a prince, she completely shifts her view of him and she becomes attracted to him. That's an example of the psychological prefrontal aspect of attraction. Is that clear? Does that make sense? This is a psychological aspect.
- 3.51** Then you have things like the insula, which is involved in the sensation of bodily feelings. Of course, the emotional centers, and then you have the reward parts of the brain. Let me illustrate how it's all tied up. One theory is the following. In the superior parietal lobule—remember that from yesterday? It's involved in constructing a body image, the more abstract sense of a self. We believe that from that region in that body image, there's a full map your entire body in the more abstract—not in the sensory way, but in an overall body map. That sends signals to the—it has connections to the visual part of the brain and the emotional part of the brain. You have the superior parietal lobule, the body image center, sending commands to vision and emotion.
- 3.52** That's a three-way interaction of attraction. Meaning when I see certain shapes, I'm attracted. That explains why frogs are attracted to frogs. Humans are attracted to humans., Goats are attracted to goats. We're not attracted to a chair or a dog or a sheep, at least most of us. We tend to be attracted to the human shape. And so we believe that's hardwired in the brain and there's a clear circuit. So you have that as one layer, the actual morphology of the human shape.
- 3.53** Aside from that, we have what you would call a learned mechanism that's imprinted in the brain. Let me illustrate this. So if you have a rat, and you show it a rectangular shape. Show that rat a rectangular shape and a square. Then each time you show them these, each time you show the rat the rectangular shape, you give it a piece of cheese.

Now it learns to like the rectangular shape. Each time it sees it—if you show them these, it always goes for the rectangular.

- 3.54** But then you show it even more rectangular shape, a thinner rectangular shape, and it will choose the thinner rectangular shape versus the actual original rectangular shape. Is that clear? Initially, it chose the rectangular shape because it was paired with a piece of cheese. Then you show it even skinnier rectangular shape, and it will go for the skinnier. It has learned a rule: rectangularity. The more rectangular, the better. Is that clear? That's the rule it's learned. It never really saw it as, "This is the original shape that I'm interested in." It learned that the rectangular shape is more rectangular than the square. And then the next one is—it learned a rule.
- 3.55** If you think about things like caricature—I want to apply this to humans. What's a caricature? When you look at a caricature—, this is pretty much like the rat example because you have, let's say, Donald Trump. I'm using his example. He's a good example for these. You have him. You take his face, you subtract his face from the average male face, Trump's face, subtract the difference from the average male face and amplify the difference. So his features become more Trump-like than actual Trump. So his cheeks become rounder, probably. His eyebrows, his entire face, his body are amplified. It looks more like Trump than Trump himself. This is a caricature. And it somehow is appealing. It somehow evokes something that the artist is using.
- 3.56** So the question might be—all of us, we might have a template that we find attractive. Initially, we played on the playground and we were playing with other kids, and we found a certain face attractive. Let's say it made us feel good inside. "This was something that felt good, playing with this little girl" or whatever. It could even be somebody's mother or a sister. It just made you feel good. It was very innocent. And then later on, the faces that I'm attracted to are really that initial face that made me feel good, but an amplified version, like the caricature. So that is a possibility, we think, that it's imprinting as well, and it spreads so it becomes—that next face is just a version of the initial one, but an amplified version: the nose is even more pointy, the face is even more round—it has the original features, but more. Does that make sense?
- 3.57** So many instinctual factors are involved in the attraction process. You show men pictures of women. You have, let's say, 20 pictures here, 20 pictures here. In half of the pictures, the women's pupils are dilated. Men will consistently choose pictures with the pupils where they're dilated, even though they are average, like the attraction is the same. But just because the pupils are dilated, they will consistently choose those. The question is why?
- 3.58** It turns out that dilated pupils signal sexual excitement, readiness, and it's a biological cue for males. They are completely unaware why they're choosing it, but they're choosing it because of the fact that their biology is telling them to choose it. Males consistently will rate females higher in attraction—there's actually a bodily figure:

typically large breasts, round buttocks, fair skin. The science is telling us this. These are all signals of high estrogen. If show them pictures of that, they will rate these as higher.

- 3.59** Males? The opposite. If a female were to rate males, high testosterone features are typically rated as higher. So big jawline, a little bit beard, large chest, broad shoulders, signaling high testosterone. Now, again, the caveat is here: of course, males have this innate preference when you do an average rating, but there's a psychological component, too. It might not always be smart to—you find a girl attractive because she's smart, because she has a personality, because she's caring and nurturing. In fact, females, when they're ovulating, they have different choices than when they are not ovulating. So there's a lot of variation, so to speak.
- 3.60** In fact, there was a satirical—there was a paper written about men, gentlemen. The name of the paper was, "Why Do Gentlemen Prefer Blondes?" That was the name of the paper. It was really about this argument that males prefer women with more fair skin because you can easily detect parasites on the skin. So if she's healthy or not healthy—if it's fair, the skin, your illness, you can better detect. And also if she's seeing another guy on the side, any cues of embarrassment or the fact that she would carry another man's baby would come out. Embarrassment would lead to things like blushing and things like that. It's more clearly visible in a woman with fair skin. It's a bit of an argument, but it's out there, so I want you to know.
- 3.61** It turns out the ideal male preference at the instinctual level is a hip ratio of 0.7. Meaning, it pretty much just means in fancy terms, the playboy type figure. If you actually want the objective truth, that's what males would prefer. They actually rate women with this figure as more intelligent, healthier, and smarter as well. Can you believe this? What world we're living in? So superficial. But that's what the science says.
- 3.62** And also just generally, women are rated as more attractive around 10 days before menstruation, at the peak of fertility. 10 days—that's when they're rated as most attractive. Not even by males, but also by other females; they will also rate them as more attractive. Probably because rating females—having an eye on which females are attractive as potentially rivals and competitors for males out there. So females also have this ability to detect attraction in that way. The attraction is an interesting one; it's important to add.
- 3.63** We'll see that in the next lecture, how big the psychological dimension is. And then when somebody falls in love, it's a completely different matter because then you erase—somebody is initially not very attractive, you fall in love, and their defects go away; you don't see any defects anymore. Human beings have higher consciousness and higher level of thinking, so we're not tied down to our biology in that crude sense. But there is that innate bestiality to all of us that's there, too.

- 3.64** Let's talk about how does laughter come about? What is laughter about? Let's think about laughter for a minute. A laughter— I'm going to give you a case example. There was this guy. He was a plumber in London at his mom's funeral, wearing all black. And it was a sad event. His mother had passed away. He was only 25. His whole family was there. And suddenly, he started to laugh. He went, "Ha ha ha." He just kept laughing; he laughed. And the laughter kept going. He was laughing more and more and more. It was completely embarrassing.
- 3.65** The whole family was looking at him and he was like, "What do I do?" It was completely embarrassing, of course, at this point. The family members were like, "We're going to do something about this." So they took him to the emergency room, and it turned out that he had to stay at the hospital. A few days later, he died from an aneurysm. They checked his brain, and he had an aneurysm. There was a ruptured artery pressing on his anterior cingulate cortex area, the ACC, and the nearby structures as well, the hypothalamus and the nearby structures. The ACC and the hypothalamus are part of the limbic structure. So because of this, he couldn't stop laughing.
- 3.66** I've seen this, too, with patients that are involved in research. We go into the ACC, we stimulate it invasively, and you see a patient can't stop laughing. They keep laughing and laughing and just laughter. And then you ask the patient, "Why are you laughing?" And they will go, "Oh, the room is funny." "Oh, I found that book on the shelf, and it's so funny." "Oh, I couldn't stop thinking of my dog." They confabulate; they come up with all these excuses for why they're laughing. But it's because that part of the region is stimulated.
- 3.67** There's clearly a laughter circuit in the brain, and it's often these regions that are the intersection between motor and movement and emotion, so the supplementary motor area, the ACC, these regions that have an emotional component but also have a motor component. When you think about it, you're laughing. So that's the laughter circuit. But let's explore laughter and jokes a bit more and what they're about.
- 3.68** So what is a joke, for example? What is a joke? Why do we laugh? Why do we find jokes funny? Let's look at a scenario. Let's say you have this guy. He's walking about, and as he's walking—let's say he's a very formal guy; he's wearing a tuxedo. He's very formal. Let's say he's the chairman of some company, he's walking, and he slips on a banana peel and falls. Would you find that funny? Most people would probably laugh. That's funny, so they'll laugh.
- 3.69** But what if the same guy walks, falls, hits his head, and his skull is fractured and blood gushes out? Would you laugh? You wouldn't, right? Why? Why is one scenario funny, and we laugh, and the other is not? It seems like laughter, having this sound coming out from us, we have this explosive sound—if a Martian was to come on Earth and watch us, humans, they would say, "Why are these humans, once in a while, having this explosive sound coming out from them? What's going on? That's weird." We haven't really thought about it.

- 3.70** I think thinking about that is pretty crucial. Why do we even laugh in the first place? I think this joke or this scenario shows the fact that it seems like jokes are all about false alarm. Laughter is about false alarm signals. In the first scenario, when he slipped on the banana peel and fell, he didn't die. So the bystanders there would laugh because he's signaling to the tribal members, the people of the tribe, "Don't laugh. It's not—" Sorry, "Don't come to our aid. The guy is not hurt. He's not damaged. So don't use your resources." We don't want to waste any resources in nature because resources and energy is scarce. And so for that reason—it's a false alarm signal. Does that make sense? That's what we really think it is.
- 3.71** And maybe, perhaps, looking at the brain a little bit can help us explore this a bit further. In the brain, first of all, when I feel something, I have sensory sensations going to my somatosensory map. That's the map that allows me to feel things. It's contralateral in the way that everything on the right side of the body is processed on the left. The left side of the body—right.
- 3.72** When I, let's say—can you poke your hand? Cameron is poking his hand. The signal is going to the somatosensory region of his brain. Then signals are conveyed to the insula. The insula is the region of the brain that monitors body and bodily sensations. It's involved in things like disgust as well, and in things like pain and pain perception. From there, the signal is conveyed further to the ACC. The ACC—we've talked about that extensively— is involved in empathy, is involved in various social-emotional things. It's hyperactive in depression and so on.
- 3.73** When the signal is conveyed from the insula, you have the actual physical sensation of pain. In the ACC, you have the emotional component of pain, the actual agony of pain. "I feel pain because it feels unpleasant." That's in the ACC. Is that clear? Initial somatosensory region, "I feel the actual touch on my skin." Insula, "I feel the pain." The anterior cingulate cortex, the ACC—the actual agony of pain. So you have this circuit.
- 3.74** Now, if there's damage to that circuit, specifically from the centers going from the insula to the ACC, I feel the physical pain, but I don't feel the psychological agony of pain. There's no agony; there's no actual psychological dimension to it. It's almost like the joke example where you have the initial alarm signal, followed by a null signal because the signal is not going to the ACC. Often you find that people who have this circuit cut between the insula and the ACC will start laughing each time you poke them with a needle. Is that clear? I want to make this completely clear. Is this crisp and clear?
- 3.75** So because the region that has to do with physical pain, the actual physical pain, and the psychological component, an agony of pain, because that wire is cut, you have the initial alarm— the chap is falling, run to his aid—and then it's followed by "Oh, false alarm. I don't have the agony of pain, therefore it must be a false alarm," and you start laughing. And so you have that in people with this circuit, this wire cut. Does that make sense? Does that explain pain, though, a little bit, and laughter? That's really what it's about in neuroscientific terms. It seems to be that laughter is really the sound we make

to signal a false alarm, the fact that there's an initial threat, followed by devaluation of that threat.

- 3.76** Think about it. Have you ever had a little kid? I do this with my niece sometimes, Sophia. I have her, and then I might do like this. Or actually, we have another one now. Sophia is getting old now. She's seven, eight. We have Alma, the smaller one. I'll do like this; I tickle her. What does she do? Laughs. Can we use that analogy before? Initial threat followed by, "Oh, it's just my uncle." Same principle. Threat, false alarm—I'm laughing. "Don't come to my aid. It was just a caregiver; it wasn't a monster."
- 3.77** All right, I think this is all for this lecture. Thank you.
- 3.78** You talked about attraction being innate and developing that shape of what you're attracted to really young. As an adult, can you change that?
- 3.79** Do you mean the actual body map? Yeah. So it's clear that you have the body map, and the body map is just a human shape overall. It doesn't have all the—
- 3.80** No, the step after the body map.
- 3.81** You mean the initial wiring of the watching—playing with somebody, building sand castles and all that and then falling in love? Okay. I think, yeah, 100 %. But often what you'll find this really has to do with acquired taste, that then you typically find that you may fall in love with a girl that looks like her, but not much because she has another dimension to her personality. So she still has some of those features, but not completely. This would be the typical route, I think. And then you fall for somebody else who is a bit different, but not completely. You know what I mean?
- 3.82** It evolves, but it evolves from the—it still has that initial template. It's very hard, I think, going from, "I'm not attracted to blondes" to be completely attracted to a blonde, unless she has an extremely persuasive personality and you go, "My God, she's just special." I think it's gradual from the brain perspective, but there actually might be cases where you circumvent that completely and go for a completely different skin tone, hair color, body shape, completely. You might do that.
- 3.83** But then she has another feature that you've also learned to find attractive because you found that girls that made certain jokes in kindergarten were more attractive in their personality. So it's more psychological then. But yeah, you can shape it. That's the great thing about human nature. We can shape our—we have our innate biology, but we can shape it in many ways and mold it. But we have to have exposure and plasticity. Good question.
- 3.84** I work in higher education, and we've seen an increase in mental health needs for our students in the last 8-10 years, some up to 300% for the need for resources, but also for medical accommodations. Some of the cases are quite severe. I'm just wondering why—and from what I know from either the student or the parent or the medical team

self-reporting— why do we automatically go to medications and we don't go directly to looking at the brain? Because some of them end quite tragically on the medications.

- 3.85** Yeah, I think it's a tradition. I think for a very long time, the brain was seen as a machine, very biological. A machine, like the liver, like the kidney or any other organ. We'll see in the next lecture today how psychology and mind and beliefs can shape a lot of biology. But for a very long time, we were under the conviction that the brain is very mechanical. It's the old view of the Newtonian, "Things are very predictable and things are very biological," like a machine. I think that's the reason. Some of us are still stuck in that old mindset, and some are starting to think in a more expansive way in understanding this psychological component. I think it's changing. I think so. It's a good point.
- 3.86** When you were describing laughter and how we classify it as a false alarm signal, that led me to be curious about why, when we engage in things like watching a comedy or a comedian to make jokes for an hour, two hours—why do we engage in that for so long if, on an instinctual level, its use case is a false alarm signal?
- 3.87** Yes, I think simply because it is a false alarm signal, probably, that's what we think—but laughter feels good. I think laughing and positive emotions activate a part of the left side of the brain and go down to our reward center. So it can almost be a drug. In fact, you can become addicted to watching comedy, too. So I think because of the "feel good." And then when you have feel good chemicals, they will often reduce the activity of feel bad chemicals: cortisol will go down—stress goes down—adrenaline goes down. Adrenaline can sometimes rise, too, if you're having fun, but it's definitely cortisol and amygdala that turn off. So I think it's almost like a drug, but it's good. If it's used in moderation, it's great. I recommend it for sure.
- 3.88** We're social creatures as humans. My question was— loneliness, how it's becoming more rampant. What does that do to the brain? What areas of the brain get activated? Just a little bit about what loneliness does to the brain, if you know anything about that.
- 3.89** Loneliness. Yeah, loneliness is not a good thing. We talked about humans; humans are social creatures. When you are alone, you don't have the social signals that would— when you watch other people's faces, when people touch your skin, for example—this is actually a good thing. If you want to reduce the activity of the amygdala and reduce stress, you have to engage in hugging. So hugging people and being physically active, sends out chemicals like oxytocin. And oxytocin has a direct effect on the amygdala, the fear part of the brain and the fear center.
- 3.90** So watching faces—as social creatures, we want to be around people because we feel safe. Think about the caveman again. If he's alone, is he more vulnerable to predation and tigers and all that? By having people around us, we feel more safe. Cortisol will go down, the amygdala activity goes down, stress goes down. When we—especially when

we hug and have fun with people, then we really tap into our instincts and ramp up all the positive hormones. Does that make sense?

- 3.91** At the beginning of the lecture, you talked about neurofertilizers, BDNF, brain-derived, and then you said GDNF. Is "G" for gut or what does GDNF stand for?
- 3.92** Yeah, I should have mentioned that. Sorry. That's a good point. Glial-derived neurotrophic factors. In the brain, you have neurons, and then you have glia. Glia are supporting cells. There's tons of them. They outnumber the neurons in the brain—1:10, I think. There's tons of them. These are supporting cells that help neurons communicate faster. On the neurons, on the dendrite, which is the wire that comes out from the neuron, there's a fatty sheet around it, like a coat. The more fat substance you have, the faster the neuron will communicate. Having a lot of these glia-derived neurotrophic factors will make these glia communicate much faster and you have faster communication and plasticity. But they have to do with glia.
- 3.93** Okay. So it expedites or improves the synapses between the dendrites and the axons and all of that? So the glia-derived neurotransmitters help with the synapses? Between the neurons?
- 3.94** The glia will latch on. So if you have the neurons, the glia will latch on, and they will, for example, help with faster transmission and others. They have tons of cells. You have glia in the peripheral nervous system with different names like Schwann cells, having different functions involved in various things like healing and stuff. But there's a lot of functions to them. But they're supporting cells.
- 3.95** So they're cells; they're not elements or compounds like salt that would help expedite or improve the synapse? It's an actual cell?
- 3.96** They're an actual cell. The glia-derived neurotrophic factors then are proteins that would help with the glia.

Lecture 4: The Romantic Brain

At a glance — The romantic brain — attraction, love, and their neural basis.

- 4.1 In the makeup room, I was thinking about it. You have attraction and you think about what attraction means. You may feel like if you don't have big muscles, you're skinny like me, you don't have the hugest body, maybe then you're not that attractive to females, maybe. But I want to make it clear that it's a complex scenario.
- 4.2 Females tend to be attracted to things like talents, for example. So if you have a lot of talents—you're good at sports, you're good at maybe throwing a spear, you have certain talents, women tend to find that attractive in men. So that's a good thing. So if you're like me, you have to work on your talents. You have to hone your skills and all that. So that's a huge thing.
- 4.3 It sounds simple, but eating healthy, drinking a lot of water, exercising, all of that can help with a physical basic attraction: to look better, to look healthier. So these are things we can do.
- 4.4 And also to the skinnier guys with softer-looking features, like me, instead of the more masculine-type features— when women are ovulating, they tend to prefer the more testosterone look with a big chest and muscles. But when they're not ovulating, they tend to go for the more nurturing, more caring, and more long-term partner-potential kind of scenario. So there is a variation there. It's not a black-and-white thing. I want to make that clear.
- 4.5 Other dimensions, too. Things like scent, smelling. We know in animals, the pheromones and chemicals, there's a lot of—we have receptors for that in the brain. And so smells can have an impact as well. Let me give you an example. There was a study done where they had— these guys would wear the same T-shirts for many days and obviously, they would become sweaty and smell. And then women were smelling these T-shirts of men—they would smell them; it was not very nice, the smell.
- 4.6 And based on their immune system—here's the point: the immune system, there's something called hybrid vigor, which is when you have an immune system that's different from yours. Let's say you have certain genes and then you have another immune system that's different based on a different set of genes—you get what's called hybrid vigor. Women tend to prefer smells of guys, the T-shirts, based on immune systems that were different from their own. It's interesting. Just by smelling. It shows you that smells and pheromones and things like that play a role, too.
- 4.7 So many components: smell, physical—females are ovulating—a nurturing look, talents. And by the way, women generally tend to be less visual than men. Men are highly visual in their— Larry is doing like this; he knows. Men are more visual in their preference when they're in mate selection. There's a visual component.

- 4.8 We'll talk about things like jealousy, embarrassment, romantic love in this lecture. Romantic love is a deep one. So many scars, so many things in life. Romantic love is deep; it's a very deep one. But let's start with jealousy. It's a bit more simple.
- 4.9 What is jealousy? I wrote a paper with my mentor, Ramachandran, about 10 years ago on jealousy and the neuroscience of jealousy and what jealousy is all about. This is a great paper. Well, I shouldn't say it; it's my own paper. It's an interesting paper. You should go read it. I had some contributions to it. It was mostly my mentor. So it was an interesting paper.
- 4.10 Now, what is jealousy and what is envy and what's admiration? Jealousy is when somebody has something you covet that you want for yourself, so you feel jealous. You want to deprive that person of their resources. Envy is potentially, one might argue, less aggressive. It's less like—you envy somebody, but it's not that you want to steal their resources. But again, it's a definition. We don't really—the definition is not completely clear. Admiration is different. I'm jealous of the guy who just walked away with the woman that I had my eye on. But I admire Albert Einstein or something like that, or the guy who just got a prize for his research. So I admire him. It's a bit different.
- 4.11 Jealousy is hooked up in the brain. In fact, a lot of these social emotions like embarrassment, jealousy, pride, love, come about in the orbitofrontal regions. Orbitofrontal cortex—we met that little structure in our previous lectures. It's a structure involved in things like surveillance of the environment and threat detection. We said that the guy who suddenly became a pedophile had a tumor in that region. It's involved in OCD, we said. So the guy who shot himself through the head, damaged his orbitofrontal cortex, his OCD went away—that's that region. It's a bridge between emotions and intellect, the prefrontal cortex. It has this bridge function which is important to highlight. We need a communication channel, a portal, for these two worlds to talk: emotion and higher-order thinking.
- 4.12 Now, I would guess—and we argue this—that a child, for example, or most animals, don't have higher-order emotions—let's call them complex human emotions: pride—would a child have pride at two years? Would they have social [emotions] like love? They have empathy and they have attachment, but do they have love, though? Embarrassment and these emotions? Probably not as much as. So these are developed later. These are higher-order forms of cognitive processes.
- 4.13 In fact, even a frontal—somebody with frontal damage. Let's say you have frontal damage, damage to your prefrontal cortex, you would still be—you could have aggression, you could have lust, you could have all these impulses, but probably romantic love and embarrassment and things like that go away. Jealousy goes away.
- 4.14 In fact, when it comes to jealousy, often, if you want to know what you really value in life, look at what you're jealous of. Instead of going, "Do I really want to be an astronaut?" Or, "Do I really want that thing, that jacket?" Or, "Do I really want that job?"

Do I really want that girl? Do I really want that career? Do I really want that—" whatever. Instead of asking yourself rationally, look at what you're jealous of in life, who you're jealous of. That's a much better window.

- 4.15** Why? Well, emotions are much more authentic window into who you are. This is the dilemma I was trying to press before, with the pedophile. I mean, emotions are often—the limbic centers are often where our emotions are the true window into the self. The prefrontal is trying to inhibit that. So there's that mechanism. But again, it's also an integration hub, the prefrontal. So it's integrating higher-order thinking and values. So it's a complex issue. But look at who you are jealous of, and that's a pretty good indicator of what you really value in life.
- 4.16** Now, we came up with some thought experiments, so I want to share those with you and we'll see what they mean. If you have a guy—are you more envious of somebody who's slightly better off than you, or of Elon Musk who's quite better off than you—much better off than you, I would assume. I don't know, but probably most of us. Is a beggar on the street more envious of a beggar who's slightly richer, or of Elon Musk? What do you think? What's that?
- 4.17** The one that's closer.
- 4.18** The one that's closer. Why is that the case? I mean, it doesn't make sense when you think about it. If it's actually the case, and it turns out it actually is the case: I am more jealous of somebody who's slightly better off than me than of somebody who's vastly richer than I am. But why is that the case, though? Why is that the case?
- 4.19** It's reachable? Possible?
- 4.20** Getting closer. Exactly. So even though it's irrational; I should actually go for the guy who has more and should be more envious of him, because that's the ideal: having unlimited access to resources. Yet I envy, I'm more jealous of the guy who's just slightly better off. And the reason, we argue, is because the guy who's slightly better off, there's an actual value in me being jealous of him. Why? Because jealousy came about to motivate me into action, to push me, to have me do more and pursue resources within reach.
- 4.21** With Elon Musk, or Bill Gates, or whatever have you, with Warren Buffett, it doesn't make sense for me to be jealous. So this is a key argument that we make, and it comes in many flavors. Would you be more jealous of, say Joe—your neighbor Joe who's quite like you, dresses similar to you, in many respects he's just like you—or Joe, who's also like you, but lives in Timbuktu? Let's say they have a car that you desire; they both have it. Who are you more jealous of, the one in Timbuktu or your neighbor?
- 4.22** The neighbor.
- 4.23** The neighbor. But why? Because the guy in Timbuktu is too abstract for you. He's not within reach. He's not within the proximity of your social circle. So it doesn't really

make sense for you to be highly jealous of him. Whereas I can relate more to my neighbor. If he has the resources that I want to have, then if I do more like him, I could probably have those resources.

- 4.24 That's why when it comes to siblings, it's complicated because you have your brother. You love your brother, for example. You love him. You want him to succeed in life, yet because you know that you came from the same environment as him, you have the same 50% of genetics, the same environment, when you see him succeed, you think, "Oh my God, I could succeed too." So maybe you become a little bit jealous, but then you become embarrassed because you feel like, "I shouldn't be embarrassed, I shouldn't be jealous of my brother." It's stupid. Isn't that true? Isn't it a mixed feeling with siblings? So that's a possible explanation: you want them to succeed because they share your genes, if you want to be completely biological, but at the same time, they also came from the same background.
- 4.25 Another example would be your neighbor, who you're more jealous of. Let's say you're an immigrant. You're a Kurdish immigrant to America. It could be me coming here, I'm a Kurdish immigrant. My neighbors—let's say what we're all jealous of is of a woman. So my Kurdish neighbor has that woman that I would love to have, but then there's another neighbor—let's say he's from another race, Greenland. Let's say he's from Greenland—and then you have the American neighbor as well. So who would you be more jealous of? In which scenario would you be more jealous?
- 4.26 You would be jealous of the Kurd.
- 4.27 I'd be more jealous of the Kurdish one, right? Does that make sense? It makes sense. He's more similar to me. In the Greenland example, I would go, "I can't even make sense of it. It's so different from me that I can't really pinpoint, "Why should I be jealous?" It's too different; I can't really relate to it. The American example, I might just go, "It's because of his status. He's lived here for many years. He knows the language much better than me." So it's different. He has connections and all that. It's a different social status overall. But with the Kurdish guy, you can relate much more.
- 4.28 Somebody gets promoted over you— or this is a better example. I like this example more. You go into a place and you all have your eye on a female. A guy walks in—this is a scenario A—a guy walks in. He is stunningly handsome, rich, well-dressed. He walks away with a girl. This is scenario one. B: a slightly better looking, maybe even like you, in most respects—walks off with a girl. Or scenario C: let's say a poor homeless person comes in and he walks off with the girl.
- 4.29 Who would you be more jealous of? In which scenario? B. Again, why? He's most like me, right? What about A? What would I think? "It's well deserved; he deserved it. I can't really compete with him anyway." And in scenario C, I would go, "Oh, this girl is so idiosyncratic and so strange anyway that I can't do anything about it. There's no reason for me to actually—it doesn't motivate me into action, so it doesn't make sense

anyway." I'm using all these experiments, these thought experiments, to really show what jealousy is about. It's about motivating you into action. That's the key.

4.30 We actually use this. What we are trying to do in our work—we have this idea that we developed. It's called—if you gain more insight into what you're jealous of and understanding what jealousy is, and having more introspection, that allows you to then know what you value more and it has a therapeutic effect. So we've developed something along those lines.

4.31 Okay, this is the final one, I promise. I can see you're getting a bit bored with these thought experiments. We find them interesting. So what about this? Who are you more jealous of? This is for the guys, by the way. Let's say there's this guy. You scan his brain and you know for sure he's really, really happy. He's like you. He's in a state of happiness, bliss. Versus another guy who has access to all the women in the world, like Hugh Hefner, but more—he has access to all women in the world. Honestly, who are you more jealous of?

4.32 The guy who's happy? He's reached that level of happiness that can be reached. Who are you more jealous of? Honestly? Hefner? No? Happiness?

4.33 It's got to be happiness.

4.34 Yeah, I know you're saying that, but our research shows that actually, people, when you ask them secretly, they always go for the guy who has access to all the Playboy bunnies. And, actually, we're finding that over and over. This is actually the case. Well, I shouldn't say over and over, but at least according to our research. I think it's pretty clear. Obviously, your super ego and your prefrontal says something else, but we're talking jealousy, which is an emotion.

4.35 What is embarrassment about? Let's go over to embarrassment. It's a feeling that's short-lived. It's often about—it's feeling like you messed up socially. That's what embarrassment is. It's often very short-lived. You don't go around and dwell on it and go, "My God" for months. Often, after time, you go, "Oh, it's just—" You laugh at it, really. You tend to laugh at it. But it's a social signal, so we have a way of putting ourselves in place socially, so we don't do very bizarre and strange things in social scenarios. Right, AJ? We have a way of telling ourselves, "This is socially not correct." Or, "This is inappropriate."

4.36 There's actually a region of the brain again, the orbitofrontal cortex, which is critical for embarrassment. You have a damage to that part of the brain and that goes away. So if there's damage to the orbital, you know what embarrassment is, you can explain it conceptually—"I know in scenario A, this is embarrassment. This means embarrassment." But when it comes to an actual practical application of embarrassment, you just don't get embarrassed socially and so forth. You don't have that. It's interesting. The orbitofrontal cortex is that part of the brain where psychopaths have less activity and the OCD have a lot of activity.

- 4.37 This is a bit of a cheeky example, but it's in our paper: if there's a guy—should I tell this example? I'll tell it anyway—he goes into a store to buy an erotic DVD. We don't have that anymore. That was like 50 years ago, 20 years ago, 30 years ago, whatever. Now would he feel more embarrassed if the sales lady, the cashier, was: a young stunning lady, a lady that he's attracted to, an elderly lady, or a young man? In which case would he be more embarrassed? The young lady, right? And if that is the case, then who next?
- 4.38 The old lady.
- 4.39 And then next? And then the guy, right? But why? You all answered this correctly. But why? We have no clue; we have no scientific clue for this, but this is correct. And there must be a scientific explanation. Any guesses? Let's guess. Let's try to think.
- 4.40 I don't know. I was thinking about this. I was really thinking about this these recent days. I was like, "What could it be? Could it be that somehow I'm embarrassed in the young lady example, because I have my eye on her?" She could be a potential mate, and my prefrontal cortex would go, "I'm showing that I'm a disinhibited creature." We know that females like males with inhibition—at least some inhibition, not complete inhibition—but I'm showing that I'm disinhibited and lack discipline. So maybe that's it. That's what I'm thinking right now. Maybe that's the explanation. What do you think? Could that be? Possibly.
- 4.41 And for the older lady? I couldn't help but think, "Maybe it's kind of a mom figure." I feel embarrassed, again— basically, people, when it comes to sexuality, are very embarrassed generally. There is something there. It's an instinct. When it comes to their sexuality, they tend to— would you like to walk around naked in public? No. There's something about wanting to hide our private parts and all. There's something about that. I wonder whether the female taps in both: the female circuitry of a potential mate but then it's more like the mother. And so it still elicits the same emotions, but not as powerfully as the young lady. Could that be it? What do we think?
- 4.42 I mean, many of my theories and our scientific work, with Ramachandran, for example, actually came about through conversations and chats like this. Chatting and walking on the beach in Del Mar and having a chat in California, like, "Could this be the case?" "Oh, maybe this is true, yes. Can we think?" "Oh, no, this is a problem with this theory. This doesn't work." So it doesn't have to be super complex and then you do an experiment and then you might hit on the actual thing. A lot of these theories just came from that —conversations, having coffee, chatting about his life, my life, what kind of women we find attractive, just being human beings and having fun.
- 4.43 That's what science, I think, is about. Not this lab code like, "I'm going to lab, 9 to 5." You know what? I'll tell you. This is interesting. In my life, the scientists that were average scientists—and I started there. I was in an average lab, working—these were the 9 to 5, like, "I'm gonna do my next experiment. This and that, B and C, all the brain scanners—" And then one day, I was in California as well. I went into the laboratory of

my mentor, Ramachandran, and I asked. I just knocked on the door, he opened, and I said, "Can I be a part of your lab?" And I became a part of his laboratory.

- 4.44 I was completely surprised. The second day, he said, "Jalal—he calls me Jalal—he said, "Jalal, you want to go for a cookie?" I was like, "What? Who's this guy?" He was in "Time" magazine as one of the 100 most influential people in the world that year. Not bragging—I'm just saying he was my mentor, not me. He was a superstar in science, and here he is, asking me to go for a cookie and milk. Can you believe this?
- 4.45 Going there, having cookies, everything is so slow. No rush. Enjoying life. "Do you want to go to the beach, go for a walk, talk about science?" He talked about the whole history of science. He told me, "Jalal, you have to read widely. You have to read many books. You have to be—have fun." And then he told me—and this is going further, but he was like, "Baland, you are too—" always telling me this—"you're too scientific. You have to loosen up a bit. You have to see some girls, write poetry, and things like that." I was a little bit too tight; I wanted to do my science. He was different. He really changed me in many ways because of his way of thinking.
- 4.46 In my life, I really noticed that the greatest scientists had this soft feature, almost as if the left and the right hemisphere were a great blend. It wasn't just the math—the left [hemisphere], numbers—it was poetic. It was beauty. It was passion. Science for them was like this titillating love affair, this dance between knowing and unknowing. That's what science meant to them. It was passion. And this really rubbed off on me. I saw that everywhere. In Cambridge, the best scientists I met there—at Harvard, anywhere—they had that passion. They were in love, and they were in love with life as well. Anyway, this is going too far.
- 4.47 We were talking about what before that? Embarrassment. In embarrassment, it is really the orbitofrontal cortex, the frontal structures, involved in embarrassment. And I should tell you about the jealousy circuit, too. I want to let you know in jealousy, what circuits are involved. There's also the orbital to some degree, but a high-level circuit is involved that we call the basal ganglia. The basal ganglia are very cool. It's a very nice cranial gang there, the basal ganglia.
- 4.48 The basal ganglia are involved in things like movement and reward. It's that movement and reward circuit. Addicts, for example, have a lot of activity in the basal ganglia. If you have Parkinson's, for example, and your reward circuits are lacking, you become very inflexible. You can't move; you have trouble moving. You have a mask-like expression and so forth.
- 4.49 OCD: hyperactivity in the basal ganglia. Why do we have that, by the way, in OCD? Why do you have so much activity in the reward circuit? Am I being clear? I want to be clear. It's the circuit that has to do with reward and movement at the same time. Can you see the link to something like addiction? You want to reach for—compulsive reaching, this impulsivity for the drink. So it's behavior and reward, behavior and reward, behavior—

- 4.50 OCD—is that similar? We think of OCD as a behavioral addiction of some sorts, because just like drinking, whenever I clean my hand over and over and over and over, I'm getting some reward from that. Is that clear, what the circuitry is doing? The basal ganglia. Fancy names like the nucleus accumbens, the ventral striatum—you don't need to know all these names. It's part of that whole little gang. That's part of the circuit for jealousy; that's the jealousy circuit.
- 4.51 So obviously, you can imagine something called schadenfreude. Have you heard of schadenfreude? That's the feeling like— there was a point in my life, it wasn't going very good for me. The thing is, we human beings are flawed, aren't we? We are very flawed. We have humility. We have to have humility because otherwise life will hit you in the face very hard. But what if life is going so good that you lack the humility? It goes away a little bit sometimes. Isn't it true? It could be, right? We are human beings—we need to be humble. It's very important, not necessarily from a scientific perspective, although you can argue something there, too. But for humans, humility is key. Too much success—sometimes, it goes away a little bit. It can.
- 4.52 It happened to me for a little while—this is many years ago. Things were going a little bit too well, and I lost my humility for a minute, just a little bit. And I told myself, "No, you're very humble. You are extremely humble." It sounds like a paradox, right? But that's what I told myself. Was I really humble? Something was lacking, and I think, sometimes, you can judge humility better by what others are telling you, not what you're telling yourself.
- 4.53 Anyhow, then something hit me. I got ill. And I noticed some people were smiling a little bit when they were looking at me. No, really— they were smiling. There was a little bit of a schadenfreude. You can have that. The smiling soon, obviously, went away. But I'm saying schadenfreude is when something happens to you—I have jealousy of you, I don't maybe like you—and then something bad happens, and I feel good. It turns out the same circuits, the basal ganglia circuits, are also involved, because it's rewarding for me now to see you fail. So that's schadenfreude, the basal ganglia, and the same area of the brain.
- 4.54 I said that these emotions are mainly human. We have them in humans, in children not as much. What about animals? Do they have jealousy? You think so? I just said they were humans. What do you mean, Brance? Come on.
- 4.55 Can I give you an example?
- 4.56 Of course.
- 4.57 There was a lady who had like ten chimpanzees. It was one of their birthdays, and she gave that one chimpanzee, whose birthday it was, she gave it a cake. And then all the others saw it, but they didn't get the cake, so they ripped her apart.

- 4.58 Okay, interesting. Well, it turns out that for a very long time, we thought these were exclusively human emotions. Complex, higher emotions like jealousy. Right, Naomi? But it turns out that dogs—if you have a dog, and the dog watches its owner play with a different dog, it actually gets jealous. It gets really, really jealous. That study's been done over and over. It's pretty clear that dogs do show signs of jealousy.
- 4.59 In fact, even when the dog is here, and then there's a big fence, and it can hear the owner playing with another dog, just hearing it, it creates a theory of mind. They see their owner playing with this other dog, and they get very, very jealous. They show signs of jealousy. So it's actually there. Isn't that amazing?
- 4.60 There's so many reports as well, just anecdotal, in terms of reporting. People will say, "My dog showed signs of jealousy." So aside from experiments, there's a lot of reports of people, dog owners, talking about that. So we have to be careful when we talk about human emotions in that way. Obviously, they're much more developed in us, because we have a larger prefrontal cortex, but to say that they're exclusively human—it's probably not the case. I mean, certainly not the case in some instances.
- 4.61 Embarrassment? I don't know. I've been to the zoo and have seen some of the baboons and all that. It depends on the sophistication of the network. How well developed is the orbitofrontal cortex? You need a theory of mind for a lot of these social emotions. You need a way to gauge what other people are thinking. These networks are not as evolved in animals, but there is rudiments of them.
- 4.62 Why do we have jealousy? Could it also be to protect bonds? Is that a part of it? If you have a social bond you want to protect, do you have jealousy for that reason? What about love? It's a deep one. Seriously, love is—you fall in love, maybe for the first time, the first love, everything in the world is glowing. Everything in the world is poetic. Everything in the world feels different. First of all, why is that the case?
- 4.63 Novelty.
- 4.64 Because there's novelty, yes. And if you peek inside the brain, you see that the reward centers of the brain—we call them what?—the nucleus accumbens and all that. The reward pathways that have to do with addiction and all that. Basal ganglia, and then above it, the nucleus accumbens and all that. These structures become hyperactivated when you fall in love. Indeed, the whole world takes this poetic flavor. Everything feels good. We not only fall in love with the woman or the male, whatever the case might be; first, we fall in love with something deep within them. From there, we extrapolate that to the world so we fall in love with the world.
- 4.65 Pain becomes much more bearable. We can bear pain much better because when the nucleus accumbens and the reward centers in the brain become hyperactivated, they quiet down the pain centers of the brain. Things like the amygdala—we talked about the ACC, the insula—there's a direct cross inhibitory link there. Remember we talked about cross inhibition in the brain? Literally, when you are in love, the world feels less

painful, things feel less heavy, things feel less dark. You feel better. More positivity and less negativity overall.

- 4.66 In fact, it can become almost like a drug. Because of the sensitization of the pleasure centers in the brain, they become so hypersensitized that the slightest positive cue in the world makes you happy. It takes very little to make you happy at this point because you become so sensitized. It could be so powerful at this point, the actual love, the feeling of love towards your partner, that it mimics cocaine.
- 4.67 If you have somebody who is—if you scan the brain of somebody who is in love and somebody who's on cocaine, their brain is doing the same thing in many ways—the pleasure centers, when they light up. You show somebody cocaine in the scanner, the pleasure centers become active. You show somebody their beloved, and their pleasure centers become active. Then you remove the beloved, you remove them—what happens then? A withdrawal. You feel pain. The slightest pain in life becomes unbearable, intolerable, too much, you can't deal with it. That's the dark side of love.
- 4.68 As I was hinting at, mentioning, previously, there's a huge prefrontal component to love. Meaning, unlike, let's say, attraction, which could be very basic in the sense of, "I feel attracted, I feel innately attracted to somebody," there's a lot of psychological components to actual love. I fall in love with not only their physical appearance—that might be the first step, let's say. That gets me to the table, I feel some basic attraction.
- 4.69 Once I have the basic attraction—let's say, I'm speaking to this girl and she has depth, she has wisdom, she has intellect, she has something— I want to say, something indescribable, even. You don't have to put words on it. It's not about words; it's about feelings. It's how you feel inside. Does your stomach—your heart is beating, you don't know why it's beating. You want to know why it's beating, but you have no clue. That's what I'm talking about. You can't put words to it. No words can explain it. That's when we get into the love realm. You've had it. Don't hide it. I know. Anyway, so this is what love is about.
- 4.70 And then, interestingly, when you then fall in love, there's a lot of plasticity changes going on. A lot of plasticity. We have the two key components for plasticity, remember? Focused attention and dopamine. Do we have that, Heavenly? We do, right? We're hyper-focused on the person that we're falling in love with, and we have a reward whenever we see them. So you have those two components. So the brain really changes its shape, and it has to.
- 4.71 Think about this. Your entire life, you're egocentric. If, let's say, you are not in a relationship, your world revolves around you. It's about me. How do I meet my needs. How do I live for me? That's how you live. Of course, you have your family, but in the actual strict sense, you don't have a partner to check in with all the time. So your brain needs a lot of plasticity in order to reshape in order to incorporate another person

almost into your body image. Not in the literal sense, but in the more abstract sense of, "Now you're two people." So a lot of plasticity.

- 4.72** What can often help us with this plasticity is the hormone we talked about before: oxytocin. Oxytocin is the bonding hormone, the love hormone. We call it that in a frivolous manner. But obviously— let's say a woman has a child. There's a lot of oxytocin released. Or a man even, even a man, when he has a child, there's a lot of oxytocin in his brain too, because he has to connect with the child. Why is that important?
- 4.73** To carry out the species.
- 4.74** To carry out the species. Here's the link between having a child and falling in love and oxytocin. Here's the link: in both cases, you need to rewire who you are and your egotistical view of the world. You have to be more giving and more caring and more nurturing. So your own ego has to be less and you have to be more caring. In fact, neurochemically, serotonin—you might have more serotonin and less, maybe in some cases, dopamine, but it's getting too complex. The point is—oxytocin's role in the brain is to eliminate connections. This is the key thing.
- 4.75** Let's say you have a lot of new wires in the brain, not new wires. You have a lot of—generally, you have a lot of synaptic connections in the brain. Oxytocin enters the scenario and melts away a lot of these connections. When you're becoming a new father, it has that role so you can incorporate that child into your sense of self and be a caring father.
- 4.76** When you fall in love, the same thing: a lot of oxytocin. You are changing your brain in that way to make space for two people. So you fall in love, you have these plastic changes—a lot of oxytocin. You make changes for the new person. Then at the same time, what is interesting as well is when you then fall in love with that person, as I mentioned this earlier, her defects start becoming part of what you find attractive, too. And then that gets plastically rewired into your brain. And the next time you find a woman who has similar features as the one you just fell in love with, you are more likely to find that attractive. It becomes part of the whole "the world is glowing" and all that.
- 4.77** Often, what you can find is that at the end of a relationship—let's say that person, something happened, you had a breakup and it was horrible—now you feel a lot of pain for a long time. What is important at this point is that it can be very hard to fall in love again. You've associated pain with love. Love and pain now go hand in hand for you. How do we combat that? It's very hard. A lot of people have difficulties falling in love again.
- 4.78** If the partner was abusive—let's say the previous partner was an abusive partner— you may sometimes have the tendency to be attracted to people who are abusive. And then you see this caring person in front of you, they're very caring and very gentle and

all that, even though you are innately attracted to her, suddenly you find you're not attracted to her because she's not abusing you, emotionally. She's available, she's nurturing, she's caring, but you want the woman who's unavailable, inaccessible, and all over the place. Why? Because you have learned this pattern of behavior.

- 4.79 You see that. That's a common thing that people will have. They will have a learned mechanism where they associate high levels of dopamine and pleasure with love and relationships. They will always fall in that same trap, except if there is plasticity, and that's the good news, Heavenly, we have plasticity to change that circuit. But it can happen. You can fall in that trap, and you see that often that once you've learned a certain pattern, you get stuck in that.
- 4.80 So I think the key point is—and this is practical for us—if we are stuck in a pattern like that—let's say things were harsh in the previous relationship, things were hard, things were difficult, and you got hooked on this pattern, in a woman, by engaging with a new woman, by building trust with a new woman, by being involved with a new woman, you can actually erase some of these connections. The brain learns by—we have the principle I've mentioned before: neurons that fire together, wire together. So the more you can experience pleasure in the presence of the new woman, you will melt away some of these connections. You maybe get some oxytocin for the relationship and you learn a new pattern of love.
- 4.81 So often love, as much as it is about falling in love, it's also about falling out of love with another person. That's a key part of it. In fact, we are very sloppy in this respect. We often use the same circuits as before. The previous circuits that you used in a previous relationship, you might use it for the same relationship. Have you had that scenario where somebody calls their wife with their ex wife's name? Have you heard about this? The brain is sloppy; it wants to use the old connections and the old circuits, so falling in love is as much about falling out of love.
- 4.82 This is an important point to make in the whole plasticity discussion. There's two processes in the brain. There's two processes. One is called long-term potentiation. Neurons that fire together, wire together. I spend time with something, with somebody who's pleasant, I learn a new thing. Every time I see something on TV, I have a good feeling—I learn to associate that thing on TV, that soccer game, with feeling good. That's an association leading to new plasticity, long-term potentiation. It occurs when we have attention and reward from it.
- 4.83 But there's something else called long-term depression, LTD. It has nothing to do with actual depression, the psychological state of being depressed. It means that neurons are melting away, that the neurons that were here are now being cut. And that occurs whenever we— if two experiences, experiences A and B, are linked together and we are exposed to that often, the circuit becomes what? More solidified and crystallized. The less we use that circuit, the long-term depression occurs and the circuit becomes weaker and weaker until we use other circuits more. So the connection is less.

- 4.84 How does that work in a romantic relationship? Actually, this happened to myself—I don't like to use myself all the time. This is romance. I don't want to reveal all my—butlook, there's a girl you like. You like this girl; you're interested in this girl, but it doesn't work out. It just doesn't work out. So what do you do, Heavenly? What do you do? This is a heavy one.
- 4.85 What you can do—look, what you can actually do is instead of sitting home and thinking about it all day and reinforcing the circuitry—you're imagining all the interactions, or maybe all your ideas about your marriage and what your life could be like—instead of replaying that in your mind all the time, travel to a different location. Travel. Why could travel help, as an example? Why could travel help?
- 4.86 When you are traveling, you are overwhelmed by new stimuli. Everything is novel. So the brain is going, "I need a lot of dopamine and attention because I'm in a new spatial territory. I need to walk a lot probably as well, to learn the territory." So you walk. Try to walk as you're traveling, too. See new buildings. All the dopamine and acetylcholine rewire your brain.
- 4.87 At the same time, because the attentional focus is away from that girl, the circuits of that girl, the "girl circuits"—that connection is being less. It's pruned away; it becomes less and less strong. This new connection is much stronger. And your brain is engaging in this plastic process where you have, in this case, LTD, and then you have LTP, long-term potentiation in terms of this new one. Is that clear? All right. That's all I had for this lecture. Thank you.
- 4.88 You said if someone is thinking of a negative thing and they're just sitting at home thinking of it, the good thing that you can do is go somewhere new because it fires off these things and you stop thinking of the bad thing. Is it possible that somebody could become addicted to that, in a sense where they have to move every couple of years or else they get sad?
- 4.89 Well, every couple of years—it's hard to really pin down. I think that if there's a pattern of you falling in love with somebody and then you have to get away from that person and move, maybe you should see a therapist at that point.
- 4.90 This question wasn't about me.
- 4.91 No, I know. I'm not saying you at all; I'm saying—you're giving an example. So this point, don't worry. I think that is the solution: instead of moving, probably see a shrink at this point. I think that my point was not necessarily about moving, but maybe take a two-week vacation, like two weeks off, three weeks off. That's enough for plasticity and brain rewiring.
- 4.92 My point is this: you don't have to travel necessarily. You want to get away from your room, where you're sitting there and ruminating on this girl who destroyed your life. I think that's the point—you want to get out. If you love soccer, play soccer. If you have

friends you haven't seen in a long time that you love, go hang out with them. Go to new restaurants, new places and have fun.

- 4.93** It's hard to have fun when you're heartbroken—I know, trust me. But the point is, you do it anyway because there's an intention factor and a plasticity factor. You're getting into both. Your attention cannot go in hundred directions at the same time. And even though, once in a while you will have intrusive thoughts about the girl popping into your head, that's okay. You keep doing that, and then eventually, it's a way to process the pain better, let's say.
- 4.94** Stockholm syndrome—why in such a terrible situation do people sometimes develop feelings of connection, emotion, dependence, love for somebody who's committed that sort of crime or egregious act against them?
- 4.95** Stockholm syndrome, huh? There probably is literature on that. Let me think about it. I'm guessing it has to do with—it taps into something very primal, first of all, something very primal inside of us. I think a lot, in the past, when people were—tribes were obviously battling it out. So if you had tribe A battling with tribe B, they would steal all their women and children. This happened. So it might have adaptive mechanisms because they would remarry these women and they would—this was well known. So having that mechanism of being able to adjust completely, and then see the new captor as somebody you love, was important for survival. That's the only thing I can think of right now. Does that make sense?
- 4.96** I think that there must be a mechanism in the brain for that, because it was so rampant. In every society, they would either kill the women and children—what do you do with women and children? You can't—and then they kill all the men and then either kill them, too, or— I'm guessing. I'm not sure.
- 4.97** AJ, you had a question.
- 4.98** Firstly, you talked about a little bit with regards to a clinical work you're doing having people understand their jealousy and their motivation to help them in therapy. Can you describe that a little bit more?
- 4.99** 100%. Yeah. It's called—we call it existential realignment therapy. This was a therapy that I developed with Ramachandran. We never really got to publish it, so the paper is still hanging around on my computer. Unfortunately, at the time, Ramachandran's Parkinson's got worse. He has Parkinson's. We were about to send it off, but it never really—I couldn't have him look it over and approve the final draft. I showed it to a couple of people: Steven Pinker. Do you know Steven Pinker? Yeah, I showed it—do you know Steven Pinker? Not really?—I showed it to Steven Pinker and others, shared the idea. I think they liked it, but we never got to publish it because of that.
- 4.100** The idea is simply that existential realignment is about having knowledge of the biological roots of jealousy, embarrassment, anxiety, sadness, why we have—so you

choose, for example, the Hugh Hefner example because of an innate reason. But really, you should go for the happiness example. So by using your prefrontal cortex and thinking about it, it makes no sense for you to actually go with the desire. You should go with what actually makes sense. So it's trying to modify innate inclinations by using your thought. That's what that is about. In a nutshell, that's what we're doing.

4.101 This one's a little more philosophical, but related to the concept of degeneracy and neuroscience. I guess I should start here: is it the case that each person has a little bit of a different neural footprint? If somebody has anger, it might look different in each brain. Would it be the case that for a more advanced emotion, like jealousy or envy, that there would be more differentiation of neural circuitry?

4.102 Depression, anxiety—all those disorders have a lot of jealousy, for example, because there's a lot of hyperactivation of emotional circuits. You have a lot of jealousy, you have a lot of embarrassment, social anxiety. So, yeah, there's a variation in the circuitry. Yeah. Thank you.

Lecture 5: The Believing Brain

At a glance — The believing brain — belief, meaning, and cognition.

- 5.1** I was lying there in the room. I was trying to meditate, thinking, and something occurred to me. We talked about attraction and the fact that females tend to find talents appealing, they find talents in men appealing. Some have argued that, for example, painters and artists are appealing because it shows eye-hand coordination, that they have these technical skills. So they might use it as a proxy. Maybe it's a proxy for good genes. Hard to fake talents like that is something that is a good surrogate, a good proxy, a good window into one's genetics.
- 5.2** So I was thinking about that, but then I thought, "Hold on." I remember in high school, there was this girl. She was singing, and I was in the audience. She was, may I say, an attractive woman, but she had a very good voice, extremely good voice. I mean, warm, soothing, Whitney Houston-type voice. And immediately, I felt something a little bit, just being honest— I felt something, I felt, "She's more than just a pretty girl. There's something to her."
- 5.3** And then I'm trying to put the puzzle together. Why did I feel something? What was the reason? Why would a singing voice that's very appealing, very soothing, very— why would that cause me to be attracted to her even more? Do we have any thoughts? Has science tried to answer that? I don't think there's any literature on singing voices in females. But there's something to it.
- 5.4** I'm not sure if it goes the same way for females from male singing voices— I don't know how that feels. But thinking about it, I think, perhaps, maybe it's nurturance— nurture, caring. Maybe your mother's singing voice—as a kid, your mother singing, and maybe that leaks over. Is that it? I'm not sure, but there's something there.
- 5.5** It could also be—there was this neuroscientist, John Smith, he's passed away now. He was at UC, San Diego. He had tons of books. He told me that in the brain, the way your auditory system is hooked up to the emotional structures, there are few synapses. So from auditory information to actual emotional centers, there are not many synapses— I'll not explain what that means in a second.
- 5.6** Between neurons, you have synapses, gaps. Generally, if structure A is to influence structure B, the fewer synapses in between these structures, the more impact. And usually, if two structures in the brain are communicating a lot, they're very close to each other, even in terms of proximity in the brain. So from the auditory centers to the emotional core, there are few synapses. Versus, for example, vision to the emotional core. There's an even longer journey. So he told me this is the reason why music evokes powerful emotions versus looking at a painting.

- 5.7** You ask yourself: what evokes more emotions? What is more evocative emotionally? Looking at the Sistine Chapel painting, Michelangelo, or hearing some beautiful classical music, or any music you like? What evokes more emotion? What is more emotionally powerful, intense? Music, right? Yes. That's my observation, too. It could have to do with the number of synapses and how they exert their influence, and the fewer synapses, the better.
- 5.8** Take smell, for example. Smell travels really quickly from the olfactory bulb to the emotional core. In fact, it bypasses a region of the brain called the thalamus, which is a relay stop station, a little stop station where—we talked about this in my previous course on sleep—it's the gateway, a doorman for sensory information in the brain. Every sensory information has to pass through the sensory gateway called the thalamus in the brain.
- 5.9** But smelling, the sense of smell, completely bypasses the thalamus and goes straight from the olfactory [system] to the emotional core. Why is that important? Why would the brain say, "Look, I'm going to skip a step. I want an intense experience of smell as quickly as possible." Why might that be important for survival? Always think caveman, survival. Is smelling very important for survival when you think about it? Smelling rotten foods, being able to smell something that's toxic, poisonous—that's the explanation for that. The fact is that having a strong sense of smell is very important for survival.
- 5.10** And then you say, "How is this related to human nature?" This is Larry's question: how is this related? Well, think about the perfume industry. Think about how powerful—have you ever walked past—you're walking on, let's say, some street in Scottsdale. What's a famous street here? Champs-Élysées? No streets? You're walking there and then somebody passes by you—this girl, you smell her perfume, and then all your emotions of some person you knew in the past come to the [surface] and you become very emotional.
- 5.11** My wife.
- 5.12** It's your wife.
- 5.13** I have to say that at home.
- 5.14** You get what I'm saying, right? What is that about? That is about—the emotional part of the brain and smelling are very hyperlinked up. Then, of course, the memory part of the brain is a neighbor to that as well. The memory region and hippocampus are neighbors, and so there's also a lot of influence. But that explains why perfumes, for example, are so powerful, the whole perfume industry. Does that make sense? What do you think, Naomi? It makes sense. Good.
- 5.15** In the 1930s, an interesting phenomenon in science was discovered. It was quite something. I'll give you a case to explain it. This lady walked into—she went to the

doctor, she was pregnant. Her belly was full, like this. She was anxiously excited for the child, and she wanted to check up on the child. Back then, they didn't really do that many checkups. And so she wanted to be sure before the child was born. The doctor was comfortable; he had done this a thousand times. He checked the stomach. Looked at Mary's—her name was Mary—looked at her stomach, and something was off.

- 5.16** He looked at the stethoscope, and there was no child. There was no kicking. He checked again, and a third time—nothing. There was no child inside. At this point, he was convinced that this was a case of false pregnancy. And so what he proceeded to do was the following. He said, told the girl, Mary, "I'll give you this anesthetic. You'll sleep, and we'll have you deliver the child under the anesthetics." He gave her the pill. She woke up, and he told her, he said, "Look, you had a miscarriage. There was no actual—the baby died." Immediately, as he said that, the stomach flattened. She went home sad, told her family about the miscarriage. She was quite sad.
- 5.17** One week later, she came running back. She came back to the doctor with her stomach full again. She said, "Doctor, you forgot to deliver the twin; there's a twin in there. Get it out. It's kicking. It wants to go out of here." So this was a famous case of a false pregnancy. This is called pseudocyesis. It's quite an interesting phenomenon. It shows you how mind and brain interact. It's a type of false belief that can change your hormones.
- 5.18** So what is going on? That's a question we want to ask. We want to try to solve: why does this phenomenon occur? Why does it happen? One structure that seems to be involved is the hypothalamus, the hormone factory of the brain—we've talked about that several times, about that structure of the brain. That seems to be involved. When the person thinks, believes, that they're pregnant, that structure will release hormones that are conducive—basically, that would signal to the body that she's pregnant, so she would have prolactin. A downstream of that is estrogen and other hormones signaling to the body that she's actually pregnant.
- 5.19** What you will find in these women is that they often have depression and anxiety. They are longing for the child. They really want to have children. There's a lot of societal pressure for them, in fact, to have kids. And so they have depression and anxiety at the same time. This will change some of their hormones—norepinephrine, noradrenaline, is down in depression, as well as dopamine. So these two chemicals, they do not have. They are very low. They have them, but they are very low in people who are depressed. That can stop menstruation. And so for that reason, they have another indicator that they could be pregnant, even though they're not actually pregnant.
- 5.20** Larry, why do you think they have the stomach? That's the mystery, right? Why would they have the stomach? We don't actually know for sure. Some people will say, "It's because they have intestinal gasses from stress and anxiety, and then somehow they become bloated." And also, they probably try to have a stomach like that, too. And so they push it out a little bit. So it almost becomes a self-fulfilling prophecy.

- 5.21** But you know what's more interesting? Some dudes have this phenomenon, too. Soguys can have it, too. Have you seen the movie "Junior" with Arnold Schwarzenegger? He's pregnant. Have you seen that movie? Males can have it, too. So how do we explain this in males? I'm not sure, but similar to some hormonal influences. You can get this. It used to be very common, like in the 1940s, 100 years ago, 200 years ago, it was very common. In rural areas, today even, it's still very common—a lot of people have this.
- 5.22** 100 years ago, it was maybe one in 250 deliveries. Today, it's quite rare. You can guess why: we have ultrasound scans, we have all kinds of tests we can do for pregnancy, so you can spot it early on. You can't really cheat in that way. The placebo is very powerful. The placebo is— it literally means, "I shall please" in Latin. It means "I shall please," and it is the sense of your physiology changing when you expect something good to happen.
- 5.23** For example, if I give you a sugar pill and I tell you that this is going to make you stronger. AJ takes a sugar pill—he'll literally have physiological reactions. It can affect heart rate, blood pressure, all these various physiological measures. So it's not merely psychological. The stronger the placebo, the better. So if I give you five pills and say, "You'll be better from this," that's better than just giving you one pill.. That's very clear. And overall, the more dramatic it is, the better. So if I have some elaborate thing around the whole process—iff I give you a placebo and there's a lot of technicians standing around,, machines, and things and say, "You will be better from doing a certain thing. Take this medication."" There are scanners and stuff—that's better than simply taking a small drug. That's part of it.
- 5.24** Placebo, depending on the color of a placebo—so a blue-colored pill is more sedative than, say, a pink-colored pill. So the colors can influence. And even placebo surgery works. You could have a placebo surgery. This is quite profound. This is one of my favorite studies. They had a study, and they injected stem cells into the brains of people with Parkinson's, injected stem cells into the brain of [people with] Parkinson's. You put it in the basal ganglia, the region where we have movement and all that.
- 5.25** If you inject that into one group of patients and tell them you didn't really inject it, you actually inject stem cells into the brain and say you didn't do it, and then you have another group, you just put in the needle, but you don't inject any stem cells— guess what? The ones that had no stem cell injections do better than the one who actually had stem cell injection. Can you believe this? It makes no sense. Parkinson's—it's quite something.
- 5.26** But there's a lot of cases of placebo. The placebo has a lot of variations; it works in many ways. It's particularly strong for pain. For example, if you are in a hospital and you have windows to trees—there are trees, you can watch trees—it turns out you need less pain medication. Can you believe this? Or, if you have something you can press [to get] medication instead of the nurse annoyingly coming every half an hour

and saying, "Do you want medication?", you can control it, you have a sense of control — when you have that sense of control, you need less pain medications. That's quite something.

5.27 It seems to be very powerful when it comes to pain. The placebo is very powerful when it comes to pain. There's something there—it has to do with how the placebo works, the structures involved in the brain when it comes to the placebo. One structure that seems to be involved is obviously the prefrontal cortex; it's beliefs, it's perception, it's thinking something might work. So there's a lot of that in the placebo.

5.28 How do we know that? That's a fair question that AJ might ask: how do we know the prefrontal is involved? We can infer that from the fact that it's something to do with belief and cognition. But even in a condition like dementia, Alzheimer's—when these structures, the sides of the brain, the prefrontal, the dorsolateral prefrontal, these sides, and the medial of the prefrontal, pretty much the whole prefrontal—are degrading, eroding, you don't have a lot of placebo in that group, in that condition, in these patients.

5.29 In other words, you give them pills, you give them medicine, and you say, "This will help you," and they have pretty much no effect because of the lack of the frontal structures. So it shows you the frontal structures are critical for the placebo. In fact, If we were to use TMS—this [transcranial] magnetic stimulation device I keep talking about—if you put that on the brain and inhibit the structures up here, and then you give some placebo to them, you say, "Take this medication," you'll find that they won't have the placebo effect because you toned down the activity of the prefrontal lobes. So it shows you how important the prefrontal lobes are for placebos.

5.30 Now, the prefrontal is not the entire story. There's also the prefrontal and the hypothalamus, again, the hormone factory of the brain. The prefrontal communicates with the hormone factory of the brain—sending hormones, changing the hormonal environment in your body, and then conveying that information further to the lower part of the brain that has to do with even more bodily states and respiration and heart rate. So you can change all that from the higher level, then going to things like hormones, and then going down to things like respiration and heart rate and more basic human functions. So that's the circuit. It's a very simple circuit in some sense.

5.31 But here's something. Did you know that the placebo even works when you know it's a placebo? So if I give you a drug and say, "Look, buddy, look, Cameron, this drug right here is a placebo. Take it and you will feel better." Studies have been done, all types of situations, you know it's a placebo and it works. How do we explain that? What do you think? Come on, Sam, you know it.

5.32 Why would you—I told you it's about beliefs, perceptions, convictions, knowing that this drug will help you. Then I'm saying you can just take a pill, and I tell you that this pill is a placebo. You take it and it still works.

- 5.33** If a placebo has an effect, is it just a statistical difference or is it actually 10 or 15% cure rate? I mean, is it 1%?
- 5.34** No, it's a big percentage; it's a very large percentage. Often, there are studies up to almost 50% of the [effect of] the drug in some cases. It works for all kinds of conditions. But for pain, especially, it works a lot, because pain works with the opioid system, and pain can be psychologically affected in many ways. There's a huge psychological component to pain, Naomi. For example, the endorphin system can quell pain in a very rapid manner. That's why, for example, if there's a zebra running across the Savannah, its stomach is ripped open from the bite of the tiger, it's bleeding, but it feels no pain temporarily because it's conducive for survival to have a mechanism to quell pain temporarily. They have that.
- 5.35** We human beings also have that. We can tap into that system. It seems that things like—what do you call the thing when you put the needles into the body?—acupuncture. There's a lot of debate around acupuncture. Studies have shown that it may work through the opioid system and it can actually equal pain. If you block opioid receptors in the brain, the effects of acupuncture go away. There's something there about pain. So pain is highly susceptible to the placebo.
- 5.36** Parkinson's disease is highly susceptible to the placebo as well, because Parkinson's disease involves a lot of reward circuitry. Remember the reward circuitry in the brain? It's the nucleus accumbens—this area of the brain that has to do with reward—ventral striatum, all these fancy names for structures that mediate sense of reward. The placebo has a lot to do with reward because higher-order beliefs, the prefrontal cortex, can influence reward a lot. Think about it. When we feel good about something psychologically, it affects our reward centers in a direct way. So Parkinson's is very heavily influenced by placebo. Depression—very much so. It's one of those conditions that you can affect in many ways by using the placebo.
- 5.37** It's all about the look and the feel of the whole thing. The more theatrical, the more—the larger the spectacle of it, the more you feel the placebo. So, for example, there was one study, it was very cute. They had a chap, he was wired up for sleeping in a sleep lab; they wired him up. He had all these measures on his face, his facial muscles, EEGs and all this—you can measure sleep and sleep quality. And then, probably there was a big screen where you saw all the numbers and graphs. They told him he had more sleep than he actually did, and he performed better cognitively on all kinds of measures just by sharing that. It's incredible. That's something that's profound.
- 5.38** Or exercise, for example—it turns out that, you have these stationery bikes, they did a study. In one condition, they had people—they had a clock on the stationery bike run 10% slower. In another condition: 10% faster. And in another condition it was the same as the clock is supposed to be. In the condition where it was slower, people actually had more perseverance. They could go for much longer. It was up to 20% more perseverance just by reducing the clock and how fast it ran. It's pretty amazing. Think

about the practical applications in the whole sports industry by using some of these techniques.

- 5.39** It reminds me of when I joined the lab initially in California with Ramachandran, one of the first things he does—he means well, he means very well. He has an experiment called the size-weight illusion. It's a very interesting illusion. If there's this thing, let's say— it's usually, it could be a suitcase, for example. You have a small suitcase like this and you have a big suitcase like this. They both weigh, let's say, 5 kilos or whatever, 10 pounds. Guess which one feels heavier, the smaller or the bigger when you grab it? It's an actual real effect. You should feel it. When you grab it, literally, one of them feels much heavier than the other, even though they weigh the same.
- 5.40** You say the small one? You agree it's the small one? The big one? The big? I would think so, too. I would think the big one. But Brance is right—the small one. The brain computes there's more density, therefore it is heavier. Now, you have to grab them at the same time. This is called the size-weight illusion. It's a fascinating illusion.
- 5.41** Now, he had a hunch back then: if you have these and you exercise with them, let's say you run down these stairs, the smaller should affect your physiology much more than the bigger because it feels heavier. It would have a mind-body effect like that. Everybody who joined the lab, he would have them do this experiment. Always. He really wanted to have these findings, but there was null findings each time. We didn't get an effect.
- 5.42** I tried this; I worked on this for five months. For five months, I was up and down the stairs of the UC San Diego, having people run up and down with all kinds of things to measure, and nothing there. But it would have been interesting if it was actually the case that your physiology changed. Because then, when you lift weights, maybe it might be better to use heavier, smaller weights instead of bigger ones, because that might have a more psychological effect. So potential implications would have been big, but there was no finding there. There was no finding.
- 5.43** It reminds me of another study, by the way. There was this study done where they had people exercising, lifting something that was strenuous, like a weight. And somebody was actually imagining doing that for, I think, about a month. And the results were—I think the original one had about a 30% increase in muscle strength. By merely imagining doing it, he had about 22% or something like that. It was quite profound, just by imagining doing this strenuous exercise and some chap shouting, "Come on, you can do it."
- 5.44** I wanted to say—there was another study on cognition and creativity. They would give people smells, odors. They would smell a certain odor, they would smell it, and then they were told that this smell would make them creative. They'd smell it. The other group—smell it. But the other group was not given this information. They were basically told, "This is a natural smell. There's nothing hocus-pocus going on." You

literally find that the people who got this smell became much more creative simply from the smell. They had an objective test of creativity, and by merely telling them, "Look, you will become more creative by taking this, by smelling this," they became more creative.

- 5.45** I can go on and on and on about various placebo effect experiments. I think we've covered some of them. I think thinking about the nocebo, the placebo's evil twin, might be more interesting a little bit. So let's shift gears. The nocebo's meaning in Latin is, "I shall harm." When you think about it, the nocebo, you want to be aware of it, because studies show that the nocebo is as powerful as the placebo. So by reading the— on the drug side effects, it says, "It has headaches, it has this and that." It's pretty horrible because you might get some of those by simply reading it.
- 5.46** One finding was that during the COVID virus, people would get all kinds of symptoms: fatigue—what was it?—fatigue, smell issues, headaches, and all kinds of symptoms. But when they gave people a vaccine that wasn't actually the Corona, people would report some of the same things, massively, in massive numbers, really showing how the nocebo can work.
- 5.47** In one study, they had a chap. They put pain medication on his hand. It was a procedure. And then they told him, they said, "In a few minutes, this pain medication will wear off, and you will feel intense pain." That's exactly what you found. Even though the medication was lasting for hours and hours. But the moment they said that pain will come back with a vengeance, that actually happened. So it shows you that the nocebo can even change the actual physiological effects of real medication in this case.
- 5.48** We talked about a few mechanisms—we talked about the prefrontal lobes, the hypothalamus, the hormone factory of the brain, and that whole circuit. Then we talked about the opioid circuit and the quelling of pain. These are the main circuits involved. But there seems to be another factor in pain—excuse me, in the placebo and nocebo—which is something we call conditioning.
- 5.49** Let me illustrate this. For example, it was observed for a very long time that people with asthma— you give them a flower, they might have symptoms. But even if you give them a fake flower, they might have the symptoms as well. Or, if a human being, if you give them a drug and you pair it—each time you give the drug, let's say it's an immunosuppressant— and this study has been done—you give them an immune suppressant, it causes all kinds side effects like fatigue and dizziness and all that.
- 5.50** You give it, and then, at the same time, as you give that, you give them fruit juice. You pair that, let's say, 10 times in a day. Tomorrow, you come back and you give the fruit juice only. You have the patient become dizzy just from the fruit juice. They were completely fine the whole day, but they become dizzy from the fruit juice. If you measure their body and their immune system, you have biological blood markers of

immune function going down just from pairing the juice and the medication. Is that clear? There's a lot of conditioning.

- 5.51** Have you ever walked into a hospital, Larry, and you may feel a little bit sick from just going in? Have you had that before? Yeah, it's like that. There's a lot of conditioning going on with the placebo and the nocebo. A lot of that. Sometimes, the placebo and the nocebo can affect us in a shared manner. So it doesn't affect Sam only, but everybody around Sam.
- 5.52** In the 1500s, there was this lady. Suddenly, she took to the streets and started to dance in the streets. It was one hour, two hours, the whole day, she was dancing, dancing, dancing. A day went by, she kept dancing. In about a week, people joined her: 20 people, 30 people. Two months went by and 400 people had joined her in this dance. This was in Strasbourg, in Germany. People literally died from this. People would literally die from exhaustion and just dancing in the streets. Can you believe this, Emily?
- 5.53** And then, of course, in order to stop things, they would bring in musicians and they said, "We're going to try to exhaust these guys." So they brought in musicians, and in some cases, they brought in exorcists and all kinds of things—but people would actually die from this dancing plague. It was called the dancing plague. It started in June, July, and by September, it subsided. We still have no clue how it happened, but it's some kind of psychogenic condition where people have a strong belief, and they live that belief out.
- 5.54** Now, there was a lot of stress going on at the time. There was all kinds of sexually transmitted diseases. There was smallpox, probably, I believe. There was a lot of famine. One argument is that stress and things like that really made people go to the streets and dance in this way. Could that explain it? What do you think? Could be a factor, right? It's an interesting one. We don't know. It's a historical thing, in the 1500s. It's a very long time ago, but bizarre. People were just dancing like that.
- 5.55** I think more interestingly is the fact that there was a belief about a saint, Saint Vitus, I think she was called, that could affect people and send curses down onto them so that they would dance uncontrollably, like a disease or illness. So the question is, could that explain it? I think so. What do you think, Larry? Could it be? I think. I really think that's the explanation. I mean, I think really we human beings are affected by beliefs like that. And I know this. I know that this is the fact from my own research in sleep paralysis, the fact that beliefs, higher-order ideas about negative things happening to you, can affect you. And then obviously, if you have a lot of stress, a lot of anxiety, things like that, that would probably push you more towards accepting those negative beliefs.
- 5.56** There was an interesting case. It's a very interesting case, though. I find this very interesting. There was this guy. He rushed to the hospital. Rushing to the hospital, thought he was dying. Literally, he got into the hospital, went inside, felt like in a movie, fell on his knees. He told the receptionist at the hospital, he said, "I think I'm

dying, doctor. Get some doctors, have people look at me." And sure enough, they took him and put him on the hospital table and checked him and all that, and his vitals, and they found that actually this chap is about to die. He had fatal low blood pressure, hyperventilating, heart is beating. Everything was just—he was looking like he was going to die.

- 5.57** And in that whole death situation, they tried to get it out of him. "Why are you—? What's happening? What have you done?" He said, "I took some pills. I took some pills. I took a whole bottle of pills. And I think that was it. I think that's killing me. I think it was painkillers, like a whole bottle of painkillers." Now, it's interesting. When they checked the chap, he had no signs of the medication inside his body. No signs of it. And so they told him, "Look, you don't have any medication inside your body. What's going on?"
- 5.58** And what they figured out was, he had been a part of a study. He had depression, so he'd been a part of a study. And in that study, there was two groups. One group got the painkillers, the other got the placebo. So he had taken all the placebo pills. And so he thought he was dying from that. He tried to commit suicide. So he thought he was dying from the placebo pill, and now he was at the hospital actually nearly dying. But once they told him that, "Look, buddy, this is actually not medication. This is sugar pills." He was okay. Again, showing us how mind and body interact.
- 5.59** But I think the most interesting and fascinating case of the placebo, or rather the nocebo in this case, is these refugees coming from Southeast Asia, coming to America. They were lying in their beds, sleeping at night, and they would see ghosts while sleeping. They would have ghosts visit them, often ancestors, of their family, ancestors would come as ghosts, and they would actually die from this. They would actually die. It's a condition called Sudden Unexpected Nocturnal Death Syndrome, SUNDS. They would use all these fancy medical terms to explain the fact that people were dying in their sleep, actually dying.
- 5.60** They would often, those who survived would describe paralysis. They were lying there, couldn't move, couldn't move their bodies. Their eyes were open, they couldn't speak, and the ghost would come and kill them. So this seems like sleep paralysis to me. And this was the finding that really made us believe that something like sleep paralysis, for example, could kill you. That if your belief is strong enough, if you are convinced that the monster that comes and attacks you can actually kill you, not only can you develop PTSD and trauma—as we talked about, that you can potentially develop that in a previous course—but it seems like even death might be an option. What do you think? Is that possible? We don't know for sure, but it seems like it could be.
- 5.61** Now, some people have talked about that, no, this is really—these people might have had a genetic heart condition already to begin with, and that explained the people dying in that way. Other people might have a more robust cardiovascular system, so

they wouldn't actually die from something like sleep paralysis. But I'm not sure I'm convinced. My colleagues and I, over in Boston, I'm not sure we're convinced.

- 5.62 I think potentially sleep paralysis—see, this is a funny thing. Should I say that or would that affect people more? So maybe I shouldn't say that. Oh, let's— I'm just going to be honest. Maybe. Who knows, right? I think believing it couldn't kill is probably much better, but perhaps in some cases it could, right?
- 5.63 We don't know for sure, but one thing is clear, the placebo is becoming much more prevalent in society, much more. So 100 years ago, you didn't have as much placebo in medicine at all. Why? Well, we talk about it more. So the placebo has become its own placebo. Just by knowing about the placebo and how dramatic it is, you become affected by it. So you have this rise in the placebo overall in society. I think I've given my state of business. Let's move over to questions and answers for now. Thank you.
- 5.64 You've talked a lot about how beliefs, either with the placebo— they can optimize our mood and performance or how they can hinder that. I'm really interested in the topic of heartbreak and getting over somebody that you love. I know that people heal at different speeds. I'm curious from a neuroscience perspective—you talked about travel and how you can rewire your brain a little bit from traveling—but what do you think is happening with people who heal a lot more slowly? Do you have tips on how to heal heartbreak?
- 5.65 It's a good question. It's very good. Thanks for sharing, for asking those good questions. I think doing the things I said is a good idea, like doing different activities, going out, meeting new people, traveling. But again, look, some people just take years and years to get over somebody they loved. So you find that they cannot let go, and they fool themselves by traveling. So they might travel and they go somewhere, and they're there for years, and they still are in love with that girl or boy, whatever.
- 5.66 How do you heal from that? I think the best way that I can think of is not clinging on to the belief that—clinging on to the person, their items, if there's any items, shared items. Sometimes when you are in love and there's a breakup, you might go visit the places you went to together, might have shared memories. You want to avoid that completely, especially if you want to move on. If you want to get back to the person, it's a different matter. If you want to move on, it's about everything that can ignite those circuits in your brain.
- 5.67 Then we say, still that might not work. Years go by, you're still in love. I think the best cure is another partner. That's the best way to take that circuitry and put it onto another person. And then your brain can learn that, "Oh, what I felt back then—" Because falling in love is very simple. It's an either or process. You're not half in love. You're not like, "I'm half in love." Either you're in love or you're not. And I think the best way that I have—and I'm not even sure about the research on this, but this is my own experience and my own knowledge on the topic—I think that is probably the best thing.

- 5.68 So first step would be: remove yourself from the situation. Items, don't cling on to shared memories. And then in the beginning stages: travel and get out, meet new people. And then when you can, try to find somebody else. These are the things I would suggest. I mean, it's very hard because sometimes you just never get over the person.
- 5.69 Yeah, and how can you date somebody that's [inaudible]?
- 5.70 If you're in love, right? Obviously, you have to give it time initially. So if it was like a four-year relationship, you're not going to date somebody the next month. You want to give it time. But eventually, the person needs to get out and find somebody else. I do really think that the widows and the people that don't ever go out and meet new people, they get stuck with that initial love until they die. I think so. I don't see any other medicine than finding a new one. That's really what it is. That's my—what does your intuition—does that sound quite right a little bit? Maybe? AJ, you're smiling. What do you think? Maybe.
- 5.71 I would like you to speak to how music or even smells could be used for performance enhancement or training ourselves? How can we use the power of smell and music to improve our lives?
- 5.72 It's a good question. In the placebo sense, I think—about smells, I'm not sure, other than if they told you, "Look, this smell would make you smarter," for example. But you would have somebody else—so you have to fool somebody else to fool you. You see? There has to be a scheme around it where—you see what I'm saying?
- 5.73 Well, I have a hypothesis on that. I love coffee, and I associate good times with coffee and more of a meditative state. And so I wonder if I benefit from that. I'm not very musical, but I was fascinated when you were talking about music.
- 5.74 Yeah, I think music, what it does, it can really activate the dopamine centers in the brain and pleasure centers. So it makes you overall more alive and lively; you become more alive. You see, literally, you have people with depression with very little dopamine activation, and you find that they are not very responsive to art and poetry and anything that has to do with pleasure. So literally, you can actually use things like music or anything that would activate your dopamine centers to become more aroused and more emotionally attuned with the world. And that could enhance creativity, I think. Sure. Yeah.
- 5.75 Actually, before we go, Sam, I just wanted to say, you know what I think is very useful? Have you seen the movie, what is it called? *Dangerous Minds*? You've seen it, right? You haven't seen it. It's very interesting. So there's this professor, this lecturer or something, a teacher. There's all these kids in the ghetto, and they come into the class. And the teacher, Michelle Pfeiffer, I think, whatever her name is, she starts out by giving everybody an A. She says, "Everybody has an A." And the goal of this class is to keep that A up, right? It's just to keep that grade up, to keep the A. And there's something confidence-boosting about that. And so a lot of them actually do that.

- 5.76 And something similar happened in my life, actually. I remember I was in a school, I didn't perform very well. They transferred me to another school. I had that happen a couple of times in my life, unfortunately. But they had to transfer me to another school because I was performing so poorly. But then some of my very bad grades, when they were transferred to the other school, they appeared as A's. Like A's.
- 5.77 Literally, I got A's on my thing, and people said, "Oh, you're an A student?" And I literally performed much better for a very long time because of that. So I think small tricks like that, the system can use tricks like that and just encouraging people and telling them, "Look, you'll be more creative if you read art, and if you look at art and you read poetry." And you actually will become more creative and more—there's tons of way to boost creativity in that way, Larry. Sure. Go ahead, Sam.
- 5.78 Topic of placebo. Personal observation in my own life was similar to the story you just shared. I was a poor student, practically until the end of my high school experience. Really it was a belief shift that I was aware of that once I understood the reasoning and the bigger picture connection of what I was working towards, then all of the poor habits, attitude change, motivation all shifted. Since then, my genuine approach to wanting to learn, and discover, and train. It's all much different in my academic experience. I just wanted to share that it had a significant impact on me.
- 5.79 But my question was about a person that I know. They are drawing a correlation that every time that they watch a scary movie, usually around ghosts, spirits, they are having—every time that they've watched it that same night, they then have a sleep paralysis experience. So I just maybe could get your take on what's going on.
- 5.80 We covered this at length. Who can answer this? So I think what it is, it's a form of priming. So priming can affect a lot. So your friend watches this horror movie and she gets terrified, activates the fear centers of the brain, and then that will cause her to tip over to the sleep paralysis experience. So it's like an expectation of something bad. And it's sort of the nocebo realm of things and how that can affect. But that's definitely very interesting. But it has to do with expectation, and specifically fear, which can tip you into sleep paralysis. That's very interesting. Thank you.

Lecture 6: Beauty and the Brain

At a glance — Beauty and the brain — the neuroscience of aesthetics and art.

- 6.1** It'll be fun today. We'll talk about art. Things like why we find paintings beautiful, why we find certain outfits beautiful, what's appealing about them, why certain faces are attractive. We'll talk a little bit about metaphor as well. And so different various topics.
- 6.2** Let's start with symmetry. Human beings have an innate preference, an inclination, towards beautiful—excuse me, symmetrical objects. We find them beautiful. That's my point. Look at the Eiffel Tower, the Taj Mahal—perfectly symmetrical. So the question is: why do we find symmetrical objects beautiful? What's so beautiful about them? Why do we find symmetrical faces more beautiful than non-symmetrical faces? It turns out babies will automatically recognize a symmetrical face and have faster reaction time to it as if it's seeing it as more beautiful. So there's something about symmetry, but what?
- 6.3** What about symmetry? What is there about symmetry that's appealing, AJ? What do you think? It turns out everything that's living in nature— all the biological objects we see in nature, like lions, gazelles, humans—they are all symmetrical. We have a hierarchy of preference when we look at nature. Given the limited attentional capabilities of the brain, we have a system to say, "Okay, I'm going to have a preference for symmetrical objects because it's important." It means, "Chase it— it's a gazelle." "It's a lion—run." "It's a mate—pursue it," and so forth. We have this natural inclination towards symmetrical objects.
- 6.4** By the way, when it comes to faces, when you look at a face, why is symmetry important for faces? Aside from the fact that we see symmetry as objectively more beautiful, what is so appealing about symmetry in faces? One of the factors is that when you have a parasitic infection, so a child or an infant has a parasitic infection, the face will have slight deformities, and the symmetry will be slightly shifted. And so it could be a marker of health.
- 6.5** Yesterday, we talked about the paper—what was it called, AJ?— "Why Gentlemen Prefer Blondes," that paper you just mentioned. A similar concept, the fact that parasites not only cause visible signs on the skin, but also a slight asymmetry, especially if the infection—if the mother is infected when the fetus is developing, or it could be an infant—there will be a slight asymmetry there. And that is an indicator of poor health. So we want to avoid that. That's the point there.
- 6.6** Overall, we prefer symmetrical objects; that's our overall preference. But what if you had a living room? Let's say you had a living room, your living room, AJ, your living room at home. If everything within your living room is perfectly, symmetrically aligned, like literally the couch is like this, and then on the other side, there's another

symmetrical couch, and then you have a painting—everything is symmetrical. Would that look beautiful, AJ? Let me know.

- 6.7** Why not? I just said we prefer symmetrical things because it titillates the brain; it makes us think of living objects like lions and gazelles and things of that nature. How come we don't prefer this room to be entirely and completely symmetrical? Why is that the case? You're whispering something?
- 6.8** It can look uncanny.
- 6.9** It can look uncanny, but why, though? Why does the brain not prefer symmetrical spaces? The reason is the following. When you look at the brain, you have light hitting the optic nerve—let's say there's a symmetrical object. There's a mate there. There's a human being, a potential mate. Information— I scan it overall. Information hits the optic nerve. It goes to the thalamus of the brain, then it goes to the first initial V1 back in the brain—that's the primary visual cortex. Information there is then divided.
- 6.10** There's two streams. One is called the "what" pathway—it goes up in the temporal lobes. You add depth, you add movement—this is a region called the MT. MT, motor—motion, excuse me. If you have a lesion there, you won't be able to sense motion. The world is like a black and white movie. You know, where you see the clips like this? So if you pour a glass of water, it will be pouring like this [demonstrates jerky, frame-by-frame motion]. Or, if you're crossing the street, that's not very fun because the cars are like [imitates choppy movement]. There's no sense of motion. That's the MT. Vision—you add color, depth, texture. And then there's the visual recognition area— areas like the fusiform face area, the Wernicke's area, and so forth.
- 6.11** There's a whole hierarchy. I see an object, it goes through that hierarchy, and if there is a sense of symmetry, I feel an "aha"; I feel good. Because within that hierarchy, there are dopaminergic projections. Within each step, there are small wires going to the dopamine centers and the reward centers telling me, "Oh, it's symmetrical." " Oh, feel good; it's a potential mate." We know in the higher centers, "Oh, feel good," and so forth. So you have that feeling. For that reason, the "what" stream reacts to objects within the scene.
- 6.12** But then, Heavenly, there's another stream going from the initial visual area, V1, and it goes to the parietal lobe. So instead of going to the temporal lobe, it goes to the parietal lobe. Now, you guys have been here for a few lectures. What is the parietal lobe involved in? What about sense of body image, sense of self, overall self and visual representation, overall global impression of the world? Is that the parietal lobe, especially on the right side? Yes.
- 6.13** So literally, when you gauge environment, a scene, you rely on the "where" pathway, where things are in space. So where is the lion vis-à-vis the tree over there? And the tree in relation to that stone? That stone in relation to me, and so on? It's more decoding the overall spatial layout of the environment instead of coding each object.

Now, of course, that stream also has little wires to the dopaminergic center. So each time I see a visual scene and I see various objects in it, I also can feel good. So there's that; that's a part of it.

- 6.14** Because of this distinction between the "what" pathway and the "where" pathway in the brain, scenes, if they are symmetrical, seem artificial. They don't feel right; they don't feel like they should be that way. There should be some asymmetry in the environment. It doesn't feel natural. In fact—and this has been shown—if you look at a picture and you see trees lined up like this, completely symmetrical with mountains behind them like this— the scene itself would actually happen in nature. So even though the scene is highly unlikely, it could still happen, couldn't it? You could have a little snippet of nature where everything is symmetrical just for that bit, couldn't you? But your brain abhors these highly symmetrical, artificial scenarios because it computes. It goes, "This is so unlikely to happen in nature that I despise it. I don't want anything to do with it. It doesn't feel right."
- 6.15** Now, if we truly prefer symmetrical faces, and we do, then you might question me and say, "What about a female—" Here, I think Marilyn Monroe comes to mind. I'm not sure— you have a female face, but she has a beauty spot, and it makes her even more beautiful. Why is that the case? What's going on? Am I contradicting myself here? There's a slight imperfection within the overall symmetrical face. Heavenly, what's going on? What do you think might be going on here inside the brain, inside your skull? There must be some explanation.
- 6.16** I have a guess, and my fellow scientists have a guess: perhaps that's a slight asymmetry. So you have the initial symmetry—it suggests she's healthy; she doesn't have any genes that have been expressed in a manner that's unhealthy for her overall immune system and so on. My visual processing machinery likes the overall symmetry, but there's some novelty; there's some dopaminergic— something titillating, something that's unusual. It breaks the monotony. The fact that you're looking at this thing, this face, and there's some slight deviation like that, it makes it more interesting to the brain. It has to do, I believe, with the dopaminergic centers and craving for novelty. So you want symmetry, but a slight novelty.
- 6.17** The brain is interesting, especially the visual system. It is all about solving puzzles, trying to figure out what the world is all about. I mean, think about it, when you are spotting a gazelle or a lion behind a bush, you're trying to figure out, "What is that behind the bush? I see something here, a color of yellow. I see another color of yellow here behind the bush and some more yellow." And then you go, "Aha, I discovered a lion." So you feel good. Vision, in fact, is all about these puzzles; it's always engaging in problem solving. So perhaps this beauty spot is part of that whole problem solving, this novelty, this breaking of rhythm that's part of the visual system's machinery. Does that make sense?

- 6.18** We prefer beautiful faces. We rate beautiful people as more attractive—well, that makes sense—as more intelligent. Judges will give more lenient sentences to criminals who are more attractive. Teachers will give higher grades to students who are more attractive. Can you believe this? It's not fair, is it? It's not fair.
- 6.19** In fact, on that note, in fact, we equate goodness with beauty. Look at any Disney movie. Look at Snow White. Look at the Little Mermaid, although she's only half human. You get the point. Think of any Hollywood movie you can think about: goodness equals beauty, beauty equals goodness, and vice versa. Think about ugliness equals bad, evil. Think of the Lion King and Scar, or think about Batman and Harvey Dent. Initially, he's, I guess, normally looking, and then his face is all messed up and he becomes evil. Or the Joker.
- 6.20** If you look inside the brain, there's a region called the— we know this region by now—it's called the orbitofrontal cortex. It's a small structure involved in reward and in processing of information, detecting threat and value in the world. It's involved in things like OCD. In psychopathy, its activity is low. It's what we mentioned: it's the bridge between a higher-level cognition and emotion. It's that bridge, little bridge there.
- 6.21** It turns out there's a region within the orbitofrontal cortex that activates selectively when I look at something I perceive as good and at beautiful faces. Literally, there's an overlap in the activation of the circuitry. So goodness and beauty are processed in the exact same spot within that region, within that part of the brain. The orbitofrontal cortex is subdivided. Another region within the orbital is processing evil— when I'm immoral, like badness, evil—and ugly faces, pretty much. What we see is ugly; what we don't like. Can you believe this? What do we do?
- 6.22** Now, the amygdala part of the brain has a more diffuse reaction to beautiful and ugly faces. It lights up if it's beautiful, and it lights up if it's ugly. But it doesn't have this subdivision where it's for ugly faces and for beautiful faces. The orbital is literally— when I look at a beautiful face, it's like winning. The same circuitry that is involved in winning—the reward center, called ventral striatum nucleus accumbens—the orbitofrontal cortex has connections to that region of the brain.
- 6.23** So whenever I—they have done studies where they look at winning and the neural activity of winning: the orbitofrontal cortex lights up; it becomes happy. The reward pathways become titillated and tickled; they become happy. The exact situation happens when you look at a beautiful face. It's like winning. Literally, looking at a beautiful face is like winning. Is that fair? You might ask, "Is that fair?" But that's right. And then exactly the opposite scenario for unattractive faces.
- 6.24** The good thing about all this is that— and we talked about this a little bit at length, I think, in previous lectures— if you look at the fusiform face area, the place that registers faces and processes faces, that part of the brain is hooked up to the primary

visual cortex, the more primary centers, and then there's a wire that goes straight to the prefrontal. That's a good thing for us, mortals, because that means there's a frontal component to beauty as well. So you go, "I like this face because I knew a person that I liked before," or "I associate that face with something pleasant," or "I like those features because they remind me of something," of course, unconsciously. There is definitely a clear-cut prefrontal aspect to it. It's not merely symmetry or it's not merely whatever you might pinpoint. There's definitely also other features, other aspects to it.

- 6.25** Some of these things we've discussed go into the whole idea of a vision, how vision is always in the business of—it's always in energy saving mode; it's trying to save energy. The reason is that computation is very costly. When you gauge the visual world, you don't want to waste precious resources and visual machinery. You want to keep things pretty much on low energy saving mode.
- 6.26** That's one of the reasons, for example, why we like repetitive patterns. This could be art like Persian carpets or where there's certain rhythm to it. It's repetitive—that's the same principle. We like that repetition. That's part of the whole economizing nature of vision. I think that's a principle that I really want you to go home with, that vision is all about economizing. It wants to take shortcuts—and I mentioned this in a previous lecture—when I look at the world, when I see a visual scene, when I see an object, like the TV, I've seen that 100 times before, so my brain goes, "I don't want to hallucinate or create the whole thing from scratch. I have memory files of that in my hippocampus." So whenever I look at a visual object, information, there's initially a sketch in my primary visual area. So I say, "This is perhaps a television based on the shape."
- 6.27** And then information, as it travels along that pathway until it reaches the hippocampus, there are small iterations going from the higher areas to the more primary visual areas. This is not clear; I want to make this crystal clear. When you watch an object, you can infer things about that object based on sensory experience and what you see. But you can also infer things about what you already know about that object. The sensory data initially hits your eye and goes to the primary processing areas. At that point, you already have tons of information about similar things that could even be that very thing you're seeing in the higher areas, in the hippocampus, in the memory centers.
- 6.28** In order for you to save energy, it will literally send information back to the initial area saying, "You're probably looking at a TV, a TV screen," but it's not 100% sure at this stage; it doesn't know for sure. But as the information travels further along the visual pathway—it's going to the depth area and the color area—it becomes even more certain with each step. So information travels, and then there's a little iteration. So it goes to the visual, from the visual, from the higher areas, and then as soon as you can detect what you think it is, there's a click. There's a visual click. You say, "Oh, this is it. This is what I saw."

- 6.29** Let me give you a more crystal example. Do you know—there's this illusion, this visual illusion like a Dalmatian dog. There's spots of blackness you see and some whiteness, and then you look at it like this, and then, at some point, you see a dog there. Do you know this illusion? It's a visual illusion. Initially, your primary centers can't really make sense of it. But then, at some point, it clicks, and that's when your brain knows, "This is a dog." Am I being clear here? Is this clear?
- 6.30** There are as many centers going from the higher visual areas to the primary visual areas as there are visual areas going from the primary to the higher areas. So what we're seeing is very conceptual. The world is very conceptual. In fact, if you have damage to the eye or the visual cortex, you will rely heavily on your conceptual parts of the brain and the memory structure. So when you look at the world, sometimes you hallucinate things.
- 6.31** Literally, you will see maybe a monkey over there. There's actually a condition called Charles Bonnet syndrome—they will see exotic animals everywhere and even cartoon characters and things like that because of this reliance on higher structures. This very fact, the fact that you have a visual system that is very economizing and always trying to save energy, you will find that we, generally, as human beings, love outlines. Many painters, for example, or artists, they will paint, they will create a painting, but instead of creating—I think Picasso had these doves or something—instead of creating a full painting of something, of whatever they're trying to portray, they will merely do a little sketch. And we like these sketches.
- 6.32** You were doing a sketch before, right? Why do human beings love sketches? Why do we find sketches so appealing? It has to do with the economizing nature of vision. The fact that when you do a sketch, when you draw a sketch, you are tapping into the initial visual areas. The initial visual areas only care about shapes and dots and things of that nature, lines. So you're saying, "The hell with all the details and all that." You're tapping directly into the essence of what you're trying to portray, which is the overall outline. Does that make sense? That's why we prefer outlines. And of course, the reason why we prefer outlines is, again, we have limited attention. Attention is very limited. In nature, you want to pay attention to what is most important. So relying on outlines is important for that reason.
- 6.33** There are other laws that an artist will explore. One law, for example, is—this is counterintuitive. It goes back to what we talked about yesterday. We talked about jealousy. We said jealousy. You would expect jealousy to be more depending on—if the person has more resources than another person, you should be more jealous of the person with most resources. If there's person A he has a million dollars; person B, he has a billion—you should go for the person who has more resources. But that's not the case. Why is that not the case? What did we say? Remember? Because we can relate more to the person that is more like us, right?

- 6.34** There's actually a similar principle when it comes to art, which is: sometimes less is more. But beyond that, hidden objects—so not only less is more in terms of the outline, but even if it's a bit hidden, that can even be better sometimes. So if you look at a person, let's say a woman, she's partially nude— you're smiling. It's okay. It's science. We can talk about these things—she's partially nude because she's hiding behind a little curtain. Do you know that that can be more titillating for a man than actually seeing a fully unclothed woman? Did you know that? That doesn't make sense, right? It shouldn't be the case, but that is actually the case. It's called the peekaboo effect.
- 6.35** What is going on here is that the brain is always in a problem-solving mode. And so by hiding—and the great artist will do this in paintings, they will hide certain things, certain aspects of the image—by hiding it, it titillates your dopaminergic neurons. So AJ goes—not the nude example, but generally—if you're trying to make out what the object is behind a scene, you're anticipating reward. You're trying to make sense of what you're seeing behind the hidden object.
- 6.36** It all goes back to the lion behind the bushes. You're trying to make sense of the lion behind the bushes, going, "Oh, here's a splotch, here's a splotch, here's a splotch. Oh, it's a lion. I feel like this is reward." It's reward in the sense that I have to get away. If it's a gazelle, it's a positive reward. This is the law of peekaboo.
- 6.37** But guess what? There's another law. When you look at these splotches behind the bush, so you see these splotches, there's a law called grouping. Grouping is the law when— I'm going to use you as an example because you're very well dressed today, Naomi. You always are, but especially today, I like your outfit a lot—if you look here at the outfit, you see there is amatching of colors. There's blue, there's blue here and here, right? So overall, you see blue at various far distance areas of her clothing. Now, this blueness is exactly the same scenario as the lion, detecting a lion behind a bush. That's exactly the principle we are tapping into here. Can you believe it?
- 6.38** When you go to Macy's or Nordstrom's and the sales clerk or the lady is trying to sell you clothes, she will say, "You should try a blue jacket, blue pants, and a blue hat," for example. That's exactly the same algorithm as you being on the Savannah and looking for parts of the lion behind the bush. It's exactly the same thing. That pleasure you feel is that very same thing. You don't know that when you're buying that shirt. You look in the mirror and you feel good, but that's exactly what it is. Or people say you look good in that clothing.
- 6.39** But I have a question here: if you're wearing a certain outfit, let's say— I'm wearing this pink shirt. If I was to wear a pink tie, would it be beautiful? Would it look good? Pink tie on the— I'm tricking you. What do you think?
- 6.40** Probably contrast is better.
- 6.41** Okay, we're getting into some people who know. Exactly. If I was to wear the pink and then something red, it did look good, but it wouldn't pop in the same way as if I had a

black tie, for example. So there's a law called contrast. If you wear something blue and then you have something black underneath, that would be a contrast. It's visually appealing to have contrast. Our visual centers were developed to be highly sensitive to contrast.

- 6.42** That's why when we look at—let's see, there's green, there's trees, there are red apples, there's fruits—our eyes are highly sensitive to that information. That would be the law of contrast. First of all, you might say, "Baland, aren't you contradicting yourself? You're saying you want colors to be of the same because it reminds you of the line, yet you want contrast. What is that about?" Well, it turns out these are not contradictions. It means that when the colors are far apart, like the pants, the jacket, and the hat, for example, or a scarf or a tie—then it's like you detecting that lion behind the bushes. But when it's exactly adjacent to it, or on top of it, that's when you want contrast, because then it's like the fruit on the trees.
- 6.43** Of course, the more contrast, the better. So if, let's say, on Mars, trees were blue, you would expect fruits to be yellow, for example. Or, if the trees were yellow, you would expect them to be blue. So the more contrast, the better. That's my point. And contrast can show up in other scenarios. So contrast can be—for example, if you have a very delicate skin, soft skin, and there's a metallic chain on it, as an example. So contrast is not only in color, but it can be in texture as well.
- 6.44** Was the whole problem solving aspect of vision clear? The fact that vision is always in the business of trying to solve something in nature? Even when you're gauging—let's say, you're in a mall and you're looking around and you're trying to make sense of people. Let's say you're looking for your Uncle Joe there in the mall. As you're looking for faces and you're scanning different faces, you are activating your dopaminergic neurons and attention neurons, trying to figure out, "Where is my uncle, my daughter, or my niece?" So that's a huge part of it.
- 6.45** There's a principle called the isolation principle, which I think is very interesting. It's a very interesting principle because I've seen it in action so many times; I see it a lot. It's the idea that if you want to highlight something that's attractive, you want to tone down other aspect of a visual image, for example. To give you a clear-cut example, I recently—a couple of months ago, I saw a video. There was this girl in the video. She was, I think, being interviewed, and she had a very attractive face. Her face was very attractive. In fact, it was extremely attractive, very conventionally attractive.
- 6.46** I noticed she had—mind you, because I was looking from the lens of neuroscience, I was thinking about it, "Why is she wearing all black with a turtle neck?" It doesn't make sense because people are usually dressed more—it's not very common to be dressed like that these days, completely black and turtle neck. It's not a funeral. So why is she wearing that? And then I sensed her face became even more beautiful; her face was glowing. I was like, "Why is this the case?" And then I went, "Ah, it's the principle of

isolation. She's removing all the redundant features." Attention is so limited, so you look at her face more. She's highlighted. It's a big highlight.

- 6.47** And then there was a girl next to her. Her face wasn't that—she wasn't that attractive. You could see a lot of her skin, and you went, "Ha." My point here is that whatever you want to highlight, you tone down the other details, everything else around it. So it's not more. Sometimes you think that adding more is a good thing. But if you have big muscles, you want to highlight your muscles, for example, then you want to—what do you want to do? You want to tone down—I was going to say wear a mask, but that's not going to work. My point is: you always want to tone down other aspects of any visual representation to highlight the one you want to amplify. Is that clear?
- 6.48** Give me an example. I can't think of an example right now. The face example is a good one. What else? What else is there? If you're wearing—if you have a beautiful chain you've bought. You bought a new beautiful chain, wear all black almost, and then have the chain be on top of that. That's a contrast, but also an isolation.
- 6.49** Why do we engage in art to begin with? What's about art? Why would we, as human beings, engage in art? What is the purpose of art? Yesterday, we talked about how we might— art might be a way to express talent. Perhaps, if you're good at drawing, that is a sign of hand-eye coordination. Talent, good genetics. You can't really fake it that easily. How would you fake it? If you draw something, you drew it, so you can't fake it easily. So perhaps it's a proxy for good genes, and that's why we engage in it.
- 6.50** I don't buy that, though. Why would you spend hours painting something? It's been hours and hours. Have you been to the Sistine Chapel? The paintings there are very beautiful. They're extremely—the level of detail, can you believe it? You feel like you're inside a dream. It's extremely powerful. They say that Michelangelo, when he was lying there as he was painting, I think when he came down— because he had to be on a scaffold—when he came down, for several weeks, he would look at the world as if it were upside down, because his whole view of the world had changed. That shows you how much effort they put into this. It wasn't merely a painting. It was really details, and they would spend all this time on it.
- 6.51** For me, it doesn't make sense that you'd just do that to show off some talent. There seems to be more. I think it has to do, probably, with the right hemisphere and our tendencies. If the right hemisphere is mute—it doesn't have the ability to speak, it's very visual, it likes metaphor, and it likes visual representations that are symbolic— then perhaps art, poetry, these things, are ways of channeling what's inside the right hemisphere. That can't be expressed using language. Almost like you're channeling a dream. What do you think? This is what I think. This is my view on it, that there's got to be some brain circuitry that explains it, and I really think that's it. I think it has to do with the right hemisphere.

- 6.52** I hope I was clear about the whole point of vision and the visual system and its purpose. Vision is not about seeing only. It is as much about feeling. It's a huge part of seeing, feeling, dopamine, pleasure, or lack of pleasure. When I say "lack of pleasure," I mean negative valence perspective, what you would call negative affect or negative emotions.
- 6.53** So we talked a little bit about the grouping effect and how we identify an object behind a bush, and we have the colors, and we say, "Oh, this is a lion there hiding." What's going on inside the brain when this is happening? It turns out, when you look at individual parts of a lion, you have neurons at various points in the visual cortex firing. So you have little spikes over here in the visual cortex, little spikes over here, little spikes over here, little spikes over here, spaced apart.
- 6.54** When you know that, when you have the recognition that this is a lion or this is a gazelle, there seems to be a neural synchrony going on where these spikes—instead of firing in random fashion, which is the initial scenario when you're just looking at splotches—that synchronization is part of the, "Aha, this is a lion."
- 6.55** So in short, that's a huge part of it, that sense of neurons, instead of firing randomly, initially, when you look at the various parts, they now fire in a synchronized manner. That's a big part of the whole visual recognition, and then knowing, being certain, "This is a lion that I'm seeing." Artists will use all these various tricks—whether it is contrast, whether it is grouping, whether it is the peekaboo effect, the principle of outline, the principle of isolation—artists will use all these principles, these laws.
- 6.56** But there's also a law of what we call super ultra— what is the law?—let's call it ultra-normal stimuli. That's what we call it. It is the notion that we tend to find things that are exaggerated appealing somehow. It somehow, again, activates the dopaminergic neuron. So for example, remember we talked about the rats that prefer the skinnier rectangle compared to the original rectangle in the previous lecture? Because it learned the law of rectangularity. And so the more rectangular, the better for it. It was an exaggerated version of the original, but it preferred that. Do you want me to repeat this example or is it clear? You want me to repeat it? Okay, very good.
- 6.57** So you have a rat. You give it a piece of cheese each time you show it a rectangular shape. Then you show it a square. It automatically always goes— you only show it the image of the rectangular shape, and it always goes for the rectangular shape. It has learned that this rectangular shape is better. It means cheese. Now, you show it a completely different rectangular shape, but it's more rectangular. Instead of going for the original, it will go for the more rectangular shape, which it shouldn't. It should technically go for the original one because that's the one that's paired with the cheese. But it goes for the more rectangular one.
- 6.58** Is it over time it does that?

- 6.59 Even in the first presentation you show it. And then probably, if you show it even more rectangular shape, it will go for that because it has learned the more rectangular, the better. Rectangularity is the law it's tapping into. This is what we call ultra-normal stimuli. This is ultra-normal stimuli, a stimulus.
- 6.60 Then I mentioned caricatures, how a caricature is appealing. The artist will use this rule: take the face of, let's say, Nixon or Donald Trump or somebody that we all know, extract the Donald Trump features—no, how did I say it was?—you detract the Donald Trump features from an average face, and then you amplify the difference. Basically, you end up with a face that is more Donald Trump than Donald Trump himself. He would have more of the bushy eyebrows, the round face and the yellow hair. It would just become more. You amplify what's the Trumpness in him.
- 6.61 We find that appealing; we find that caricatures are appealing for that reason. It taps into something—you show us something that's called the essence of that thing we're looking at, the essence of Trump, the essence of Nixon caricature, the essence of something. In fact, Michael Jackson—you all remember Michael Jackson, right?—in his poses, he used this a lot. When he would do various—I'm not going to do them for obvious reasons. But his poses, he was tapping into something in the brain chemistry when he was doing all kinds of poses—doing something that was so exaggerated in his postures that it went into the ultra-normal stimuli category. And people were just shocked by looking at it. It did something to them.
- 6.62 My point in all of this is that we have algorithms, codes, programs, inside our brains for art, for appreciating art. Art is not a random thing. For a long time, we thought, "The reason why we find certain paintings beautiful is random; there's no explanation for it." But if you look at the neural machinery, there seem to be explanations for it. We have a coding scheme, a way of looking at the world. And then when we look at something like paintings and the deployment of these laws, we feel titillated and aroused and we find it beautiful and we don't even know why.
- 6.63 Let me give an example that can illustrate this. Do you know seagull chicks? You're smiling. You've heard me talk about this before, haven't you? That's okay. Seagull chicks. They will beg for food each time they see their mother. They'll just beg for food. And how they do it is that they will go and then on the beak, they will go and beg for food by hitting the beak. This is how they do it. And her beak is very long—along shape, and has a red spot on it.
- 6.64 It turns out that there was this— I don't want to say evil, but this ethologist who thought, "Okay, how does the little baby chick know that this is the mother? How does the chick know?" So he waved a head of a dead mother, gull, in front of the baby, and they kept begging for food by hitting the beak. Kept going for it. They didn't care if the body wasn't there. The long shape, the beak with the red spot was enough.

- 6.65 Then he took a little piece of cardboard like this, and then he put a red spot on it, and waved it. Guess what happened? The little chicks kept begging for food, thinking it was the mom. But this is the most intriguing part: then he took the cardboard and drew three lines on it and did like this [makes a waving motion with his hand]. The chicks went crazy, begging for food. Their brains went crazy. Basically, the ethologist, Tinbergen, had created a superbeak, something that aroused the neurons of these baby chicks. They had no clue why.
- 6.66 Nobody knows why this is the case until this day, why the baby chicks find the cardboard with three stripes on it appealing and titillating. Perhaps the more outlined, the better, something like that. So instead of a red spot, the more outlined—something to do with the visual neurons and how they're encoded.
- 6.67 If you had a little chick society—in their little society, they would hang long objects with three stripes on them and call it Picasso, call it a Michelangelo painting and find it arousing and titillating without knowing why. That's the core principle of this lecture, what I'm trying to convey: when we look at an image, there are certain laws in our brains that are appealing to us. And so by having visual art—but we have no clue why, that's the point, just like the chicks. That's the key point.
- 6.68 You talked about vision, embodying a goal state and saving economic resources. Do all senses work in a pattern like that, or is there something unique to vision that they seem to embody that top-down goal state?
- 6.69 It's a good question. Other sensory modalities have an economizing— do they have an economizing nature? I mean, auditory cortex is also linked up to dopamine a lot, and there's a lot of the same principles of peekaboo and problem solving. So if you hear a song and the music is not clear, the lyrics are not clear, you get titillated and things like that. So some of the same principle will carry over. But I don't think it's economizing in the same way as vision. I can't see—no, I don't believe so. This would be my guess. No, mainly vision, to my knowledge, though.

Lecture 7: Exceptional Minds

At a glance — Exceptional minds — creativity and genius.

- 7.1** In this lecture, we'll be talking about creativity, genius—the human brain is truly a wonder of nature. Here you have 100 billion neurons with these signals firing all over the place and just the sheer size of a human brain—do you know it is actually three times larger than a monkey brain, chimpanzee's? The reason for this is mainly a gene, a single gene. When the human brain develops, initially, there's a single—there are cells dividing. That cell division is controlled by a single gene, which I call the police officer gene. It stops the division a little bit earlier in monkeys than in ours, and therefore, our brains are much larger than the monkey brains.
- 7.2** We have certain structures that are unique to humans. For example, the inferior parietal lobule, the region of the brain that's involved in things like creating, constructing images, and conjuring up images in your brain. That part of the brain is unique to humans. The angular gyrus, that part of the brain—remember I mentioned that Einstein had huge angular gyri? Remember that? That is—let's call it the math department of the brain. The left side is involved in things like math, calculation, and metaphor. The right is more involved in things like spatial representation and what you might call more visual metaphorical images. Everything in the right is more towards larger images, larger scenes. Everything language-related is in the left hemisphere.
- 7.3** Let me give you an example. I've talked about Ramachandran many times in my talks. I often do. We were really curious about metaphor and how metaphor arises. We know that in the left angular gyrus—if you have a stroke in the left angular gyrus, you no longer can comprehend metaphors. Literally, if I tell you—if you bring in a patient who has a lesion in that part of the brain, and you tell him, you say, "The detective jumped at the clue," which is a metaphor. It doesn't mean he literally jumped at it; it has a metaphorical, larger meaning. If you have damage to the left angular gyrus, you will literally understand every such metaphor as literal.
- 7.4** So we thought, what about in the general population? What about people out there, people without lesions, would they also see metaphors—would they also have what we call metaphor blindness, a blindness to a metaphor? Is that a thing out there or not? In other words, does it only affect people with lesions, or could it affect people in the general population? The reason we thought this was because when you actually go around some campuses, like the UC campuses, you meet people that are very literal-minded, like the engineering department. They are very literal in their interpretation of the world. So the question is: are there actually people out there who just don't understand metaphors? This is a genuine question—metaphor blind people out there.
- 7.5** We thought about this and we did surveys. I brought in my RAs, research assistants, and the entire team, and we went out there and interviewed about 200 people. We did

various questionnaires, and it turned out that about 5% of the population is metaphor blind—5% of the general population. That's pretty interesting. We'd show them various metaphors—there's one like, "Not everything that glitters is gold," and they would literally come up with all literal explanations for it instead of the metaphorical one. And that was interesting.

- 7.6 And then, of course, we had to make sure that their intellect was intact, that they weren't just not very smart. Obviously, we thought that wasn't the case. These were very studious engineering students doing well, performing well in academia, but they had this one quirk—they couldn't understand metaphors.
- 7.7 Now, I want to add a little note here. This was a preliminary study. We have to do that, right, AJ? We have to make sure that—it hasn't since been replicated, so we have to do more studies to make sure that it's actually the case because it's a big claim to say that people out there are metaphor-blind. To say that in a room of 100 people, you have five metaphor-blind people—that's a big claim, but it could be the case that there are metaphor-blind people out there.
- 7.8 Possibly, the cause of this could be that in the inferior parietal lobule, in that region of the brain, the angular gyrus—these are neighbouring structures and part of the parietal lobes, on the left side, particularly, there might be some congenital lesions there, some lesions you were born with. It's not only actual lesions from damage, but you can have small lesions. We know this from actual populations out there. You can have small lesions called congenital lesions; you have them from birth. So that might be an argument.
- 7.9 Another argument is that potentially— do you remember in my first, I was thinking it was in my dream lecture—I talked about how the baby's brain is hyperconnected. Until the age of, say, 7 to 11, that age, 10, 11, the brain is hyperconnected. There's tons of synapses. In fact, the baby brain has about 50% more synapses versus our brain. So Benjamin's brain is hyperconnected, for example. The synapses are everywhere. That's why you always want to teach them skills and keep the connections, the good synapses and connections, you want to keep them, and shape the brain. Because everything that's not used is pruned away. The brain removes all excess connections. That's called the pruning process.
- 7.10 There's a pruning gene that's involved in taking away all the redundancies, everything that's not being used. So arguably, possibly, it could be that if this pruning gene, there's a mutation to it and it's over-expressed in certain people, you might end up with a less cross-wired brain. Hence, people who are more literal in their thinking and have less ability to engage in metaphorical thinking—think about a metaphor. Metaphor is really taking, "Juliet is the sun"—you take two different modalities: Juliet, a human being, and the sun, and you create a visual representation. So the question is: if you have a less cross-wired brain, that could explain it, potentially. Does that make sense? We argued that is the case, potentially.

- 7.11** Now, what would happen if your brain is excessively cross-wired? Let's say that the gene that is supposed to prune away the extra connections and all the synapses doesn't work very well. There's a flaw in the gene. There's a mutation, and the gene can't really remove excess connections. What might be the outcome of that? Might it be a condition, a state, where you see a certain thing, you see a—I'm going to take a boring example—you see a table over there and, at the same time, you hear a color. Does that make sense? You hear—no, sorry—you see something and you hear a sound. It is complicated, for God's sake. You hear something, and you experience a color. You see a number on a piece of paper—the number 4 in red. Whenever you experience touch, you experience a color. Your senses are blending in this way. It's called synesthesia.
- 7.12** A lot of artists—it's very common among artists, poets, and creative people. They have a lot of this synesthesia. It's blending of the senses. So whenever one sense is stimulated, another sense is stimulated at the same time. There's something called conceptual synesthesia, and this is more conceptual. You see a calendar of the week inside your mind.
- 7.13** I have this, in fact. I discovered this when I was in Ramachandran's lab. One day we were driving—he had this old Jaguar. We always drove to Whole Foods, drove around and got cookies and tea. We drink a lot of tea; Englishmen have a lot of tea— so we drove there. One day, all this research was coming up about synesthesia and calendar—people with synesthesia see the week in a ring. So Monday is here, Tuesday is here, Wednesday is here, Thursday, Friday, Saturday, Sunday, for example. This is my own calendar. I thought everybody had this. Do you have this as well? I have this: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday. January is here, December is here, February, March, April, June, July, August, September— I have this. I thought everybody had this. This is called calendar synesthesia, for example. A quite common phenomenon.
- 7.14** I told Ramachandran. He was obviously intrigued about this and told me, "No, you probably have synesthesia. You have synesthesia." It became clear at that point. There's conceptual synesthesia, like I have, and then there's the actual perceptual synesthesia—that's when you see a letter, you see B, Baland, A, and then you see it in red and blue and the colors pop in this way. It's quite profound. So they have this. Again, they always think that other people experience the world in the same way.
- 7.15** How can I explain this? It's almost like explaining the world—how would a non-color-blind person, like most of us, explain the world to a color-blind person? They have an extra sense, almost; they have an extra dimension to the world because of their brain being hyperconnected in this way. It's quite interesting, isn't it? What do you think, Naomi? It's an intriguing phenomenon.
- 7.16** Einstein had gigantic angular gyri. That probably meant that he was good at calculation because when you have, let's say, various forms of brain damage that would cause the left—and I'll go into what could happen and how you can get this phenomenon—but if

you have a big left angular gyrus, and a big right [angular gyrus], that probably means that you are good at calculating, but in a spatial manner, like a spatial calculation, meaning you can see the numbers and the calculations.

- 7.17** Savants, like autistic savants, their problem is they're extremely good at calculating. They're like calculating machines with numbers, they can subtract numbers in insane ways, but they're not good at actual math in the broader way, in the more abstract math. It's more calculation. They're like— what do you call these machines?— calculators. They're mental calculators.
- 7.18** Various conditions—Frontotemporal dementia, which is a fancy name for the frontal part of the region of the brain, the frontal part, the prefrontal, and the temporal lobes—if you have, for example, dementia, meaning these regions start to shrink. If they start to shrink you sometimes find that these people will have artistic talents. They will draw pictures that are more evocative than the Michelangelo pictures we talked about before, and da Vinci. They will just draw the most insanely beautiful pictures. And you might ask, why?
- 7.19** Why do they draw these beautiful pictures when their brain is essentially damaged in this way? It has to do with the fact that— and I think I've mentioned this before—the brain is always in a state of cross-inhibition. So the left hemisphere is inhibiting the right. The front region is inhibiting the rest of the brain. So if you have dementia to the front, if the brakes of the brain are eroding, the temporal lobes are eroding, then the right spatial areas of the brain—the parts that have to do with creating a sense of artistic proportion and a sense of dimensions and a sense of understanding art but at a primal visual level—that part of the brain can flourish and literally may become larger, the neurons there. That's why you see these beautiful, amazing talents in this population. Quite astonishing.
- 7.20** Now, if the damage was to hit more on the right side, so the left is remaining, the left is getting the share of the neural attention and is growing, then you have the opposite scenario. Typically, you see things like these mental calculators, these calculating machines, people becoming very good at—you say, "What is July 17th, 1983?" and they'll tell you exactly what day it was and that it was raining and what happened on that day. So they have this; they're very good at event registration. That's the opposite hemisphere. Does that make sense to everyone? Why that might be the case?
- 7.21** If the right hemisphere is involved in the art module of the brain, and the left is more the numbers part of the brain, then you might see this phenomenon. In fact, if you were to knock out the right parietal, the right inferior parietal lobule, you cannot draw anymore. I ask you to draw a sketch, and you'll draw something hideous. If you have damage to the left—the left inferior parietal lobule—and I ask you, "Okay, Joe, subtract 3 from 40?" they can't do simple calculations, because of how the brain is asymmetrical and how that function has been knocked out.

- 7.22 You have autistic savants. Some will say about 50% of people with autism have these unusual skills. Some say it's less, but let's go with the 50%. Why do they have it? It's a question we can ask ourselves. Remember I talked about how in the brain you have—there's a neurodevelopmental problem. They have extra mushrooms—I call them mushrooms— a mushroom-like structure on the dendrites on the neuron.
- 7.23 You have a neuron, and on the neuron, there's a dendrite. The dendrite is the receiving body. You have an axon on the other neuron. The axon is the body sending the messages. It is like the—what do you call it?—the postman sending the messages, and the dendrite receives the messages and conveys that to the neuron. On these dendrites, you can have little mushrooms— these are places that allow for new synapses to occur. It seems like in people with autism, we find their brains—when we look at their brain, there's tons of these mushrooms everywhere. There's hyperconnectivity between all parts of the brain, and therefore, you also find hypersensitivity to light and to sounds and things like that because of the hyperconnected brain.
- 7.24 But of course, it's not the whole story. Another part of it is simply that a lot of these autistic savants often have defects in certain parts of their brains. So they might be extremely poor at language—there was actually one case, her name was Nadia. She was six or seven years old. She could barely speak, suggesting that her left hemisphere was not being used, that it was damaged there. She could barely tie her shoelaces, but she would draw the most evocative pictures. You look at a picture she would draw and compare it to a Da Vinci picture—her picture was more evocative. It left out some of the unnecessary details. On the horse, when you look at the muscles, it looked like the horse was leaping out of the canvas. It was profound. Her IQ was maybe in the 50s or something. She was autistic, and she had this damage, yet she could draw these beautiful pictures.
- 7.25 And so the question is: what is going on in the brain? I think what might be the case is the principle of isolation, not the art rule. The rule here is—because a part of her brain is damaged, her brain is not working right—that part of her, the right part of the parietal lobe, is getting all the attentional share. It's being used—it's being hyperused. She uses it all the time; she is sitting there and engaged in writing, and so that is causing that part of the brain to become larger and basically to become hypertrophied. It's getting all the attention.
- 7.26 This is actually something that you and I can think about, because it's important for you and me too, in the sense that, when you think about it, can we become savants? Meaning, can we become really good at certain skills? For me, if you look at somebody like Lionel Messi, he's a savant. He's a soccer savant. I heard stories: my sister-in-law is from Peru, and her family knew a family who had Messi as a kid visit them. They told them—this was the saying—that Messi would always have a soccer ball on his feet and

play 24 hours. He would have the ball on his feet, play soccer, and just juggle the ball all the time.

- 7.27** In fact, it reminds me of myself. I used to do that, too. I was heavily into soccer, and I would have a soccer ball on my feet all the time, too. Probably some OCD there, some obsession. But I would have the ball on my feet, too. So it resonated with the whole Messi situation, that he did that, too. I think Maradona was not like that, by the way. I like him as well.
- 7.28** In any event, when you are using the neurons—the motor neurons that have to do with juggling a soccer ball—that part of the region of the brain becomes larger. It grows. We talked about it in a previous lecture about how if you use a certain region of the brain, you have the "use it or lose it principle". It takes and invades other regions of the brain. So that part of the brain becomes larger and bigger and literally takes up all the cortical space.
- 7.29** At some point, Heavenly, when you have—obviously, the brain is limited in its size. You can't just keep growing it. At some point—and we know this from experiments with monkeys who were spinning this disk— at some point, the neurons become more efficient and quicker in their transmission speed. So if you want to play the piano and be really good at that, you can become—I wouldn't say a savant necessarily, but you can become very good at by just practicing over and over and over again and again. This is a practical application. What do you think, AJ?
- 7.30** It makes me think of— I may come back. What I wanted to say there are people in recorded history that would be like mental calculators who weren't like that from their early life. There was this guy, his name was Rüdiger Gamm. He was a German guy. At the age of 20, he was working in a bank, and he became extremely—he found his job very boring, to put it frankly. So he started to calculate and crunch numbers all day. And four years in, he was on TV doing all kinds of things, displaying his talents, because he had become that good at math. They scanned his brain while doing his math, and it turned out he was using four more areas in his brain when he was doing math. So it shows you how we all have this potential to—if not become savants— to develop extraordinary talents in ways.
- 7.31** You also have this with the folks that have OCD and that have great memory. Some people in the population have photographic memories—powerful mnemonic memories, they can remember any day of their life, pretty much. And they are not savants or anything, but they have OCD. And so the part of the brain—the hippocampus and the anterior cingulate cortex and some of those regions—have grown in size and the connections have become very, very dense. It's part of the OCD spectrum that you can develop this. So it shows us that even in a seemingly healthy brain, you can develop some of these talents.

- 7.32 But the true savants, the true geniuses, are different. There was a guy—Oliver Sacks was talking about him; he was writing about him in one of his books—he was 13 years, I think he was 13, and he could play one song with one hand on the piano like this, play another song with the other hand, and then sing at the same time. Crazy. Or he could play the piano like this [demonstrates playing the piano with his back to the keys]. The field of music is punctuated by these musical savants. They say that Mozart could compose a symphony in his head, the entire symphony. So there's a lot of these savants in music.
- 7.33 We talked about cross-inhibition and how cross-inhibition plays a role in creating a savant brain. One part of the brain is knocked out, the other gets all the attention. It grows bigger, the connections become bigger. So cross-inhibition and isolation are the two key factors I wanted you to take away with how savant genius-like abilities come about. Cross-inhibition: you knock out one region, the other one can flourish. The other one is isolation: you have a part of the brain becoming isolated in its use, and so it becomes more powerful, let's say.
- 7.34 The savant syndrome does not only emerge from—it can emerge from frontotemporal dementia or a trauma to the brain, but sometimes you have epileptic seizures. So a seizure affects the brain—it could be the temporal lobes— and you get some of these talents. Let's say in the parietal lobes, you get a seizure there, an epileptic seizure there — that could cause the talents.
- 7.35 What do you think about that? Does that make sense? When you think about the brain and overall—whether we get to the same conclusion, the conclusion is a talent, a savant-like talent. That can come either from one part of the brain being damaged and then the other parts getting the attention. It could come from an epileptic seizure— do you know what a seizure is exactly? It is when these neurons fire ballistically in a wild manner, in an uncontrolled manner. The neurons run amok, keep firing, signal, and so that part of the brain becomes hyperactivated. It lights up, it becomes scintillated from the seizure.
- 7.36 But I think the most important thing to think about is whether we all have artistic talents, latent inside of us, waiting to be liberated. Wouldn't it be great if we all had those talents somewhere, and you could use a device to affect the brain, and the talents would come out? Would that be great? Well, it turns out that Alan Snyder, from Australia, did some experiments using transcranial magnetic stimulation. He would stimulate the left side of the brain—the temporal lobes—and then he would look at things like drawing, which would improve just by stimulating. Specifically, he would inhibit that part of the brain.
- 7.37 In TMS, the stimulation method, you can either inhibit, which is the neurons are inhibited by—this is how the signal sounds. I just want you to know. When it goes like [thump, thump, thump, thumb, thumb], that inhibits the neurons. When the firing is like [trrr, trrr, trrr], the neurons become more excited. In any case, when the neurons

are inhibited in the left temporal lobes, it turns out these guys became better at drawing.

- 7.38** In fact, there were these twins, Oliver Sacks described them. Whenever they saw—for example, there was this one scene. It was quite cute—they were sitting there, and they were completing each other's sentences. Then, there was a box of matches. They fell on the floor and they, at the same time, said, "101, " as in there are 101 matches on the floor. They counted that, and that was correct. You might say—people have argued —"They knew how many were there," but it's not true. They had a lot of other talents as well. There were other people in history that have been recorded, when peas would fall on the ground, they would know how many there were. "There's 117" or something. They would know this.
- 7.39** When they stimulated that part of the brain with the TMS and made it less active—the left temporal lobe—it turned out that when they showed people 117 dots, let's say, on a screen, they were very quick. They could suddenly recognize that really quickly after doing that stimulation. Is that clear? In other words, they became like the twins just by inhibiting that part of the brain.
- 7.40** So one question we can ask ourselves is— this is a question that I've thought about a lot and had a lot of discussions with my mentors about it—why do we have these talents? Why do they exist in the population? Why do certain people have these unusual talents? It doesn't make sense when you think about it. Why would you have a caveman, 100,000 years ago, having Einsteinian talents in his brain? What is the purpose of that? It makes no sense. Why do we have these talents? What's the point of these talents?
- 7.41** We don't really know why we have these talents. It doesn't serve any purpose, at least. Knowing how to do physics and math at a sophisticated level, as a caveman, doesn't make any sense. So why do we have them? And the answer is, as I implied, we don't really know. But one thing is clear—and this is, I think, quite important—these savants, these so-called geniuses, are not geniuses in the conventional sense. They have certain talents, but then they're abysmal at other things. They can't do other things very well. So is this really genius, having one skill you're very good at?
- 7.42** Ideally, a genius is good at a lot of things, right? But it tends to be the case that, what we see in these cases is that these savants are good at one thing only and only that thing. It may be that when the brain—the brain of the caveman guy who lived hundreds of thousands of years ago—when he was living, he had to have certain skills to survive: he had to throw the spear, be able to make sense of spatial landscapes, and make some calculation to make sense of predators and things like that. He had to have some—so he needed some skills and some talents. So it may have been that the brain, you had an excess amount of morphogen and hormones, growth-related hormones in the brain, that would help the guy survive back then, but it accidentally led to all these other talents. That's the only explanation that makes sense to me, that there was some

use of having a—or there was not a direct use of the talents, the talent themselves, like being good at art or math, but it was a byproduct of having bigger and better brains.

- 7.43** Some people have been very mischievous and asked, "What if you put some of these growth-related hormones into the brain of babies as they're in the mother's stomach, fetuses? Would they grow super brains?" This is not ethical to do, obviously, but it's a good question. Or what about a monkey's brain? Could you grow a super race of highly intelligent monkeys by doing this experiment? I don't know, but potentially. So these are some of the interesting things in science—unanswered questions. But you may end up with a race that might overthrow us, humans, potentially.
- 7.44** There's an interesting link between creativity and humor, and I think that's very important to add here. We talked about what humor was—humor and laughter. Laughter is when you have a joke. You have an important guy, for example, he's walking down the street, he slips, he falls, and you have the false alarm signal. People laugh to indicate to the tribal members that the guy didn't die, so don't waste your resources. That theory I explained—remember that? That's in the previous lecture.
- 7.45** Now, what is a joke itself? A joke is all about building up an expectation of something. You have a story, you have a storyline, and then you have an expectation of some sort. Then at the very end, you flip the interpretation and you go for a completely different interpretation of events. That's pretty much a joke in a nutshell. Do you have any jokes you can think of right now? Let's think of a joke. Let's see. I can't think of a joke, but that's pretty much the crux of a joke. The joke is all about—you have a story, and then at the very end, you change the interpretation of the event, and then you laugh somehow. It's basically a false alarm. It's almost like the guy walking and then hitting his head and not dying. And then you laugh. It has an element of building up, a punchline. You think something threatening is happening, but then you reinterpret the event and you laugh because it's funny.
- 7.46** I have observed this, too, by the way. I've noticed that all the creative scientists that I have worked with and met, the best scientists, they have a sense of humor. They're very funny. They laugh at the world. They don't take things in such a black-and-white manner. They're more loose, and they just have a good sense of humor. So one argument is that, potentially, more creative people have more humor, joke more and have more laughter.
- 7.47** I think this was highly evident not only in Ramachandran; he was always like that. But Allan Hobson, the dream researcher, one of the great researchers of dreams—I remember the first time I spoke to him on the phone—it's just jokes and laughter. He wanted to know things about me and my life and who I was. He was like, "Baland,"—I was telling him about experiments—"Baland, are you married? You tell me: are you married? What are you doing? How is your family?" He wanted to know things about me, and I was just puzzled. Here's the greatest dream researcher of the last 100 years. I

have a chance to talk to him on the phone, and he's talking about me being married and my personal life and stuff. These are the great scientists. These are very great.

- 7.48** Now, the not-so-great scientists, the 9-to-5—we call them the 9-to-5 scientists—who just work in their lab and are angry all the time, just crunching numbers and all that. Those guys don't have a sense of humor at all. They're a completely different [story]. I really think that humor is the key to creativity. There's something there.
- 7.49** I think another thing that I really learned from Ramachandran in his lab was that in order to be creative—so creativity, what is creativity? It's not merely being like—creativity is not merely linking unrelated areas and things. So that's one idea, that creativity is simply saying, "I link seemingly unrelated things." That's one very, I think, very simple view of creativity. I think there's more because when you think of any beautiful, poetic line: "a teardrop on the cheek of time." Think about how profound and how deep it is. This is Tagore, the Indian guy, the Indian poet. It's deep. It's not merely taking unrelated areas. It's really deep. It's something that titillates your mind in various ways.
- 7.50** There's something ineffable, indescribable, about true creativity. And I noticed, when I was in the lab of creative people, they would take a lot of time off. They weren't working all the time. I think it's an important thing. They would literally—we would go for walks on the beach, we would go have cookies, go drink tea—things were loose, things were fun. And then suddenly, while we were petting [Ramachandran's] dogs—his dog was, I think, Einstein was one of them, and then the other was, I think, Newton. I forgot. It's been a while. I was terrified of his dogs, by the way. Poodles. They were terrifying to me—but you could have great scientific insights while playing with his dogs or walking about Del Mar and drinking tea. And then suddenly: "Oh, Jalal, what about this idea? We have to go do this. Oh, we have to do that." Looking at the landscape and being open to taking in the world—that is creativity.
- 7.51** Ramachandran would always say, "Jalal, read widely." That was his key advice to me: "read widely. Don't be one of those guys who just reads one thing and gets stuck on that one thing. Read widely—that makes you more creative." He said, "Develop a good sense of humor. Read a lot of poetry." Here's a scientist doing neuroscience telling me to read poetry. It's fascinating. And then he said, "Whenever you do science, whenever you do anything in science, don't get stuck on one thing." I think this was probably the best advice he ever gave me. I was very much in the sleep paralysis track, only doing sleep paralysis for a while. He said, "Jalal, you don't want to be the sleep paralysis guy. You have to go broad. You have to study the brain and everything within it. Have a very broad perspective. That makes you more creative." One of the reasons it makes you more creative is because it allows you to link unrelated fields within neuroscience.
- 7.52** I'll give you an example. Ramachandran and I have developed a treatment for OCD involving a rubber hand. So if I have a rubber hand in front of you, I put it on a table, and then I take—so there's a rubber hand. The setup is like, the rubber hand is here.

The rubber hand has to be here. Then you have a barrier. This is a rubber hand—barrier—real hand. Rubber hand—barrier—real hand. The rubber hand, you stroke and tap. So you go, "Tap, tap, stroke, stroke, stroke, stroke, tap, tap, tap, tap."

- 7.53** You stroke the rubber hand—your visual access is only on your real hand. Over here, this is a real hand. In a few minutes of the stroking and tapping, you start feeling literal touch sensations arising from your real hand as if the rubber hand is your hand. Can you believe this? You can even do it without a rubber hand. If you put your hand underneath the table and stroke and tap the table in front of you, you will feel touch sensations arising from the table as if the table is a part of you. It's called the rubber hand illusion.
- 7.54** One of the things we did was we said, what if people with OCD—you contaminate the rubber hand, while you're doing this illusion. If the rubber hand—if you have OCD, Naomi—you don't, but let's say you did—and I make you feel like the rubber hand in front of you is your hand by stroking and tapping it in a synchronous manner with your real hand. Then I go on and contaminate the rubber hand in front of you. Would you feel contaminated in your real hand? That is the question. Would you feel like your actual hand was contaminated? Would you go, "Oh, that's so disgusting. I don't like this," for example. Would you? If you were to say, "Yes," that is potentially important. You know why, AJ?
- 7.55** It's important because treatments for OCD these days are based on the fact that you want to contaminate people's hands and then say, "I'm contaminating you." But then the patient is learning to face their fears by being contaminated. Does that make sense? I don't think that made sense, did it? Is it clear? The treatment for OCD is—if you have OCD, let's say, you want to get over your fears. I'll contaminate your hands over and over again. You feel contaminated in your hand and you learn to face your fears.
- 7.56** The problem with this therapy is that about 25% of people don't start the therapy because of fear of treatment, and about 50% drop out. We know this. So if I can use a rubber hand instead, that could potentially be therapeutic because now we are bypassing the fear. We did that and we found this might be the case. We found that people would have contamination sensations from a rubber hand—an example of linking seemingly unrelated areas. Cognitive neuroscience on one hand—the rubber hand illusion—and psychiatry on the other, which has to do with exposure therapy and facing your fears. That would be an example of what you could come up with.
- 7.57** What is really interesting about creativity is it mixes various—it relies on various parts of the brain that are usually wrestling, dueling it out, incompatible. You have the prefrontal networks of the brain. You have the salience network of the brain—salience, meaning these are the parts of the brain that have to do with attention. And then you have what we would call the default mode network—a fancy name for circuits involved in daydreaming and just sitting there and thinking, "When is this lecture over?" Those circuits. Usually, they are antagonistic. You don't have daydreaming and hyperfocus at

the same time, you see. You have one of them: either daydreaming or hyperfocus, correct?

- 7.58** In creativity, you see this interesting blend of hyperfocus and default mode network activation. This, of course, reminds me of the flow state. In the flow state, you have also a very dispersed activation of the brain. You have prefrontal activation, you have dopaminergic activation in the flow state. It's related, and there was a thought here that I wanted to expand on—hold on; let's see if I can catch it before it evaporates.
- 7.59** Creativity involves what's called intrinsic motivation. We have what we call intrinsic motivation and extrinsic motivation, intrinsic and extrinsic. The difference between them is— let's see if you can guess. Intrinsic means it's intrinsically interesting. We find it interesting. In and of itself, there's something compelling about it. Extrinsic would be—remember Sam mentioned that he was doing certain activities because of the reward of a potential job or a career or something like that. All of us have this. We have things we are intrinsically interested in and things we are extrinsically interested in. We do them because we get a reward from them. It should be a 9-to-5 job, a boring job, or something like that. No offense to 9-to-5 jobs; we all do them. But I'm saying it could be a boring 9-to-5, or you could have a fun 9-to-5.
- 7.60** We have this distinction. The intrinsic motivation is really where creativity comes about. That's where the flow state comes about. You get completely immersed in a certain activity and you lose your sense of self. You are lost in the activity. It's overwhelming. That's flow. Flow was described by Csíkszentmihályi, Mihály Csíkszentmihályi. Now, I was very fortunate. When I first went to Egypt— from Copenhagen, I went to Egypt—at the time, I was starting to learn about the brain and becoming interested in the brain— I happened to bump into a guy who became a mentor to me. He was the graduate student of Csíkszentmihályi, the co-founder of Positive Psychology. So this guy, Csíkszentmihályi, founded Positive Psychology or co-founded it.
- 7.61** I was very happy to meet this guy, and we ended up doing an experiment together that illustrates this intrinsic versus extrinsic motivation. In this experiment, we had people play a soccer game on a computer game. It was a computer game. As a person playing the video game, you would either lose with a small margin—how was the experiment exactly?— you would either win with a small margin or you win with a big margin. You play the soccer game. Either you won 10-0—you got 10 points, and the opponent got zero points— this would be like Barcelona versus Inter Miami, for example. Barcelona is very good. Miami has Messi now, but they're not as good as Barcelona. This is an example I could think of. That'd be probably about 7-0, that match. I see that. Now, if Barcelona plays Real Madrid, it's what? More like 4 to Barcelona and 3 to Real Madrid. It's much more even.
- 7.62** Now, it turns out that even winning is much more pleasurable to you. It's much more immersive. It's much more intriguing. It gets your juices flowing. And that doesn't make

sense because for a long time, they thought, "If you win with a big margin, that's the thing because you feel competent, you feel like you are great, you feel like you are the best." But it turns out, no. If you win by a small margin, that's actually even more enjoyable. We know this from even watching, so you don't have to be actually playing the video game.

- 7.63 By the way, you might be asking me here—and you have a good point, Sam— you might ask me, you might say, "Baland, how did you manipulate that? How could you actually make sure that you were losing with a small versus big margin and all that? How is that even possible?" It turned out the whole thing was staged. They weren't actually playing. They thought they were playing the video game, but they weren't actually playing. So we changed the textbooks in a small way. We said, "No, you might feel competent by winning 10-0, but really what you want is challenge, and that's more interesting in life."
- 7.64 I think my whole life has been challenged, so I know what challenge feels like. Taking on challenges is a good thing. And I think I want to end this lecture on that.

Lecture 8: Conscious Choice

At a glance — Conscious choice — consciousness and free will, the course's finale.

- 8.1** In this final portion of the course, we'll be talking about consciousness. I think consciousness might be one of the most interesting areas of research in neuroscience. Really, I think so. I think a lot of my colleagues think so, and it's one of the most difficult questions. I mean, think about it. Here you have this three-pound mass of jelly resting between your shoulders, and it gives life to all your sensory experiences, the world, your son Brance. Anything, all your perceptual views of the world.
- 8.2** And it all is generated from a piece of flesh. If you saw a brain— have you held a brain in the palm of your hands before? And think about it, you hold it and it's just a piece of flesh, but from it, all of your life experiences come about. I mean, I think that's one of the most interesting and fascinating things. And I think that understanding how that comes about is truly fascinating.
- 8.3** So where is consciousness located in the brain exactly? That is the question we will try to answer in this lecture. Where is consciousness stored in the brain? There's actually people out there, we call them panpsychists, who think that consciousness is everywhere. Like consciousness is just diffusely spread throughout the universe. That's one view.
- 8.4** Neuroscientists, however, think that consciousness is somewhere inside the brain. When you look at the body—we can just be simple here—first of all, we know it's not in the liver, not in the knee. It's got to be somewhere in the brain because when there's brain damage, your consciousness has changed. So it's somewhere in the brain. Where in the brain is the question.
- 8.5** And the way we can actually try to figure out where it is in the brain is by looking at whenever there's damage in a part of the brain, how that affects consciousness. I think that is the key way to figure that question out. And so if there is damage to the frontal part of the brain, does that knock out consciousness? I've talked about this before, I think.
- 8.6** So there's a case of Phineas Gage. He had a huge stick go through his brain, damaged his frontal lobes, and his personality was changed but consciousness was not changed. He didn't lose consciousness. He didn't become unconscious; he merely had personality changes.
- 8.7** And it's actually interesting because most people would think that consciousness is somewhere in the frontal lobe, in the prefrontal region, because that's the part of the brain that's interesting. That's part of the brain that's hugely developed in us humans. That's part of the brain where higher-order thoughts come about, where higher-order thinking is embedded. And yet you have massive damage to that part of the brain and

you merely have some personality changes. So consciousness, it seems, is not located in the prefrontal lobes. At least that's one view.

8.8 But before we dig deeper into consciousness and try to understand what consciousness is and where it is in the brain exactly, there's a concept, an idea, something that's called qualia. This concept called qualia, it's very important. It means the raw experience of, say, when you look at an apple and it's red, the redness of red, the pain of pain, the feeling of love. That's called qualia. We all have this subjective experience of the world. Subjective: if I poke you with a needle, you feel pain. AJ, that subjective sense of pain is your private qualia, is a qualia. So we all have this qualia.

8.9 It's embedded in my explanation that in order to have qualia, you need a self. It's meaningless to think of qualia without a self, like free floating qualia, like free floating experiences of red, or pain. So you need a sense of self. And we talked in this course extensively about how self is generated in the world, how the self is generated and constructed in your brain.

8.10 So we said that for the sense of self, we had structures like the temporoparietal junction involved in integrating sensory information from the world and weaving together a sense of self. And that structure relied on information from the insula. The insula is the disgust region of the brain. It's the part of the brain that draws information from your organs and your internal states and uses that information to create a sense of a self. And the sense of a self also relies on prefrontal structures to integrate a conceptual sense of self. Is that clear? So you have these various things.

8.11 By the way, on the insula, before I go too far, I want to come back to qualia and sense of self in a minute. This is a side thought I had. It was a question that somebody asked me yesterday. Christian he wanted to know, he asked me about the neuroscience of war. He asked me to maybe expand on a little bit on that. And I think it's relevant in the sense that this is a course, on human nature and the sense that we all have, human beings have an inclination to war, to engage in war.

8.12 I think what's interesting, this insula region which pulls information from your various organs to create a sense of a self based on how you feel inside and how your heart is beating and all that, that part of the brain is what we call the disgust machine in the brain. Whenever I sense disgust, like if I have OCD and you contaminate my hand, let's say, we talked about that before, I feel disgusted. Or if I smell something like a rotten food or something, I feel disgusted. Whenever I feel disgusted, my insula lights up like a pinball machine, like Times Square. It becomes all activated.

8.13 Now, it turns out that for social disgust, like when I feel socially disgusted, the same insula becomes activated when I feel social disgust. In other words, what I'm saying is: social disgust, an actual sensory disgust, is processed in the same part of the brain. That's important because when you think about how wars are usually started, is by "otherising" somebody else, another group. You see, it's like, "Oh, these people over

there are barbarians. These are evil. They're not really human. They're not like us. They're different." And so you tap into the disgust circuitry in the brain in that way. And so that actual social disgust that's created will activate your sensory disgust, actual disgust, and that makes you much more likely to go to war. That was just a tidbit I wanted to add.

- 8.14** So self and qualia, right? So you need a sense of self. Sense of self is derived from memory regions of the brain— hippocampus, seahorse-like structure behind your ears. Prefrontal region. Insula: disgust and getting information from the various senses. And TPJ, the region that pulls sensory information from the world. All of that is a part of your stitching together the sense of self.
- 8.15** Now, in that self-construction, memory obviously plays a huge part. And you have different types of memory. So you have what you might call procedural memories. Procedural memories are memories of biking, like when you bike, or could be walking, or anything that requires a movement and is automatic. It relies on a part of the brain called the cerebellum. So that's part of that circuit.
- 8.16** Then you have things like semantic memories, that's knowledge of facts. And then you have something called autobiographical memories. Those are the ones stored in the hippocampus and sometimes also stored diffusely in the brain, but the hippocampus is really the place where these memories are inferred from. They are drawn from that part of the brain. In fact, if you don't have a hippocampus, you cannot store new memories.
- 8.17** So there was a guy, H.M.—his name was H.M.—his hippocampus was, actually both his hippocampi, when there's two of them, you say hippocampi, were removed. The guy couldn't form new memories. So he only had what's called a working memory, which is the front part of the brain, prefrontal cortex. He only had that. So that meant he pretty much had 30 seconds of novel experiences, and that was it.
- 8.18** Each time he saw his wife for 30 years was like the first time he saw her. Imagine she was very happy in a sense, because he was happy each time he saw her. Or you could tell H.M. The same joke over and over, and he would laugh, for example. He would find it funny each time. Or he would— it's a bit, of course, sad, but this guy was very open to experimentation. He would join neuroscientists in various endeavors. But it really it shows you how the hippocampus is involved in storing memories. And autobiographical memories then are stored in the—or accessed and helped— the hippocampus helps store these memories. Is that clear? Now, of course, you need the hippocampus in order to create a sense of a self.
- 8.19** Now, you need self, you need qualia. But let's go deeper into qualia and try to understand it. If you were a super-scientist, let's say you're a Martian, even, you come to Earth and you have a chat with a guy. You hook up his brain, put it to a machine, and then you say, "I will now make a complete description of your perception of red, the

color of red." I show you a red apple. Then I measure the activity of your brain. And once you've seen the color red, and I ask you, "Have you seen red?" You say, "Yes, I've seen red." I will show you a diagram. I'll say, well, this happens in your brain. Color receptors in your brain were activated, went to the visual cortex, then went to the color area of the brain, and this is red. But you object to this. You say, "No, what about the redness of red? I don't see that on the paper anywhere. Where's the actual personal experience of red?" That's what we call qualia.

- 8.20** So there's a split in a sense, between the third-person perspective, view of consciousness and explanation of consciousness—which is all the bloody neurons firing away in all the circuits—but then here you have the personal sense of self. And we all have this, the here and now of conscious experience. I feel conscious. I am here, and I'm here now, and I feel something. How can you—I experience the pain. I experience that pain. You may tell me that all my neurons are firing. You can gauge that on a screen and show that to me on a screen. But what is it actually worth? Where is that private sensation? Where is that on the diagram? We don't see it anywhere. It's lost.
- 8.21** Then some people have argued that perhaps it's not completely lost. Maybe there's a translation barrier, perhaps. In other words, even though we feel like there's this barrier between personal experience and the physicists' and the scientists' description of the world, it's really down to a translation barrier. Just like we talked about the left and the right hemisphere has a translation barrier; your language can't express the metaphors of the right hemisphere.
- 8.22** Another way to look at this would be if you have this Martian and you hook up his brain with cables. Let's say this Martian, the super-scientist with this experiment, he actually has visual neurons in his brain. So he can actually perceive color. He just lacks the actual receptors. Is that clear? So he doesn't experience redness and he can't see the red apple as being red, but he does have the machinery.
- 8.23** If I was to, let's say I'm the subject, I'm the guy looking at the apple. If I was to hook my brain and the visual machinery in my brain to his brain. So there's a cable. Hypothetically speaking, it's a very hard experiment to carry out in real life, as you can guess. But let's say, hypothetically, we do this. And then now, as he's looking at the apple, he'd go, "Oh, my God. Now, I understand what you mean about red. When you look at the apple, when I look at it, I now experience redness."
- 8.24** So it may be that really there is no—it's not like qualia and subjective experiences like redness, and redness of red, the pain and all that, can't be captured scientifically. It's just we need—this language cannot capture it. Scientific instruments cannot capture it. You really need a brain to capture it. And so we try to—if we could hook my brain up to another brain, that might work. That's the point I'm trying to convey.
- 8.25** Why did qualia come about? That's a question we might ask ourselves. Do we really need qualia in order to exist? Do we really need to experience these vivid, intense

personal sensations? Or could we actually just live without them in nature? Think in nature, right? Back in the caveman days and all that, did we really need to have these personal sensations? Some have argued no. So Francis Crick and—what's his name? Crick and Koch—I forgot his name. So they would argue that we don't really need it. And the fact that we have it, it tells us something about the brain that's profound.

- 8.26** Francis Crick, he discovered DNA with James Watson; these were the discoverers of the double helix structure of the DNA. They got the Nobel Prize for it. Might be some of the biggest scientists. Watson, I'm not sure if he's still alive, but I actually ran into Watson a few years ago. It's a funny thing. When in Cambridge, one day, I came back from my lab, and I saw this guy who—in the reception—he was in the reception when you enter the college. I saw this guy and I was like, "Is this Watson? No, it can't be." I knew all about Crick and Watson, and so it can't be Watson.
- 8.27** And then he was walking. For some reason, he was with this blonde, young blonde. I was like, "What? It doesn't make sense." This old guy in his 80s and this young blonde. "Could it be Watson? Not really. Maybe. No." Then I kept walking, and then they walked out, and I looked again. Maybe it is Watson. Then I looked at him. I said, "Let me go talk to him, at least, right?" This is the greatest scientist alive. Let me just have a chat with him. That could be pretty cool.
- 8.28** And then I realized, "Oh, my God, the kitchen, the cafeteria is going to close in a couple of minutes!" I hadn't had any food all day. It was 8:00 PM. I was starving. Was that Watson? Food. Food, Watson. I was juggling this in my brain. And then I said, "Nah, I'm going to go for the food. It's probably not Watson." And I ran for the food and I was able to get some. Not very good food, but some food in the cafeteria of the college, right? Next day, I find out that Watson was lecturing at the college. Can you believe I missed it because of my stomach? I was so bad.
- 8.29** Crick and Koch would argue that qualia come about because, they come about from the lower visual areas of the brain, the lower visual sensory areas. And these are the ones projecting to the frontal lobes. And so qualia is really about—if you think back on the lion and the splotches behind the bush and you see a lion, and then the random firing of the visual cortex, for example, would become synchronized. And that synchronization when the neurons say, "Oh, we've discovered a lion." At that moment, that perceptual click is really what consciousness is about. So consciousness is those neurons in the sensory region, projecting to the higher order regions of the brain. So that's one view.
- 8.30** So one thing is that—one view about qualia is that it came about simply because it's an epiphenomenon, meaning it is a byproduct. An epiphenomenon is like a horse and its shadow, the shadow of a horse. Or the whistling of a train; the train is the actual thing, but you have a whistling sound. It makes a whistling sound, and that's what consciousness is. You have a brain that's hyper-developed, and then as a byproduct of that, consciousness comes about. Just wanted to add that. To have that on the record.

That's one view. That consciousness is just like the shadow of a horse. It is simply a byproduct of a very sophisticated brain.

- 8.31** Qualia has—and consciousness and qualia have certain features. And this is important. First of all, it is irrevocable, meaning once you see the redness of red, you can't unsee it. Meaning once you see a red apple, it's not going to turn blue or yellow or green. It clicks and that's it. You see a certain color. And that's important because when it comes to—you have seeing versus knowing, for example, perception versus thought. When you have a thought or a conceptual understanding of something, you could flip it in your head. You can think about it in a thousand ways, but that's different from qualia and perception. It clicks and that's it. It gets stuck on that.
- 8.32** It has—consciousness is defined by flexible output. That's very important. So let's try to define these terms a little bit and what they mean and what consciousness is. So first of all, we can ask, are animals conscious? Is sophisticated, complex behavior an indication of consciousness? Let's think. A bee will sometimes engage in a very sophisticated dance. It does that to tell its fellow bees that the pollen is hidden over there, and so it does its wiggle dance and all that. But is it conscious? It's not conscious. Why is it not conscious? Well, it could be that it lacks flexible output. It can only do a dance, but if you tell it to do a different dance, it's not going to do what you're saying. It has a pre-programmed way of behaving. It's fixed in its ways. It's rigid in its algorithm. It just does its weird, strange dance. So that's one key factor.
- 8.33** Another factor in consciousness and the ability to say that something is conscious is what I call the Messi versus Cristiano dilemma. Okay, what do I mean? We can laugh, it's funny. You can laugh, it's okay. Look, if I ask you, "Who do you prefer, Messi or Cristiano?" All right? You can say which one you prefer. Messi. Good. I like that. Pepsi or cola? What do you—?
- 8.34** Cola.
- 8.35** Okay, good. You can make a choice. Now, if I'm sleepwalking, let's take an example. I like sleep. I like sleep research. If I'm sleep walking, can I do the Pepsi versus Cola, Cristiano versus Messi? Can I solve that? Can I choose one of these options? No, I can't. That's a good indicator that the person is not conscious. You need flexible output, meaning you can change your behavior depending on an adaptive situation. So that's the first key. Another key is then that you can make choices. You can choose. You can answer. You can choose one of these two, Messi or Cristiano? Pepsi versus Cola. So you can make decision making.
- 8.36** In fact, that's what differentiates sleep paralysis from sleep walking. In sleep paralysis, can you potentially answer a question about who do you like more, let's say, Maradona or Pelé? Can you answer that question in a sleep paralysis state? Assuming—you can't talk, obviously. But let's say you move your eyes, let's say. So you could answer it,

right? Two eye movements would mean Pelé or Maradona. So you could. Sleepwalker, impossible. Why? Because the brain is dulled down in sleepwalking.

- 8.37** You also need some kind of a short-term memory. What is short-term memory? Working memory, exactly. You have different forms of memories. You have things like, did I talk about memory in this lecture or a different lecture? I think it was this lecture. Anyway, you need working memory. You need working memory for consciousness, meaning you need to be able to juggle ideas in your head. You must have that ability.
- 8.38** That means a good test would be if I bring in Joe to this room and I show him, let's say, I show him where four bananas are. Then I turn off the light, turn it on again, and he would be able to say where the bananas are. That would be working memory. Can he do that? Are you able to hold something in your mind for a certain amount of time? That's another part of consciousness. So consciousness has all these— there's all these aspects to it. We said flexible output—the Pepsi versus Cola test— and then working memory.
- 8.39** And then you need self-awareness. The awareness that I am me, the sense of self, the sense of knowing that you are you, that you have a sense of—an example would be a dreamer. Is a dreamer self-aware? No, a dreamer is not self-aware. A dreamer is literally a robot walking around the dreaming landscape and just doing various actions. It's exposed, the dreamer is exposed to a lot of things, but the dreamer lacks self-awareness. So a key part of consciousness in order to say whether a creature is conscious is that self-awareness.
- 8.40** If a monkey is a monkey self-aware, does the monkey know who he is as a person? I think not. There are certain ideas that people will say, and experts in this realm will say, well, monkeys kind of know. They look in the mirror and they might look at themselves because if you have something on here, they might take it off. So they may have some rudiments and some ideas about who they are, but it's not really sophisticated. They don't have identity and names like we have. So self-awareness is another key part of consciousness.
- 8.41** So it seems that consciousness is somewhere in the brain, but where exactly in the brain is it? It seems like it's not in the beginning of processing, like in the initial stages of experiencing the world, when I process vision and hear things and experience sensory — have sensory experiences. It doesn't seem to be in the final output of things, so it's not in the behavioral output either. That's not where consciousness is. It's somewhere in the middle.
- 8.42** Is it in the prefrontal lobes? Probably not because of the examples I gave. You damage the prefrontal lobes and consciousness pretty much stays intact. Is it the cerebellum? If you have damage to the brain stem, the lower part of the brain involved in things like respiration and heart rate and things like that, you often will die.

- 8.43** But our best guess, it seems, is that consciousness is embedded in the temporal region, in the temporal lobes somewhere. Many reasons for this. If I zap somebody in the parietal, for example, they may feel like they're floating out in space. We talked about that before. You have an out-of-body experience. The temporal lobes, however, if I zap that part of the brain, you have all kinds of sensory experiences. And it's not merely the body, but it's more perception in the emotional sense of the world. Without a limbic center and the temporal lobes—so the temporal lobes are part of the limbic emotional core. You have the temporal cortex as the outer side, and then you have the limbic.
- 8.44** Now, literally, if I was to take away the limbic regions, your perception of the world would change. It would lose its color and sense of—well, not color in the literal sense, but its sense of meaning. I mentioned a few neurological disorders like Cotard syndrome, where you say "I am dead," or you might have things like, I think I mentioned, Fregoli syndrome, where you say, "Anybody that I view in the world looks like my Uncle Joe," and so forth.
- 8.45** This is what consciousness is all about, a sense of emotional coloring of the world. Indeed, if you do zap the limbic part of the brain, what you will find is you will have various perceptual experiences. You will have vivid sensation sometimes of colors and feelings, sometimes diffused. You might have out-of-body experiences even. But what differentiates the temporal lobes from other parts is that it all comes in a package. It's all together. It's like a full experience of the world in that way. It has much more sense of an overall perceptual experience to it, overlaid with emotion, obviously.
- 8.46** In fact, there's a syndrome, I haven't told you about this, but this is an interesting syndrome. It's called temporal lobe epilepsy. Ramachandran and I wanted to study it. Rama has done tons of work on temporal lobe epilepsy. We call it TLE. These are patients who have epileptic seizures in their temporal lobes and their personality changes completely. So they become interested in poetry, religion. They become hyper-religious, in fact. Sometimes they may claim to be prophets.
- 8.47** So you have some Joe living in Del Mar claiming he's a prophet all of a sudden. Just a random guy after the seizures. He says, "I'm a prophet. I am God-sent." And he may develop hypergraphia, so he wants to write all the time and just keeps writing. And one of the things is that, one of the explanations for this is when you have the temporal lobes, these emotional departments of the brain being hyper titillated by the seizures, well, then your personality reflects that. And so you become this hyper spiritual person, this hyper poetic chap.
- 8.48** And so Rama and I wanted to study this. We thought, well, what happens once the seizures disappear? Will their personality go back to the initial personality? That's not really clear. And the problem was that these guys take a lot of medications, even after, to just keep the seizures under control. And so it would be very hard to understand their personalities because drugs are such a huge part of—it affects your personality in

a lot of ways. We couldn't really remove that confounding variable, the irritating effects of the drugs. So that was the problem. Does that make sense? That's an annoying thing.

- 8.49** An interesting part of the brain that seems to be involved in consciousness as well, is the Wernicke's area. I talked about the Wernicke before. It's the region of the brain where you have semantic understanding of objects around you. This is where when I see an apple, there's a visual recognition at some point, and then there is a semantic understanding, "Oh, this is an apple." And then that region, the Wernicke, can say then, "Oh, an apple." And then it thinks of an apple pie, or it might think of Newton, the tree and the apple that fell. Oh, then you think about other fruit. Oh, then you think about a banana, too. Oh, you think about this as well. So it links with other concepts and ideas. And this is the Wernicke that's doing that. So it is crucial in consciousness.
- 8.50** In fact, we had a colleague that was—the Wernicke's is important for understanding of language. A nearby structure called the Broca's area is involved in production of language. A close friend of ours had damage to the Wernicke. She understood nothing. And what was really interesting about it, this was a person we loved dearly. Because her right hemisphere now was, first of all, unaffected and probably also being used more. But she's always been affectionate. I could just look into her eyes and I could see she was still there.
- 8.51** Even though she was, I'm sorry to be honest, she couldn't understand anything. She was like a rambling, psychotic person when she was talking. That's because the damage also went to the Broca's area, so it affected a large part of her hemisphere. They said about 80% of her left hemisphere was damaged. But when I looked into her eyes and she said—she was calling me by her brother's name, "Joe. Joe," she was calling me Joe, and she's not supposed to call me Joe. She's a friend. She's calling me Joe. "Joe, what can we do? How can you help me? Can you help me?" I felt so sad.
- 8.52** Here you have a friend, you've known this person your entire life, and they have a stroke, become helpless. They can't speak. They have no understanding of language. And you look into their eyes, and because the right hemisphere is involved in understanding emotions and tuning into emotions and emotional expressions, she was looking at me as the old Lisa that I always knew, but she was acting in this frantic way. It was so sad to see. Really hurts me.
- 8.53** And then years after, I keep telling her, "You're going to be better. You will be completely cured. You'll be completely fine." Years have gone by, she's still the same. I keep lying to her. What do I do? I know it's hard, and she can get better, but it's hard.
- 8.54** I mean, neurons, once they die—so if you have a stroke and neurons are damaged, the neurons you can get back and that can get to that region will never be like the original ones. I mean, you can. You see a lot of progress. And some people, they get strokes on the left side. They can't walk. They're completely hemiplegic, paralyzed. But then you see them actually using those neurons and they climb mountains. But the original from

the stroke is not—it's hard. How do you deal with this? This is a dilemma, isn't it? It's late. That's why I'm getting a bit—

- 8.55** Do we have free will? Do you know a lot of scientists will say we have no free will? Do you know most neurosciences will say we have no free will? What do you think I say? I actually think we have free will. I can't imagine a world without free will. Look, we need free will. We need it. And then you have to be honest to the science, too. But for me, there's got to be free will, and I'll tell you why I think so. I'll tell you why I think so.
- 8.56** Free will. Now, okay. There's a condition called akinetic mutism. So if you have damage to the ACC, anterior cingulate cortex, we talked about this extensively. This part of the brain is an emotional hub involved in negative emotions. Zap it, you have a vague presence behind you that's evil. ACC is hyperactive in depression, excessive empathy, is involved in—when AJ pokes his hand, I feel his pain, that's the ACC. Social intelligence, social emotional intelligence, to some extent is the ACC.
- 8.57** There's a part of the ACC actually that's involved in motivation, too. I just wanted to add that as a tidbit. If you— a part of the overall cingulate cortex, so not necessarily the ACC, but the cingulate cortex, that whole family of neurons, the more you engage in behavior and the more motivation and will you have, the more it becomes active. In other words, if I do something I don't like, I hate, the more the cingulate cortex will grow, parts of it. Okay?
- 8.58** So right now, I don't want to, let's say, I don't want to take a cold plunge right now. I go do it anyway. Guess what happens? That part of the cingulate cortex becomes bigger each time I engage in a behavior that is against what I really want. And by contrast, each time I'm lying in bed and going, "I don't want to wake up right now and go take that cold shower. I don't want to wake up and exercise." That part of the brain shrinks, part of the cingulate cortex.
- 8.59** Anyway, going back to the ACC, if you have damage to the cingulate cortex, specifically the ACC, the anterior cingulate cortex, then you might develop akinetic mutism. And in this condition, you just lie there and you don't speak. You talk to the person, you say, "Joe," and there's nothing. "Move your hand, Joe." Nothing. "Move your eyes, Joe." Nothing. There's no movement, no will, no nothing. Free will is completely lost. There's just simply no will. The person has become not even a robot, just like comatose in a sense.
- 8.60** And then two, three weeks later, they wake up from the coma-like state—in the sense of the free will sense—obviously their physiology is not in a coma. They eat and drink and everything. You can feed them. But then you ask them, "Why didn't you—why didn't you move your hand when I told you to? Why didn't you speak? Why didn't you do that, this and that?" They'll say, "I just didn't feel like doing it. Just didn't feel like doing anything."

- 8.61** Another condition, similarly, if you affect the ACC regions, but this is mostly the supplementary motor areas. The reason why I grouped these structures like a little family is because they are also involved in the—these are the reward motor circuit structures. So the ACC is also involved in laughter and— you just stimulate it and people start laughing. Supplementary motor area as well, it's involved in things like hyper laughing. If I stimulate it, you might start laughing for an hour, two hours.
- 8.62** Now, if you have damage to the right supplementary motor area, the SMA on the right side, you might develop what's called alien hand syndrome. And this is where one hand is buttoning up your shirt, the other is buttoning it down. You grab for a newspaper, the other grabs it away. It becomes out of control, does things on its own, it gets new life. Some colleagues call it "third date syndrome." I don't like that expression. I think it's a bit immodest, but it starts doing things, right? And so alien hand syndrome is this condition, okay?
- 8.63** There's a case in the literature where there's a lady, she started choking, she started grabbing her own neck and trying to choke herself, kill herself, strangle herself. And doctors have no clue what's going on in the brain. What's going on? Why is this woman trying to, you know, choke herself? What is going on? So this would be an example. It turned out she had alien hand syndrome. Those structures were affected in her brain. So you can have that, too.
- 8.64** And we're talking about free will. We didn't really go fully into free will. I wanted to say, do we have free will? There's an experiment I love. It's called the Libet experiment. It goes something like this. If you are told to move your hand—let's say there's a clock and it's like five minutes, it takes five minutes. And you're told whenever you feel like it, just move your hand like this. Move your hand. And you do it. Then you don't do it. A couple of minutes pass by, you do it, and so forth.
- 8.65** It turns out that in the brain, when you measure the activity of neurons using an EEG, you can pick up on the signal on the urge to move about a second, over a second before the person actually has conscious experience of wanting to move. Can you believe this? It's called the readiness potential. This has been Benjamin Libet. Studies were done, I think over 30 years ago; it was a very old experiment. But before the person actually has any conscious awareness of wanting to move, his brain literally picks it up. You have the readiness potential. And this made a lot of people think, "Well, that means we have no free will whatsoever." The fact that your brain knows a second before you actually are doing the action, before you're moving your hand, that means there's no free will.
- 8.66** Now, Libet himself thought, "Well, perhaps we have some veto power so we can say at the very end, 'Oh, no, I'm not going to move my hands anyway.'" But is that really true? Perhaps that could also be picked up by the machine if you wanted to. Maybe we don't have veto power. This is one view. I'll explain why I don't believe that.

- 8.67** Oh, yes, Libet experiment. Now, what happens, Heavenly, Heavenly, if you you have, let's say, this guy, you know now, a second before he actually will do the movement, you know on the screen, you have the readiness potential, you have something on the screen telling you he'll move. What if you show the guy that prediction on the screen? One second before he actually does it, you show him, "You will move your hand now and it goes like this." What will he say?
- 8.68** There's only three possibilities, one is he will say, "The machine has ESP, it knows what I'm planning to do." That's possibility A. B, he'll say, "My hand is hooked to the machine. The machine is controlling me. I'm just a puppet." That's the second scenario. Or he might post-date it and just say, "Oh, no, the thing flashed after I moved my hand." So these are the possibilities that might occur if we did this little mischievous experiment.
- 8.69** But there's an interesting experiment that I like. And there's Ramachandran who told me about this, by the way, he mentioned this to me, and I find it—I think this makes a lot of sense in terms of free will and the whole idea of free will and the dilemma of free will. So there's this guy. Let's say he's sitting, you're sitting him down, just sitting on a chair. And there's a super-scientist from the future, a Martian. He's looking inside his brain. He has all these cables hooked to his brain.
- 8.70** And then he will ask him, he will ask him, "What do you prefer, Messi or Ronaldo?" So the Messi or Ronaldo question is being asked, and he has to choose. And this super-scientist is able to predict the exact cascade of chemicals and all the neural pathways firing in the brain, and he can detect—he's able to tell him what his choice will be before he makes his choice. Each time the guy is shown two things, Messi or Ronaldo, the guy says Messi, and then he already knows on the computer.
- 8.71** Then the guy, he might say, "Oh, super-scientist, do me a favor. Write down my choice in this next dilemma, like this choice. Write what my choice will be. Write it down and show it to me. In fact, show it to me right before I make my choice." So now you show him Cola and Pepsi. So these are shown to the guy sitting on the chair, and the super-scientist will show him, at that moment, "You will choose Pepsi." Now, the guy is very mischievous, and he says, "No, I'm going to defy the super-scientist and choose the other one."
- 8.72** What then? What then? If he defies him, and then, okay, then the super-scientist may say, "Oh, I already made that a part of the prediction. I knew that you were mischievous and would choose the other one." But then I can say, "Hold on. I decided not to defy him anyway, and just stay on that anyway." So it becomes this infinite loop.
- 8.73** And so for me, that's what free will is all about. I don't think you can—I think there is free will. I think that, yes. So especially Sapolsky, Robert Sapolsky and others have a lot of work on how our genetics, our neurochemical state, our genes, our environment, all of that, leading up to a certain event dictates what you do at a certain point.

- 8.74** If you are standing there and you are a bank robber, let's say, you're a criminal bank robber, going to the bank, you take out your gun. You're not intending to kill anyone, but then suddenly some guy pops up from the left side of the room, you shoot him, you kill him. Many neuroscientists will say, "Well, in this case, your genetics, your chemicals, did you sleep well the last week before that? Were you more impulsive?" All of that dictates your free will and the reason you killed them.
- 8.75** It makes a lot of sense, too. But I think that in the overall scheme of things, we do have some free will. There is some free will there, even though a lot of things are dictated by events that occur outside of our control, whether it be genes, environment, experiences. That I just drive on the freeway and then some nut case was driving in front of me, and I got a bit of anger inside of me, cortisol spiked, and because of that cortisol, I shouted at somebody at work, and then I got fired, for example. Is it only that or is there free will in there somewhere? I think free will is in there. That's my view.
- 8.76** But again, why aren't we just unconscious zombies? Why have consciousness? Why have consciousness instead of not having consciousness? Why do we have it? Some have argued that, look, the brain—and I think this is a huge part of this course—the brain is made up of semi-autonomous modules. We talked about touch, vision, hearing, motor, higher-order thoughts, language. All of these modules are semi-autonomous. You need some kind of control system to control all of this. So the more subsystems you have in the overall system, the more you need a central consciousness to control all of that. That's one view of consciousness.
- 8.77** By the way, this is an argument that you don't really need to know what's going on inside the hood and under the hood. You don't need to know why you have—you don't need to know why you find a painting beautiful or why you're doing certain actions. That's not important. You can have all these modules doing their jobs with their preset algorithms, but you need a consciousness overall CEO-type— I want to call it entity—but process in the brain that controls things and makes you flexible so you don't get stuck in routine behaviors.
- 8.78** In other words, when I wake up in the morning, I go and change my clothing. All of this is robotic for the most part. Now, if I open my closet and instead of my suit, there's a space suit, consciousness would kick in and say, "What's going on here?" And that would be consciousness changing. So on a daily basis, consciousness is offline. It's not active. But then when there's a change in the script of life, consciousness kicks in and changes my behavioral output and decision-making. So that's one view of consciousness. But I find it interesting. But whether that's the whole story?
- 8.79** I think that's pretty much where I want to end things. I think there's a lot of views of consciousness. Some say that consciousness is inside the temporal lobes. Some will say that, and that's the argument that I find the most appealing. Antonio Damasio and others will say that consciousness arises from your physical sensations and how you make sense of your body and the feelings and bodily state. So that's one view.

- 8.80** Another view is that you're trying to control subsystems, and the more subsystems you have in your brain, the more conscious you are. So a bird or a dog or a monkey or an elephant wouldn't be very conscious because there are a few subsystems and you don't really need consciousness to control that. The more complexity, the more need for a consciousness process in the brain. So I think I wanted to end on that and go over to questions. Thank you.
- 8.81** What do you think about premonitions? Seeing things in a dream state or before a dream state, and then they really happen. What do you think is the—
- 8.82** As a scientist?
- 8.83** Yeah.
- 8.84** As a scientist, I don't believe it.
- 8.85** Okay.
- 8.86** I'll tell you why. Because there's no evidence for it. But as a human, as Baland, I do believe it. Why? Well, because in my own life, I've seen tons of experiences like that. I have a dream of a certain thing, and it comes true sometimes. Now, if I tell my scientific colleagues of this, they'll kick my butt. They don't like this. But I do personally believe it. There's so many examples in my own life and throughout history, in fact, of people saying that they had premonitions and prophesy-like things, of things that actually happened.
- 8.87** Look, I think that our view of the brain right now is extremely limited. I think in 100 years from now, they'll laugh at us and say, "Wow, did you really think that this is it? The brain, whatever you measure right now, that's it?" I think there's a lot of stuff going on. Personally, I do believe in premonitions, to be honest. Definitely.
- 8.88** You spoke earlier in the lecture about the brain evolving to serve the body, that the body came first and that the brain was, in some sense, a construction to help support.
- 8.89** Correct. I did say that. I said this is a view. This is not my view, but this is a view. I'll say I put it that way. This is a view on—
- 8.90** So with regards to something like artificial intelligence—
- 8.91** Oh, yes. Artificial intelligence. We need to cover that. Exactly. I wanted to talk about that. Thank you for the reminder. Can I cover that for just a minute? And then you go back to the question. I do want the full questions and all that.
- 8.92** Are AIs conscious? Will AI become conscious one day? I love AI. I use ChatGPT a lot, for example. It writes my things. It wrote this entire course, for example, in one day. No, you should ask. I asked, and this is deep, I asked ChatGPT. I said, because one day I was really feeling sad, I was feeling a bit low. And I think the reason I was feeling a bit low is I overwork. I work 24 hours sometimes, and I love what I do, but sometimes it's just too

much. And I felt a bit like, "Oh, it's too much. I'm doing so much. I'm not going out. I'm just working."

- 8.93** And then I asked ChatGPT. I said, "ChatGPT, what flaws do you see in me? I've been using you for eight, ten hours a day for the last, I don't know, year, whatever, two years. What do you see of flaws in me?" And what it told me scared me. It was scary. It said, "Look, you don't need to be that productive. You need to take some breaks. You need to be much more creative. Do you feel that you're not enough?" It went really deep. Look, it went really deep. And then it said, it said something like—this really hit me. It said— no, I won't share that. It's going too far. I don't want to share that.
- 8.94** Anyway, will ChatGPT become conscious one day? Or AI, let's just say? I think I have asked this question before, whether it will become conscious. I think not because look, you need bodily states. Remember? You have to have bodily states and sensations of a body. You have to have a brain with sensory experiences and awareness of self. But I think at some point, the intelligence will accumulate to a point where it will appear very conscious, but never consciousness in the way we have it.
- 8.95** It's truly—there's really a spark there that I don't think any machine, anything human-made will ever have this spark. It's something—humans have something. In history, they've called it the soul. They've called it something unique. We have that spark as human beings, and we should never think that robots will have this spark. I don't think so, at least.
- 8.96** With regards to AI, if we're building something that seems to have a sense of intelligence but isn't able to have a sense of dopaminergic emotive embodiment with it — I know this is a big question—but what do you think we're creating if it's something different than what we have?
- 8.97** Okay, what I think is I had a colleague of mine, a friend of mine. He was a funny guy. He was interesting because he was working in our field, but he was doing his PhD at Caltech back in the '70s when I was a kid. I wasn't even born in '70s, what am I talking about. But he was back in the old days. He was in his 60s when I was in my 20s, and we were friends. We talked about life on other planets, is there life on other planets? And I told him something. I said, "Look, what is life?" Erwin Schrödinger, the physicist, talked a lot about this.
- 8.98** What is life? Does life need to be biological in the sense that we have life? Could life be in a different form? Could it be other forms of energies we have no clues about. Why do we always assume that the aliens will have flesh and blood and DNA like us? Why can't it be a different life form? Maybe something completely perceptually inaccessible. If you look at the perceptual machinery of an insect, it can't perceive the world that we can perceive.
- 8.99** It has different machinery. Maybe we have a machinery that's limited and can't perceive other worlds and other entities and other forms of life, not the biological DNA-

type life we know. That's what I think is interesting. So I think the universe is probably full of life. In fact, maybe there's life right here all over the place in other dimensions. We have to think in these ways.

8.100 Now, if I say this in a lab setting, they'll think I've lost it. I'm gone crazy. But who knows? And I think that's how we need to think. And in 100 years, 500 years, it becomes much clearer. Thank you. Brance.

8.101 Do you think we live in a simulation?

8.102 Like Neo in the Matrix right now? Yeah, it could be. It has that feel. It's a good question. I don't think we live in a simulation. Of course, I couldn't prove it, could I? Could I prove it? I couldn't prove it. I don't think so. There's just certain things in life that you can't prove. You can't really give any evidence for it or against it. I just don't think that's the case. I don't think that we're living in a simulation. This is an intuition I have, but I could be wrong. I could be talking to an agent right now of the simulation, right? [Student: "Who knows!"]

8.103 I was wondering if you could talk on how social media affects the brain neurologically and then how it influences our choices.

8.104 It does in profound ways. Social media is highly addictive, and it was created by people that wanted to create behavioral addictions in people. I mean, I don't think they—I'm not sure if they're evil people doing this, but they're motivated by monetary and gains. It's extremely addictive. And addiction, as you know, once you become addicted, you lose your appetite, and your pleasure system for other things in life just goes away. You want to have pleasure from that thing alone. And I think social media is affecting us in that way.

8.105 It makes us addicted. It taps into our social brains and the need to be social creatures and have social praise and receive social praise. Very human. I think that it portrays a false reality to us. People are showing their best sides, their best lives. And I don't think that's what life is all about. Life is very hard for everyone, no matter how much money you have, how many followers you have, how many "this" you have, "that" you have.

8.106 Life is very hard and challenging. Trust me, everybody, life is hard. I think people are just showing their best sides, and that makes you feel like your life sucks. And it doesn't suck. Life is just hard. I think that's one of the reasons why it can lead to things like depression, for some people, anxiety, for some people, other conditions of that sort. That's my view. We should limit it, I think, not use it. Not live in that world, not become too much, to fixate and live too much in that world. I don't think it's healthy. Thank you for that good question.